IVAN OVINNIKOV

REINFORCEMENT LEARNING FROM DEMONSTRATIONS: METHODS AND APPLICATIONS IN DIGITAL TWIN SIMULATIONS

DISS. ETH NO. 30388

REINFORCEMENT LEARNING FROM DEMONSTRATIONS: METHODS AND APPLICATIONS IN DIGITAL TWIN SIMULATIONS

A dissertation submitted to attain the degree of

DOCTOR OF SCIENCES
(Dr. sc. ETH Zurich)

presented by

IVAN OVINNIKOV
MSc. ETH in Information Technology and
Electrical Engineering
ETH Zurich

born on 06.02.1990

accepted on the recommendation of

Prof. Dr. Joachim M. Buhmann, examiner
Prof. Dr. Andreas Krause, co-examiner
Dr. Raimundo Sierra, co-examiner

2024

Ivan Ovinnikov: *Reinforcement learning from demonstrations: methods and applications in digital twin simulations,* © 2024

DOI: 10.3929/ethz-b-000708755

To my family

ABSTRACT

Modeling the decision making behaviour of agents embedded in interactive environments is crucially important for achieving truly intelligent artificial agents capable of interfacing with the real world. The elicitation of meaningful adaptive behaviours occurs by specifying a control loop that minimizes an error functional describing the task objective. Adaptive agents possessing such a loop are capable of learning meaningful behaviours via optimization of quantities extracted from the environment they interact with. In the formalism of *reinforcement learning*, this training signal is described by the reward function of a Markov decision process. Often, the reward specification is difficult or impossible to obtain. Such a challenge can be alleviated by adopting the *learning from demonstrations* formalism. This formalism provides training signals from examples of previously demonstrated behaviours. This thesis aims to investigate how behavioural patterns can be inferred via imitation of experts in diverse environments.

The first part of this thesis studies the specific setting of surgical simulation where we develop an algorithmic pipeline for performance evaluation of surgical practitioners based on training virtual reinforcement learning agents. Digital twin representations of real world settings provide a fertile ground for reinforcement learning based approaches due to their capacity for near-unlimited interaction and the inherent ease of data acquisition. The results suggest a novel, principled way to evaluate and assist users of surgical simulations.

In the second part of the thesis, we present two algorithmic improvements of the proposed pipeline. The first tackles the issue of learning from diverse demonstrations, a problem which arises due to varying modalities of measured expert behaviours. We address this learning task by leveraging tools from causal inference to derive causally invariant representations of reward functions, thereby improving generalization properties across a range of environment dynamics. The second algorithmic addition extends the distribution matching perspective underlying the learning from demonstrations framework through the lens of optimal transport. The result is a novel imitation learning method based on sliced Wasserstein distances which significantly improves sample efficiency of online imitation learning methods.

ZUSAMMENFASSUNG

Die Modellierung des Entscheidungsverhaltens von Agenten, die in interaktiven Umgebungen eingebettet sind, ist von entscheidender Bedeutung für das Erzielen wirklich intelligenter künstlicher Agenten, die in der Lage sind, mit der realen Welt zu interagieren. Das Hervorrufen von sinnvollen adaptiven Verhalten erfolgt durch die Spezifikation eines Regelkreises, wodurch eine Fehlerfunktion minimiert wird, die das Aufgabenziel beschreibt. Im Formalismus des Reinforcement Learning (z.Dt. bestärkendes Lernen) wird dieses Trainingssignal durch die Belohnungsfunktion eines Markov-Entscheidungsprozesses beschrieben. Oftmals ist es schwierig oder unmöglich, die gewünschte Belohnungsspezifikation zu erhalten – eine Herausforderung, die gemildert werden kann, indem der Agent aus erfolgreichen Demonstrationen lernt, was durch den Formalismus des *Lernens aus Demonstrationen* beschrieben werden kann. Das Ziel dieser Arbeit ist es, das Verständnis von Verhaltensmustern zu erweitern, welche durch Nachahmung von Experten in vielfältigen Umgebungen erlangt werden.

Der erste Teil dieser Arbeit untersucht das spezifische Setting chirurgischer Simulationen. Hier entwickeln wir eine algorithmische Pipeline zur Leistungsbewertung von Chirurgen in Ausbildung, die auf der Schulung der virtuellen RL-Agenten basiert. *Digital Twin*-Darstellungen von Analoga aus der realen Welt bieten einen fruchtbaren Boden für Ansätze, die auf Verstärkungslernen basieren aufgrund ihrer Kapazität für nahezu unbegrenzte Interaktion und der Einfachheit der Datenerfassung. Die Ergebnisse deuten auf eine neuartige, prinzipielle Methode zur Bewertung und Unterstützung von Benutzern chirurgischer Simulationen.

Im zweiten Teil der Arbeit stellen wir zwei algorithmische Entwicklungen vor, die die obengenannte Assistenz-Pipeline erweitern. Die erste befasst sich mit der Frage des Lernens aus diversen Demonstrationen – ein Problem, das aufgrund unterschiedlicher Modalitäten der vorhandenen Demonstrationen entsteht. Wir gehen das Problem an, indem wir kausal invariante Representationen der Belohnungsfunktionen herleiten, was in verbesserten Verallgemeinerungseigenschaften resultiert. Die zweite algorithmische Ergänzung schlägt eine neuartige Nachahmungs-Lernmethode vor, die auf geschnittenen Wasserstein-Distanzen basiert und die Abtastungseffizienz des Online-Nachahmungslernens erheblich verbessert.

ACKNOWLEDGEMENTS

I am deeply grateful to Prof. Dr. Joachim M. Buhmann for granting me the opportunity to pursue a doctoral degree under his guidance. It has been a privilege to explore diverse research directions and engage in enriching discussions throughout the years. I am also thankful for the chance to contribute to a project within the medical industry alongside the dedicated team at Virtamed, led by Dr. Raimundo Sierra, who generously served as a member of my committee.

I extend my sincere gratitude to my co-referee, Prof. Dr. Andreas Krause, for his encouraging feedback and the insightful exchanges with his research group. I would also like to thank Prof. Dr. Markus Gross for the valuable time spent at the Computer Graphics Laboratory and for serving as chair of my examination committee. Additionally, I extend my appreciation to my colleagues at the Institute for Machine Learning and, in particular, to Rita Klute, whose steadfast support has been a source of calm during challenging times.

To my parents, Valery and Natalia Ovinnikov, and my brother, Serguei, I owe my deepest gratitude for their constant support and unconditional love. I am also immensely thankful to my friends for their encouragement over the years, especially Nils Wenzler, for sharing this journey in countless ways.

Lastly, I express my profound love and appreciation to my partner, Giulia Martinelli, for her kindness, strength, and unwavering support. I look forward to continuing to build our family together and embracing the journey ahead with our son, Nikita.

CONTENTS

NOTATION

FREQUENTLY USED SYMBOLS

$\mathcal{M}$	Markov decision process (MDP)
$\mathcal{S}$	State space
s_t	State at timestep t
$\mathcal{A}$	Action space
a_t	Action at timestep t
ξ	Trajectory
r	Reward function
μ_0	Initial state measure
γ	Discount factor
τ	Transition dynamics of an MDP
π	Policy
ρ_π	State occupancy measure of policy π
$\mathcal{D}_E$	Dataset of expert demonstrations
ρ_E	State occupancy measure of the expert
$\mathcal{B}^\pi$	Bellman operator under policy π
G^π	Empirical return of a policy π
V	Value function
Q	State-action value function
g	Discriminator function
ψ	Reward function parameters
ϕ	Value function parameters
θ	Policy parameters
D_f	f-divergence measure
$v_{\mathcal{F}}$	Integral probability metric
$\mathcal{W}_p$	p-Wasserstein distance
$\mathfrak{G}$	Structural causal model (SCM)

ABBREVIATIONS

MDP	Markov Decision Process
RL	Reinforcement Learning
IRL	Inverse Reinforcement Learning
TD	Temporal Difference
DP	Dynamic Programming
PPO	Proximal Policy Optimization
SAC	Soft Actor Critic
GAN	Generative Adversarial Network
GCL	Guided Cost Learning
BC	Behavioural Cloning
IL	Imitation Learning
GAIL	Generative Adversarial Imitation Learning
AIRL	Adversarial Inverse Reinforcement Learning
SWIL	Sliced Wasserstein Imitation Learning
ITD	Information Theoretic Divergence
IPM	Integral Probability Metric
OT	Optimal Transport
JKO	Jordan Kinderlehrer Otto
RKHS	Reproducing Kernel Hilbert Space
SCM	Structural Causal Model
FAST	Fundamentals of Arthroscopic Skills Training

1

INTRODUCTION

Intelligence is the ability to adapt to change.
— Stephen Hawking

Modeling adaptive decision making agents marks a cornerstone of artificial intelligence research. The adaptive control of dynamical systems using the *reinforcement learning* (Sutton and Barto, 2018) formalism has seen a significant amount of progress over the last few years due to advances in neural network training (Kingma et al., 2014). These advances have enabled the translation of adaptive decision making algorithms into high-dimensional domains allowing the elicitation of highly performant behaviors in a variety of demanding tasks such as playing complex games (Mnih et al., 2015; Silver et al., 2017; Vinyals et al., 2017), various robotics tasks (Billard et al., 2008), maintaining a dialogue with humans (Ouyang et al., 2022) or even controlling a nuclear fusion reactor (Degrave et al., 2022).

In order to successfully train and deploy such agents, a simulated environment is typically required. This restriction arises due to cost associated with two factors: (i) the control loop is fundamentally limited by real-time constraints; (ii) during training, the agent may perform actions which are unsafe, leading to damage or irreversible deterioration of the physical system it is interacting with. A simulated version of the environment alleviates these restrictions by providing a way to gain computational efficiency via parallelization and faster than real-time processing as well as providing safe interaction capacity, thus avoiding negative consequences of unsafe actions.

The advent of widespread digitalization and the pervasive nature of computation in today's world have enabled the development of sophisticated simulations, often denoted *digital twins*, of various real-world domains. Examples include virtual replicas of physical objects (Kalidindi et al., 2022), systems (Y. Jiang et al., 2021), or processes (Van der Valk et al., 2020). This concept is gaining traction across various industries, including manufacturing, healthcare, automotive, and urban planning, by improving the understanding and management of complex systems. The medical domain particularly benefits from simulations due to its safety critical nature. It features notable examples of digital twins such as oncological or cardiological models (Croatti et al., 2020; Laumer et al., 2023; Katsoulakis et al., 2024) resulting in a viable alternative for a number of downstream tasks, including analysis, diagnostics and training of human practitioners.

This type of simulation provides a robust platform for deploying and studying learning agents within a controlled environment. In the broader field of machine learning, where the availability of data is essential, these systems consistently generate a stream of interaction data that can be utilized for learning. Such platforms are particularly valuable in contexts like medical decision-making, where safety is paramount and the exploration of the behavioral space can be offloaded to a simulated domain to mitigate risks. This thesis explores the surgical simulation platform as a particular instance of digital twin technology, that replicates human anatomy in a digital format, allowing for detailed study and testing of interactive agents within that environment.

The elicitation of specific, complex behaviors such as surgical procedures within a control loop of a digital twin largely relies on a detailed specification and meticulous finetuning of the learning objective. As the complexity of the simulated environment increases, so does the necessity to specify a more complex objective. This growth in complexity is challenging as accounting for all the intricacies of a complex task becomes an exercise in identifying corner cases of the induced behavior. The design of such a learning objective, i.e. cost or reward functions, poses a significant challenge on the path to a broader adoption of the obtained adaptive behavioral models in practice.

To overcome the obstacle of hand-crafting such functions, we turn to the *learning from demonstrations* framework (Schaal, 1996) to derive a training signal for adaptive agents from datasets of reference behaviors. This domain encompasses a broad scope of techniques ranging from supervised behavioral cloning (Pomerleau, 1991) and preference learning approaches (P. F. Christiano et al., 2017; Brown et al., 2019) to imitation (Ho et al., 2016; Fu, Luo, et al., 2018) and inverse reinforcement learning methods (Ratliff et al., 2006; Ziebart, J. A. Bagnell, et al., 2010) . The framework of *inverse reinforcement learning* is of particular interest, since it allows to derive the reward signal from a dataset of expert demonstrations for downstream tasks such as evaluation of novices or training of assistance agents in a transfer setting. In this thesis, we primarily focus on the distribution matching formulation (Kostrikov, Nachum, et al., 2019) of such algorithms and investigate the resulting elicited behaviors.

1.1 MOTIVATION

As an example of such adaptive behaviors, we study the acquisition of surgical skills in simulated environments. Teaching complex surgical skills in an efficient and rigorous manner will have important implications for the quality of overall health care. Some of the benefits that can result from applying quantitative data driven methods are a reduced number of injuries, shorter procedure times and an improved understanding of surgical decision making. In order to approach this sophisticated problem from a machine learning perspective, we model the behavior of surgical novices in order to adapt the teaching process accordingly. A suitable way of modeling this behavior is to consider a sequential decision making agent learning from expert demonstrations.

The resulting algorithmic pipeline for learning such behaviors shows a number of appealing properties, making it an appropriate candidate to derive feedback signals for teaching. The central feedback mechanism when learning from demonstration utilizes the proximity of the novice behavior to the expert reference. To quantify this proximity, a number of metrics can be used, for instance various distribution matching methods (X. Nguyen et al., 2009; Sriperumbudur et al., 2009), that are defined globally over a dataset of demonstrations. However, in order to provide a meaningful assistance to a novice, a fine-grained feedback might be needed on a per-decision-step basis. The reward function and imitation agent policy obtained by using IRL methods provides a principled way to generate such feedback for any state the novice might find themselves in. Another benefit of employing this algorithmic choice is the data efficiency with respect to the expert demonstrations, which are typically expensive to obtain. Due to the continuous interaction with the environment enabling the statistical support estimation of the expert policy, IRL methods allow one to obtain successful behavioral policies from a very low number of demonstrations (Arora et al., 2021).

THESIS GOALS In this dissertation, we develop an algorithmic pipeline for simulated surgical environments based on sequential decision making agents which learn from demonstrations. For this purpose, we formalize surgical skill development in the context of inverse reinforcement learning algorithms. We build the foundation for deploying decision making agents in these virtual environments with the purpose of assisting human users of such systems by converting a benchmark suite of surgical tasks for RL training. The main goal of this benchmark is to train virtual agents which are capable of assisting surgical novices by providing intra-operative feedback. This goal requires to insert the knowledge of an expert practitioner into a virtual agent capable of performing the task as close as possible to the expert behavior. Once trained, we obtain access to fine-grained feedback mechanisms given by (i) the inferred intent of the expert, represented by the reward function of the MDP, which allows evaluation of trajectory subsets and (ii) the inferred imitation policy which is trained to optimize this reward, allowing queries of trajectory suggestions for assistance purposes.

1.2 DISSERTATION OVERVIEW AND CONTRIBUTIONS

The dissertation is structured in two parts. The first part presents the algorithmic pipeline that employs reinforcement learning agents for the surgical simulation digital twin. Here, we establish a series of benchmark tasks within two domains of minimally invasive surgery. The first is the domain of arthroscopic procedures focusing on foundational skills necessary for instrument manipulation. The second is the laparoscopic domain, where we study the diagnostic tour problem of the abdomen. We demonstrate the utility of the suggested pipeline by presenting an evaluation protocol for a number of human trainees of the simulated domains.

In the second part, we focus on two algorithmic extensions of the proposed pipeline. The first extension solves the challenge of learning from diverse experts, where we present a method based on the principle of causal invariance which extends the maximum entropy IRL framework in both tractable and approximate settings. The second extension deals with directly obtaining the policy mimicking the experts by leveraging a novel, more sample efficient way of performing imitation learning based on generalized sliced optimal transport distances.

In the following, we provide an overview of the individual chapter contributions:

- Chapter 2 lays the groundwork for the thesis and presents the frameworks used therein. We give an overview of the foundations of reinforcement learning techniques as well as an introduction to the distribution matching perspective on imitation learning and reward inference.

- Chapter 3 presents the algorithmic pipeline for surgical teaching assistance in simulation. Our contributions are as follows: (i) we develop a set of benchmark tasks based on two minimally-invasive surgical simulation scenarios. The first deals with instrument manipulation in an arthroscopic setting, which serves as a prerequisite course for training surgical novices. The second scenario focuses on the laparoscopic setting, where the goal is to perform a diagnostic tour of the abdomen using an endoscopic camera. This scenario features a more complex topology due to the safety critical aspect of avoiding contact with the anatomy of abdominal structures. In order to develop meaningful assistive behaviors, we propose an algorithmic pipeline based on modern RL algorithms to train virtual agents capable of assisting the teaching procedure. As a first step, we develop a number of heuristic reward functions for the proposed benchmark tasks to enable the elicitation of the desired behaviors in simulation. In a second step, we employ the inverse RL framework to infer a reward functions from a dataset of human demonstrations, thus obtaining the reward functions specific to a cohort of expert practitioners. By training a policy using this reward, we effectively transfer the expert knowledge of the cohort to a virtual agent's policy, which can be queried to provide suggestions. We validate the proposed evaluation protocol on a dataset of human demonstrations, where we recover the ground-truth heuristic ranking.

This chapter is largely based on the following publication:
Ivan Ovinnikov, Ami Beuret, Flavia Cavaliere, Joachim Buhmann
FASTRL and beyond: a reinforcement learning pipeline for fundamentals of arthroscopic surgery training
International Journal of Computer Assisted Radiology and Surgery (IJCARS), `https://doi.org/10.1007/s11548-024-03116-z`

- Chapter 4 describes the first algorithmic extension of the proposed pipeline. More often than not, the IRL assumption of truly optimal behavior is violated and it is desirable to effectively leverage sets of demonstrations of suboptimal or heterogeneous performance, which are easier to obtain. However, the diversity of experts due to different sources of heterogeneity such as individual expert preferences or varying recording circumstances poses a challenge on the way to obtaining succinct representations of the task rewards. We tackle this issue from the perspective of expert label noise arising from the pooling of diverse demonstrations under the same optimality label. In order to distill task-relevant information into the inferred reward, we propose a regularization strategy for IRL algorithms based on the principle of causal invariance. We demonstrate the improved generalization performance of the obtained rewards in a transfer setting across a range of environment dynamics.

This chapter is based on the following preprint:
Ivan Ovinnikov, Eugene Bykovets, Joachim Buhmann
Learning Causally Invariant Reward Functions from Diverse Demonstrations
Transactions on Machine Learning Research (TMLR), in review

- Chapter 5 describes the second algorithmic extension of the proposed pipeline. In this chapter, we focus on sample-efficient policy learning via direct imitation as opposed to the full IRL loop which requires the estimation of a reward. Imitation learning methods typically rely on an adversarial approximation of f-divergence measures. Information theoretic divergence measures between probability distributions do not generally reflect the geometry of the underlying problem. Moreover, they rely on density ratio estimation, which can be brittle. To alleviate these deficiencies, we recast the imitation learning problem as state occupancy matching via minimization of sliced Wasserstein distances. We provide a novel way of deriving a per-timestep reward function for policy learning by leveraging a local approximation of the gradient of the sliced Wasserstein functional in the space of occupancy measures. The resulting algorithm demonstrates significant improvements in terms of both asymptotic performance and sample efficiency on a benchmark set of robotic locomotion tasks.

  This chapter is based on the following preprint:
  Ivan Ovinnikov, Alexander Terenin, Joachim Buhmann
  Imitation Learning via Generalized Sliced Wasserstein Distances
  Transactions on Machine Learning Research (TMLR), in review

2

BACKGROUND

Our virtues and our failings are inseparable, like force and matter. When they separate, man is no more.
— Nikola Tesla

In this chapter, we lay the groundwork by defining the essential terminology and concepts that are utilized throughout the thesis. We start with an introduction to the reinforcement learning framework, that plays a key role in analyzing adaptive sequential decision-making agents. An overview of the fundamental principles of reinforcement learning is provided, followed by a discussion of various specific model-free methods employed in this work.

Subsequently, we introduce the framework of learning from demonstrations, specifically focusing on imitation and inverse reinforcement learning modalities, central to the algorithmic pipeline described in chapter 3. We primarily concentrate on the maximum entropy inverse reinforcement learning framework, which provides the foundational methodology for the algorithms explored throughout this thesis. We illustrate the link between the approximate solutions of the maximum likelihood problem for a Gibbs distribution of trajectories and a class of information theoretic divergence measures. Following this, we present the distribution matching approach using f-divergences, which forms the basis for several adversarial imitation learning methods further developed in chapter 4.

We continue by discussing distribution matching techniques based on a number of optimal transport methods, and we assemble a concise overview of their implementation in Markov decision processes. These optimal transport methods are foundational for the discussion of a novel distribution matching method introduced in chapter 5. Lastly, we provide a brief introduction to the principles of causal inference, which are employed in chapter 4 as a regularization strategy for inverse reinforcement learning methods. Specifically, we advocate the *causal invariance principle*, a concept from causality that enables improved integration with contemporary machine learning methodologies compared to other causal-graph-based approaches.

2.1 REINFORCEMENT LEARNING

The framework of reinforcement learning (Sutton and Barto, 2018) formalizes the interplay between an adaptive sequential decision making agent and an interactive environment. The high-level interaction is described using the following loop:

- The agent, described by some state at timestep t takes an action according to its behavioral policy and incurs a reward signal.
- The environment advances the agent to the next state by using the transition dynamics function which takes current state and action of the agent as input.

MARKOV DECISION PROCESS This interaction can be formally described by a *Markov decision process* (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \gamma, \mu_0, R)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{T}$ is the family of transition distributions on $\mathcal{S}$ indexed by $\mathcal{S} \times \mathcal{A}$ with $p(s'|s, a)$ describing the probability of transitioning to state s' when taking action a in state s using transition dynamics $\tau(s, a)$, γ is the discount factor, μ_0 is the initial state distribution, and $R : \mathcal{X} \to \mathbb{R}$ is the reward function. Here, space $\mathcal{X}$ denotes an element of the following sets of spaces: $\{\mathcal{S}, \mathcal{S} \times \mathcal{A}, \mathcal{S} \times \mathcal{S}, \mathcal{S} \times \mathcal{A} \times \mathcal{S}\} \ni \mathcal{X}$.

TIME HORIZON One can further distinguish between finite-horizon MDPs and infinite-horizon MDPs. The former introduces an additional variable in the description of the MDP called *horizon* T which denotes the maximal number of timesteps the agent can interact with the environment. The latter considers the case where $T \to \infty$. In the scope of this work, we focus on the finite-horizon setting, also referred to as *episodic RL*.

AGENT GOAL The goal of the agent is to maximize the total amount of reward it receives over the course of its interaction with the environment. To achieve this goal, the agent must select a *policy* function π, which returns optimal actions a for a given state s. More formally, a stochastic policy π is a conditional probability distribution of actions $a \in \mathcal{A}$ given states $s \in \mathcal{S}$. The agent's objective is then defined as follows:

$$\max_{\pi} \mathbb{E}_{\tau}[G_t^{(\pi,\gamma)}] = \max_{\pi} \mathbb{E}_{(\pi,\tau)}\left[\sum_{t=0}^{T} \gamma^t r(s_t)\right] \tag{2.1}$$

Equation (2.1) describes the fact that the agent aims to maximize the discounted sum of rewards, denoted as the return $G_t^{(\pi,\gamma)}$, obtained over the course of an episode of length T under expectation over its own policy π and transition dynamics $\tau(s,a)$.

In order to adapt the policy according to eq. (2.1), it is necessary to define look-ahead functions, i.e. functions that indicate the *expected return* of a policy π interacting with dynamics τ when starting from a particular state s and following that policy for some time horizon T. Canonically, such functions are denoted as *value functions*.

VALUE FUNCTION The *value function* $V_\pi : \mathcal{S} \to \mathbb{R}$ of a state $s \in \mathcal{S}$ is defined as follows:

$$V_\pi(s) = \mathbb{E}_{(\pi,\tau)}\left[\sum_{t=0}^{T} \gamma^t r(s_t)|s_0 = s\right] = r(s) + \mathbb{E}_\pi[V(s')] \tag{2.2}$$

The *state-action value function* or *Q-function* $Q_\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is an analogous function defined over the state-action product space $\mathcal{S} \times \mathcal{A}$:

$$Q_\pi(s,a) = \mathbb{E}_{(\pi,\tau)}\left[\sum_{t=0}^{T} \gamma^t r(s_t,a_t)|s_0 = s, a_0 = a\right] = r(s,a) + \mathbb{E}_\pi[Q(s',\pi(s'))] \tag{2.3}$$

The value functions are related to each other as follows:

$$V_\pi(s) = \int_{\mathcal{A}} Q_\pi(s,a) \tag{2.4}$$

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s) \tag{2.5}$$

where $A_\pi(s,a)$ denotes the *advantage function*. Both value functions defined above admit a recursive formulation as indicated by the second equality sign in eqs. (2.2) and (2.3). This formulation arises from the Bellman equation, which we will describe next.

BELLMAN EQUATION The Bellman equation is the discrete analogue of the Hamilton-Jacobi-Bellman (HJB) equation which extends the Hamilton-Jacobi formulation of mechanics to a Hamiltonian obtained via Bellman's principle of optimality (Bellman, 1957). The HJB equation describes the evolution of the potential of state s at timestep t, $V(t,s)$, as follows:

$$\partial_t V(t,s) + \max_a \{r(s,a) + \partial_s V(t,s)\tau(s,a)\} = 0 \tag{2.6}$$

where $r(s,a)$: denotes the negative cost (reward) function and $\tau(s,a) : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ is the action-conditional function describing (stochastic) transition dynamics.

For a time-homogeneous process, the discretization of eq. (2.6) yields the Bellman operator

$$\mathcal{B}^{\pi} V(s) = r(s) + \mathbb{E}_{s' \sim (\pi,\tau)}[\gamma V(s')]. \tag{2.7}$$

where the expectation is taken over the combination of a stochastic policy $\pi(\cdot|s)$ and stochastic dynamics $\tau(s,a)$. The *Bellman optimality operator* is defined analogously:

$$\mathcal{B} V(s) = r(s) + \max_{a \in \mathcal{A}} \mathbb{E}_{s' \sim \tau(s,a)}[\gamma V(s')]. \tag{2.8}$$

The Bellman operator is a fixed-point contraction operator in V solving the *Bellman optimality equation*:

$$V^*(s) = r(s) + \mathbb{E}_{(\pi,\tau)}[\gamma V^*(s')]. \tag{2.9}$$

A *greedy policy* is characterized by the following relation w.r.t. the value function $V(s)$:

$$\mathcal{B}^{\pi} V(s) = \mathcal{B} V(s) \tag{2.10}$$

The *optimal policy* is then defined as follows:

$$V^*(s) = \sup_{\pi} V^{\pi}(s) \quad \forall s \in \mathcal{S} \tag{2.11}$$

EXACT SOLUTION For a finite state space of cardinality $|\mathcal{S}|$, the exact solution of eq. (2.9) can be obtained by writing out the linear system of equations as follows:

$$\mathbf{v} = (I - \gamma P^{\pi})^{-1} \mathbf{r} \tag{2.12}$$

where $\mathbf{v} \in \mathbb{R}^{|\mathcal{S}|}$ is the vector of values for all $s \in \mathcal{S}$, $P^{\pi} \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ is the transition matrix under the (greedy) policy π and $\mathbf{r} \in \mathbb{R}^{|\mathcal{S}|}$ the reward vector. While this approach has a limited applicability due to the restrictions on state-space cardinality and prohibitive computational cost of the matrix inversion, it enables the RL problem of appropriate scale to be solved by a wide array of *linear programming* algorithms (Ozdaglar et al., 2023).

2.1.1 *Dynamic programming*

Approximate solutions of the Bellman optimality equation in eq. (2.9) are typically obtained using *dynamic programming* approaches by leveraging the recursive property of value functions. Two such complementary methods are the *policy iteration* and *value iteration* algorithms.

POLICY ITERATION The policy iteration algorithm uses an alternating scheme consisting of two steps: (i) policy evaluation and (ii) policy improvement. In the first step, the Bellman operator in eq. (2.7) is applied to all states in a finite state space under the current policy to evaluate the value function. In the second step, the policy is updated by choosing the action which greedily maximizes the evaluated values.

VALUE ITERATION To alleviate the computational burden of policy evaluation steps, which might require multiple passes over the state space, value iteration limits the number of policy evaluations to a single step. It can be seen as a direct application of the greedy maximization to the result of the Bellman operator in eq. (2.7). Both policy iteration and value iteration provably converge to the optimal value function in discounted finite MDPs (Sutton and Barto, 2018).

The methods outlined above are generally only applicable to finite state-action spaces of modest dimensions due to their poor scalability, often referred to as the *curse of dimensionality*. Another limiting factor is their reliance on the explicit knowledge of the transition model, which is rarely available in practice and requires approximation using *model-based* approaches. Their importance lies in the fact that they build the foundation for more sophisticated methods used in a function approximation setting. We will discuss a number of approximate methods for solving the Bellman optimality equation next.

2.1.2 *Model-free function approximation setting*

Contrary to model-based methods involving planning over a learned model of the MDP dynamics, model-free reinforcement learning methods focus on learning the policy from direct interaction with the environment. In order to make such algorithms tractable, most modern methods rely on function approximations of RL functions such as the policy or value functions. This adaptation has largely been enabled by recent progress in neural network training approaches (Kingma et al., 2014; Mnih et al., 2015). We begin by considering function approximation analogues of the policy iteration method, resulting in on-policy policy gradient methods and subsequently introduce the off-policy formulation.

POLICY GRADIENT METHODS Policy gradient methods (Williams, 1992; Kakade et al., 2002; Schulman, Levine, et al., 2015) assume a differentiable policy parameterized by a parameter vector θ. The policy gradient theorem (Sutton, McAllester, et al., 1999) derives an explicit gradient of the *expected cumulative reward* objective in eq. (2.1):

$$\nabla_\theta G_\pi(s,a) = \mathbb{E}_{(\pi,\tau)}[\nabla_\theta \log \pi(a|s) Q^\pi(s,a)] \tag{2.13}$$

The gradient estimator arising from this formulation (Williams, 1992) generally suffers from high variance and does not perform well in practice. Schulman, Moritz, et al. propose to use an approximation of the advantage function $A^\pi(s,a)$ (eq. (2.5)) instead of $Q^\pi(s,a)$ to improve the bias-variance trade-off. The resulting *generalized advantage estimator* is the de-facto standard approach to introduce a bias term used in (Schulman, Levine, et al., 2015) and (Schulman, Wolski, et al., 2017).

$$\hat{A}^{(\gamma,\lambda)}(s_t,a_t) = \delta_t + (\gamma\lambda)\delta_{t+1} + ... + (\gamma\lambda)^{T-t+1}\delta_{T-1} \tag{2.14}$$

where $\delta_t = r(s_t,a_t) + \gamma V(s_{t+1}) - V(s_t)$.

CONVERGENCE GUARANTEES When working with highly parameterized policy models, it is necessary to further limit the parameter update magnitude to attain guaranteed convergence. An initial convergence bound for policy gradient updates was presented in (Kakade et al., 2002) for a limited policy class and extended to more general policy parameterizations in (Schulman, Levine, et al., 2015) as described by theorem 1.

Theorem 1 (Schulman, Levine, et al., 2015)**.** *Let $G(\pi)$ be the discounted empirical return of policy π.*

$$G(\pi') \geq L_\pi(\pi') - \frac{4\epsilon\gamma}{(1-\gamma)^2}\alpha^2$$

where

$$\epsilon = \max_{s,a} |A_\pi(s,a)| \quad \alpha = \max_{s} D_{TV}(\pi_{old}(\cdot|s)||\pi_{new}(\cdot|s)$$

$$\text{and } L_\pi(\pi') = G(\pi_{old}) + \sum_s \rho_{\pi_{old}}(s) \sum_a \pi'(a|s) A_\pi(s,a)$$

Algorithm 1 Proximal Policy Optimization

Init: Initialize K actor networks π_θ^k, critic network V_ϕ
for iteration $n \leq N$ **do**
  for actor $k \leq K$ **do**
    Use policy π to interact with dynamics τ for T timesteps
    Compute advantage estimates $\hat{A}_{i,i=1:T}$ using GAE estimator (eq. (2.14))
  Optimize following loss w.r.t. parameters using gradient descent:

$$\mathcal{L}_{PPO}(\theta) = \mathbb{E}_{\pi_\theta}[\mathcal{L}_{clip}(\theta) - c_1\mathcal{L}_{VF} + c_2\mathcal{H}(\pi_\theta)]$$

  where $\mathcal{L}_{clip}(\theta) = \min(\frac{\pi'}{\pi}\hat{A}_t, \text{clip}(\frac{\pi'}{\pi}, 1-\epsilon_c, 1+\epsilon_c)\hat{A}_t)$,
  $\mathcal{L}_{VF}(\phi) = ||V_\phi - \hat{V}||^2$ and $\mathcal{H}(\pi_\theta)$ is the entropy of the policy π_θ
Return: Trained policy π_θ, value network V_ϕ

This theorem guarantees suitable step size for the policy update to achieve provable convergence. The proof is provided in appendix A.1.

TRUST REGION POLICY OPTIMIZATION The resulting algorithm, *trust region policy optimization* (TRPO) (Schulman, Levine, et al., 2015), introduces a constrained formulation of eq. (2.1) by restricting the Kullback-Leibler divergence between the parameters of the current policy π and the parameters of the updated policy π':

$$\max_\theta \mathbb{E}_\pi[G^\pi] \text{ s.t. } D_{KL}(\theta, \theta') \leq \delta \tag{2.15}$$

To enforce the constraints, the method utilizes an approximation of the natural gradient in the parameter space. Under suitable conditions, such as adequate step sizes and appropriate trust region definitions, the TRPO algorithm is proven to converge, leading to near-optimal policy solutions.

PROXIMAL POLICY OPTIMIZATION Proximal policy optimization (PPO) (Schulman, Wolski, et al., 2017) is an adaptation of TRPO which simplifies the implementation using various heuristics. In particular, solving for the constraint in eq. (2.15) requires the use of a second-order optimization method, which can be costly due to the computation of the Hessian-vector products. PPO is one of the two main algorithms to solve forward RL problems used in this thesis and is summarized in algorithm 1.

2.1.2.1 *Off-policy reinforcement learning*

Policy gradient methods such as PPO are considered to be sample inefficient due to the fact that they only utilize samples drawn from the rollout buffer of the current policy to update the policy and value networks. A compelling alternative is presented in the form of *off-policy methods* which maintain a buffer of transition samples obtained from previously updated policies to improve sample-efficiency.

TEMPORAL DIFFERENCE LEARNING Temporal Difference (TD) learning methods (Tesauro et al., 1995) build the foundation for off-policy methods that learn directly from environment interaction. TD methods use a *bootstrapping* method to update their estimates.

One particular example of such a method is *Q-Learning* which iteratively updates the state-action-value function Q using the L_2-loss between the current estimate and the observed one-step empirical estimate of the Q-function, denoted as $TD(0)$, where η is the learning rate:

$$Q_{k+1}(s,a) = (1-\eta)Q_k - \eta\underbrace{(r(s,a) + \gamma \max_{a'} Q(s',a'))}_{TD(0)} \tag{2.16}$$

In order to translate this objective into the *deep RL* setting, Mnih et al. proposed a number of additions such as the target network for bootstrapping and the use of a replay buffer, resulting in the DQN algorithm used to achieve superhuman performance on a large subset of Atari games.

One of the main restrictions of TD-Learning methods is the greedy policy formulation. In particular, such a formulation is an issue in the case of continuous action spaces, where an optimization routine to determine the maximum needs to be solved at every iteration. Similar to on-policy methods, this limitation is resolved by introducing actor-critic architectures which alternate between policy and value update steps.

SOFT-ACTOR-CRITIC Soft-Actor-Critic (Haarnoja et al., 2018) (SAC) is a popular off-policy RL method which solves the forward *maximum entropy* RL problem regularized using the policy entropy. It incorporates a number of algorithmic additions such as double Q functions to overcome overestimation bias (Fujimoto et al., 2018) and improved exploration properties due to an adaptive policy temperature.

Algorithm 2 Soft Actor Critic

Input: Number of iterations N, number of policy steps M, number of gradient updates G
Init: Initialize policy π_θ and two state-action-value function networks Q_ϕ and $Q_{\phi'}$. Initialize target networks $\bar{Q}_\phi$ and $\bar{Q}_{\phi'}$ and replay buffer $\mathcal{D}_{RB}$
for iteration $n \leq N$ **do**
  for policy step $m \leq M$ **do**
    Sample action $a_t \sim \pi(a_t|s_t)$
    Sample transition from environment $s_{t+1} \sim p(s_{t+1}|s_t, a_t)$
    Add transition to replay buffer $\mathcal{D}_{RB} = \mathcal{D}_{RB} \cup \{(s_t, a_t, r_t, s_{t+1})\}$
  for gradient step $g \leq G$ **do**
    Update the value networks using $\nabla_\phi J_Q(\phi)$
    Update the policy network using $\nabla_\theta J_\pi(\theta)$
    Update the Boltzmann temperature using $\nabla_\alpha J(\alpha)$
  Update the target networks $\bar{Q}$ using a moving average of previous estimate and current state-action-value function parameters (ϕ, ϕ')
Return: Trained policy π_θ, Value function Q_ϕ;

The Q-functions are updated using the standard empirical $TD(0)$-estimate:

$$J_Q(\phi) = \mathbb{E}_{(s_t,a_t)\sim\mathcal{D}_{RB}}\left[\frac{1}{2}(Q_\phi(s_t,a_t) - (r(s_t,a_t) + \gamma\mathbb{E}_{\pi,\tau}[Q_{\bar{\phi}}(s_{t+1},\pi(s_{t+1}))])^2\right] \tag{2.17}$$

The maximum entropy formulation yields a Boltzmann policy $\pi_Q(a|s) = \frac{\exp(\alpha Q(s,a))}{Z}$. To circumvent the computation of the partition function Z, the policy is parameterized using a neural network with parameters θ and trained using the forward Kullback-Leibler divergence as the information projection:

$$\pi^* = \operatorname*{argmin}_\theta \mathcal{D}_{KL}(\pi_\theta(\cdot|s)||\pi_Q(\cdot|s)) \tag{2.18}$$

By plugging in the Boltzmann policy, we obtain the following policy objective:

$$J_\pi(\theta) = \mathbb{E}_{s_t\sim\mathcal{D}_{RB}}[\mathbb{E}_{a_t\sim\pi_\theta}[\alpha \log \pi_\theta(a_t|s_t) - Q_\phi(s_t,a_t)]] \tag{2.19}$$

The resulting algorithm is summarized in Algorithm 2.

RL SUMMARY We have presented a number of foundational principles to perform training of RL agents, as well as two model-free methods PPO (algorithm 1) and SAC (algorithm 2), which are used extensively in the context of this work. Up to this point, we have assumed a reward function to be provided as part of the MDP description. However, this is often a significant assumption, and we will now shift our focus to scenarios where the reward function must be inferred from data.

2.2 INVERSE REINFORCEMENT LEARNING

The reward function is fundamental to the question of specifying the control loop. However, specifying rewards for complex tasks is challenging, and a data-driven approach is often selected.

The field of inverse reinforcement learning (IRL) (Ng, Russell, et al., 2000) tackles the learning scenario where the agent has no access to a ground truth reward signal. Instead, such methods aim to infer a suitable reward signal based on a dataset $\mathcal{D}_E = \{\xi_i\}_{i \leq N}$ of trajectories $\xi_i = (s_t, a_t)_{t \leq T}$ assumed to have been sampled from an expert policy π_E of an agent which performs near-optimally in the given MDP $\mathcal{M}$.

The main challenge of IRL is the fact that the problem is *ill-posed* (Ng, Harada, et al., 1999) – there exist many possible rewards for a given policy. A number of algorithms have been proposed to alleviate this challenge including (Ratliff et al., 2006; Abbeel et al., 2004; Ziebart, Maas, et al., 2008; Bashiri et al., 2021). In this thesis, we focus on the *maximum entropy* approach to IRL (Ziebart, Maas, et al., 2008).

2.2.1 *Maximum entropy IRL*

One way to resolve the ambiguity problem in IRL is to employ the maximum entropy principle (Jaynes, 1957). The application of this principle to IRL has been introduced in (Ziebart, Maas, et al., 2008) and is described by the following constrained variational optimization problem:

$$\max_{\psi} \mathcal{H}(p(\xi|\psi)) \text{ s.t.} \tag{2.20}$$

$$\mathbb{E}_{\xi \sim p(\xi|\psi)}[\varphi(\xi)] = \mathbb{E}_{\xi \sim \mathcal{D}_E}[\varphi(\xi)] \tag{2.21}$$

$$\int_{\Xi} p(\xi|\psi) d\xi = 1; \; p(\xi|\psi) > 0 \tag{2.22}$$

In order to learn the linear reward representation $r_\psi = \psi^T \varphi(\xi)$ for some given trajectory features $\varphi(\xi) \in \mathbb{R}^d$, the problem seeks to identify the distribution $p(\xi|\psi)$ that maximizes the entropy $\mathcal{H}(p(\xi|\psi)) = \mathbb{E}_{p(\xi|\psi)} \log p(\xi|\psi)$ of the distribution subject to moment constraints. The first constraint corresponds to the feature expectation matching between the trajectory distribution $p(\xi|\psi)$ induced by the policy trained on the current reward estimate and the expert trajectory distribution given by the expert dataset $\mathcal{D}_E$. The second constraint corresponds to the normalization constraints of $p(\xi|\psi)$. The solution of this variational problem yields the Gibbs distribution over trajectories:

$$p(\xi|\psi) = \frac{1}{Z_{\varphi,\psi}} \exp(\psi^{\varphi}(\xi)) \tag{2.23}$$

To find the optimal parameters of the Gibbs distribution in eq. (2.23), one needs to solve the following maximum likelihood problem:

$$\max_{\psi} \mathbb{E}_{x\sim\mathcal{D}}[p(\xi|\psi)] = \max_{\psi} \mathbb{E}_{\xi\sim\mathcal{D}} \left[\frac{\exp(\psi^T \varphi(\xi))}{Z_\psi} \right] \tag{2.24}$$

In tractable settings (e.g. tabular state spaces), the gradient of eq. (2.24) can be computed in closed form. In large state spaces, the partition function Z presents an estimation challenge and an approximation scheme is typically employed.

MAXIMUM ENTROPY DUAL More generally, the Gibbs distribution obtained via the maximum entropy principle belongs to the exponential family of distributions:

$$p_f(x) = p_0(x) \exp(f(x) - A(f)) \tag{2.25}$$

where $A(f) = \log \int_\Omega \exp(f(x)) p_0(x) dx$ is the log-partition function and p_0 is the base measure, that we assume to correspond to the Lebesgue or count measure depending on the continuity of the underlying measurable space. Due to the difficulties of computing this function in high-dimensional spaces, a dual formulation of the maximum likelihood problem is derived. Leveraging the strict convexity of the log-partition function $A(f)$ in f, the Fenchel dual (Rockafellar, 2015) of $A(f)$ is given by

$$A(\psi) = \max_{q\in\mathcal{P}} \langle q(x), f(x) \rangle - \mathcal{D}_{KL}(q||p_0) \tag{2.26}$$

By plugging in the resulting dual of the log-partition into the maximum likelihood, we obtain the saddle-point objective:

$$\max_{f \in \mathcal{F}} \min_{q \in \mathcal{P}} \mathbb{E}_{x \sim \mathcal{D}_E}[f(x)] - \mathbb{E}_{x \sim q}[f(x)] + \mathcal{D}_{KL}(q || p_0) \tag{2.27}$$

where $\mathcal{P}$ is the space of sampling distributions. The dual objective in eq. (2.27) is closely related to the variational dual of f-divergences (Csiszár, 1972; Donsker et al., 1976) and can be optimized using a number of *generative-adversarial-network*-like (Goodfellow et al., 2014) methods. These methods belong to a broad class of adversarial imitation learning methods which we will introduce below. To introduce these methods, we procede by giving an overview of distribution matching approaches.

2.3 DISTRIBUTION MATCHING

In this section, we introduce a number of mathematical tools to perform distribution matching between state occupancy measures defined over MDPs. This presents a unifying perspective on a number of imitation and IRL methods that we investigate in the context of this work. By aligning the distributions of state occupancies, we effectively ensure that the learned policies not only mimic specific actions but also replicate the overall state-visitation patterns of expert demonstrations. This approach bridges the gap between purely behavior-cloning strategies and those that infer underlying reward structures, enhancing both the efficacy and robustness of the learned models.

2.3.1 *Occupancy measures*

We begin by describing the notion of state *occupancy measures* that we would like to compare. Intuitively, the occupancy measure of a policy corresponds to the stationary distribution induced by rolling out a policy $\pi(a|s)$ using dynamics $\tau(s,a)$ and starting with initial state measure μ. More formally, a (discounted) state occupancy measure $\rho_\mu^\pi : \mathcal{X} \to \mathbb{R}$ is defined on the appropriate measure space $\Omega(\mathcal{X})$ where $\mathcal{X} \in \{\mathcal{S}, \mathcal{S} \times \mathcal{A}, \mathcal{S} \times \mathcal{A} \times \mathcal{S}\}$ as follows:

$$\rho_\mu^\pi(s,a) := \mathbb{E}_{s_t \sim (\mu,\pi,\tau)} \left[\sum_{t=0}^{\infty} \gamma^t P(s_t = s, a_t = a) \right] \tag{2.28}$$

where $P(s_t = s, a_t = a)$ denotes the probability of the agent being in state s and choosing action a at timestep t. The occupancy measure plays an important role in RL because of the following one-to-one correspondence with the policy (theorem 2, Syed et al., 2008).

Theorem 2 (Syed et al., 2008). *Suppose ρ is the occupancy measure of $\pi(a|s) = \frac{\rho(s,a)}{\sum_{s',a'} \rho(s',a')}$. Then, π is the only policy whose occupancy measure is ρ.*

This correspondence between π and ρ_π allows us to perform distribution matching in the measure space for state occupancy by optimizing the policy which induces the measure.

BELLMAN FLOW CONSTRAINT. The Bellman flow constraint describes the evolution of the occupancy measure $\rho(s,a)$ under the influence of a policy π and dynamics τ and constitutes a Bellman equation defined for occupancy measures of an MDP.

$$\rho(s,a) = \pi(a|s)[(1-\gamma)\mu_0(s) + \gamma\rho(s_0,a_0)\tau(s_0,a_0)ds_0da_0] \quad \rho(s,a) \geq 0. \tag{2.29}$$

where μ_0 denotes the initial state measure. It can be shown that the occupancy measure $\rho_\pi(s,a)$ is a unique solution to the Bellman flow constraint (Syed et al., 2008)[Lemma 2]. This result defines the value function as the following expectation:

$$V^\pi(s) = \mathbb{E}_{\rho_\pi}[r(s)] \tag{2.30}$$

In practice, we resort to the use of empirical measures defined over samples from the occupancy measures we would like to compare. Formally, for a sample of n transitions, $\{x_i\}_{i\leq n}$, $x_i \in \mathcal{X}$ drawn from an occupancy measure ρ_π by rolling out a policy π over dynamics τ, the empirical measure $\hat{\rho}_\pi$ associated with this sample is defined as:

$$\hat{\rho}_\pi = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i} \tag{2.31}$$

where δ_{x_i} represents the Dirac delta function centered at x_i. The empirical measure $\hat{\rho}$ is thus a probability measure that assigns a probability of $\frac{1}{n}$ to each observed data point.

2.3.2 *Measuring discrepancy between distributions*

We will now describe two classes of approaches that yield comparisons of occupancy measures. The first class contains information theoretic divergences called f-divergences (Csiszár, 1972); the second class summarizes integral probability metrics (IPM) (Sriperumbudur et al., 2009).

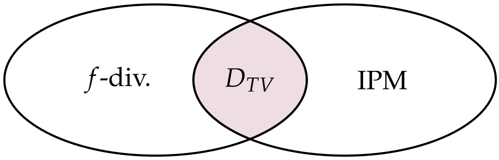

FIGURE 2.1: **Classes of f-divergences and integral probability metrics.** The total variation distance D_{TV} is the only divergence measure that is both an f-divergence and an integral probability metric.

INFORMATION THEORETIC DIVERGENCES One of the widely studied classes of divergences between probability measures is the family of f-divergences (Csiszár, 1972), defined as:

$$D_f(P, Q) := \begin{cases} \mathbb{E}_Q \left[f\left(\frac{dP}{dQ}\right)\right] & P \ll Q \\ +\infty & \text{otherwise} \end{cases} \tag{2.32}$$

where $P \ll Q$ denotes that P is *absolutely continuous* w.r.t. Q.

Table 2.1 summarizes a number of statistical divergence measures broadly known in literature with the corresponding convex functions f.

Divergence	**Function** $f(t)$	**Definition**
Kullback-Leibler	$t \log t$	$D_{KL}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx$
Jensen-Shannon	$-\log t + t - 1$	$D_{JS}(P\|Q) = \frac{1}{2} D_{KL}(P\|M) + \frac{1}{2} D_{KL}(Q\|M)$
Hellinger	$(\sqrt{t} - 1)^2$	$D_H(P, Q) = \frac{1}{2} \int (\sqrt{p(x)} - \sqrt{q(x)})^2\, dx$
Total Variation	$\frac{1}{2}\lvert t - 1\rvert$	$D_{TV}(P, Q) = \frac{1}{2} \int \lvert p(x) - q(x)\rvert\, dx$
χ^2 Divergence	$(t-1)^2$	$D_{\chi^2}(P, Q) = \int \frac{(p(x)-q(x))^2}{q(x)}\, dx$

TABLE 2.1: **Variants of f-divergences** for different convex functions f

The formulation in eq. (2.32) admits a variational dual representation (Donsker et al., 1976):

$$D_f(p\|q) = \mathbb{E}_q \left[f\left(\frac{p}{q}\right)\right] \geq \sup_{g:\mathcal{X}\to\mathbb{R}} \left(\mathbb{E}_p[g(x)] - \mathbb{E}_q[f^*(g(x))]\right) \tag{2.33}$$

where f^* is the Fenchel dual of the convex function f.

INTEGRAL PROBABILITY METRICS An alternative way to measure distances between probability distributions is the class of *integral probability metrics* (Sriperumbudur et al., 2009), defined as:

$$v_{\mathcal{F}}(p,q) := \sup_{f\in\mathcal{F}} |\mathbb{E}_p[f] - \mathbb{E}_q[f]| \tag{2.34}$$

By imposing different constraints on the function class $\mathcal{F}$ over which the supremum is taken, various statistical distances known in literature can be recovered. An overview is presented in table 2.2. We emphasize the Lipschitz smoothness constraint because of its crucial role in introducing the Kantorovich metric, an integral probability metric (IPM) that is closely associated with the *optimal transport* formalism, which we will introduce below. Formally, a function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be Lipschitz smooth if there exists a constant $L \geq 0$ such that for all $x, y \in \mathbb{R}^n$, the following inequality holds:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \tag{2.35}$$

Metric	**Function Class Constraints $\mathcal{F}$**
Total Variation Distance	$\mathcal{F} = \{f : \|f\|_\infty \leq 1\}$
Kantorovich Metric	$\mathcal{F} = \{f : \|f\|_L \leq 1\}$, Lipschitz functions
Maximum Mean Discrepancy	$\mathcal{F} = \{f : \|f\|_{\mathcal{H}} \leq 1\}$, functions from RKHS $\mathcal{H}$
Dudley Metric	$\mathcal{F} = \{f : \|f\|_\infty \leq 1, \|f\|_L \leq 1\}$, bounded Lipschitz functions

TABLE 2.2: **Integral Probability Metrics (IPM) and their function class constraints** (RKHS denotes the Reproducing Kernel Hilbert Space)

Compared to f-divergences, IPMs exhibit a number of appealing properties such as metricity, improved estimator convergence rates and consistency due the issues arising from support mismatch in the case of f-divergences (Sriperumbudur et al., 2009).

Both f-divergences and IPMs have an innate relationship to binary classification problems (X. Nguyen et al., 2009; Sriperumbudur et al., 2009). This connection is particularly useful in the context of learning from demonstrations, as it allows us to formulate the distribution matching between state occupancy measures as a two-player zero-sum game arising from the min-max optimization objective. Table 2.3 provides an overview of the correspondence between a number of optimal risk functions and the respective

f-divergences. In the case of IPMs, the margins of Lipschitz and bounded Lipschitz classifiers are related to the Kantorovich and Dudley metrics as follows (Sriperumbudur et al., 2009):

$$\frac{1}{||f||_L} \leq \frac{W(p,q)}{2} \qquad \frac{1}{||f||_{BL}} \leq \frac{\upsilon_D(p,q)}{2} \tag{2.36}$$

Risk Function and Formula	**f-Divergence and Formula**
Squared Error Loss: $\mathbb{E}[(Y-\hat{Y})^2]$	Squared Hellinger: $H^2(P,Q) = \frac{1}{2}\int(\sqrt{p(x)} - \sqrt{q(x)})^2\,dx$
Logarithmic Loss: $-\mathbb{E}[\log \hat{p}(Y)]$	Kullback-Leibler: $D_{KL}(P\|Q) = \int p(x)\log\frac{p(x)}{q(x)}\,dx$
Exponential Loss: $\mathbb{E}[e^{-Y\hat{Y}}]$	Rényi: $D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\int p(x)^\alpha q(x)^{1-\alpha}\,dx$
Hinge Loss: $\mathbb{E}[\max(0, 1-Y\hat{Y})]$	Total Variation: $D_{TV}(P,Q) = \frac{1}{2}\int\lvert p(x)-q(x)\rvert\,dx$
Cross Entropy: $-\mathbb{E}[\log\hat{p}(Y)] + \mathbb{E}[\log\hat{q}(Y)]$	Jensen-Shannon: $D_{JS}(P\|Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M)$

Table 2.3: **Optimal risk functions** and corresponding f-divergences

Connection between maximum likelihood of maximum entropy problem and dual of f-divergences As we can see in table 2.3, the logarithmic loss corresponding to the maximum-likelihood problem in eq. (2.24) is asymptotically equivalent to the minimization of the KL-divergence $\mathcal{D}_{KL}(p(x|\theta^*)||p(x|\theta))$ (Andersen, 1970). By leveraging the dual form of f-divergences in eq. (2.33), we can rewrite the minimization of $\mathcal{D}_{KL}(p(x|\theta^*)||p(x|\theta))$ in its dual form:

$$D_{KL}(p(x|\theta^*)||p(x|\theta)) = \sup_{g:\mathcal{X}\to\mathbb{R}} \Big(\mathbb{E}_{p(x|\theta^*)}[g(x)] - \mathbb{E}_{p(x|\theta)}[f^*(g(x))]\Big) \tag{2.37}$$

The corresponding KL-divergence representation is known as the Donsker-Varadhan (Donsker et al., 1976) lower bound formulation of the KL-divergence and is given as follows:

$$D_{KL}(p(\xi|\theta^*)||p(\xi|\theta)) \geq \sup_{g:\mathcal{X}\to\mathbb{R}} \Big(\mathbb{E}_{p(\xi|\theta^*)}[g(\xi)] - \mathbb{E}_{p(\xi|\theta)}[f^*(g(\xi))]\Big) \tag{2.38}$$

$$= \sup_{g:\mathcal{X}\to\mathbb{R}} \Big(\mathbb{E}_{p(\xi|\theta^*)}[g(\xi)] - \log\mathbb{E}_{p(\xi|\theta)}[e^{g(\xi)}]\Big) \tag{2.39}$$

The dual formulation of the maximum entropy problem in eq. (2.27) corresponds to the entropy-regularized dual of the *total variation distance* $D_{TV}(p,q)$ if the class of critic functions $\mathcal{G}$ is constrained to the following set of functions: $g \in \{g : \mathcal{X} \to \mathbb{R}, ||g||_\infty \leq 1\}$. This results in the following dual divergence objective:

$$D_{TV}(p||q) = \sup_{||g||_\infty \leq 1} \left(\mathbb{E}_p[g(\xi)] - \mathbb{E}_q[g(\xi)]\right) \tag{2.40}$$

SOLVING SADDLE-POINT OBJECTIVE USING GANS. The saddle-point objective in eq. (2.27) can be solved using the adversarial optimization framework underlying the *generative adversarial networks* (GAN) (Goodfellow et al., 2014) method. By leveraging the relationship of optimal risk functions to divergence measures (X. Nguyen et al., 2009), a discriminator function is defined to approximate the density ratio between the two measures. This function is then trained using the standard binary cross-entropy loss applied to the conditional label distributions corresponding to the two measures. In the case of IRL, we can approximate the first expectation using samples from the empirical expert measure as it is representative of the optimal parameter distribution $p(\xi|\theta^*)$. The second expectation is approximated using a sampling distribution (Finn, Levine, et al., 2016). We choose the sampling distribution q to correspond to the trajectory distribution in eq. (4.2) under imitation policy π_{r_ψ} and MDP dynamics $\tau(s, a)$. The total variation distance constitutes a special case of f-divergences in that it is the only f-divergence measure which is also an integral probability metric (Sriperumbudur et al., 2009; Devroye, 1985). By constraining the function class accordingly, we can perform the optimization of eq. (2.27) in a similar fashion to other adversarial imitation learning methods. Specifically, the *hinge loss* is utilized as the classifier loss function.

2.3.3 *Adversarial imitation learning methods*

The GAN interpretation of the distribution discrepancy measuring functions defined above describes a number of adversarial imitation learning algorithms. Imitation learning methods strive to directly estimate a policy π by learning from a dataset of demonstrations. A general imitation learning objective can be defined as follows:

$$\min_\pi d(\rho_E, \rho_\pi) \tag{2.41}$$

where $d : \Omega(\mathcal{X}) \times \Omega(\mathcal{X}) \to \mathbb{R}$ is a function describing the discrepancy between the expert occupancy measure ρ_E and the occupancy measure ρ_π induced by the policy π.

GAIL Perhaps the prototypical adversarial imitation learning method, generative adversarial imitation learning (GAIL) (Ho et al., 2016) casts the problem of learning an expert policy from demonstrations from a distribution matching perspective. The method asymptotically realizes the entropy-regularized Jensen-Shannon divergence between the empirical expert occupancy measure and the policy occupancy measure by playing a two-player zero-sum game similar to the game realized by GANs.

$$\min_{\pi} \left(D_{JS}(\rho_E, \rho_\pi) + \lambda \mathcal{H}(\pi) \right) \tag{2.42}$$

The discriminator $g_D : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ tries to distinguish between trajectories produced by the policy network and those by the expert, while the policy network aims to generate trajectories that are indistinguishable from the expert's. The GAIL objective is stated as the following min-max problem:

$$\min_{\pi} \max_{g_D} \left(\mathbb{E}_{\pi}[\log g_D(s, a)] + \mathbb{E}_{\pi_E}[\log(1 - g_D(s, a))] \right) \tag{2.43}$$

where $\mathbb{E}_{\pi}$ and $\mathbb{E}_{\pi_E}$ denote the expectations over state-action pairs generated by the policy and the expert, respectively.

AIRL Adversarial Inverse Reinforcement Learning (AIRL) (Fu, Luo, et al., 2018) mimics the optimization formulation of GAIL but imposes additional constraints on the structure of the discriminator function. Specifically, the discriminator function $g_D(s, a, s') : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ has the following form:

$$g_D(s, a, s') = \frac{\exp(u_\psi(s) + \gamma v_{\psi'}(s') - v_{\psi'}(s))}{\exp(u_\psi(s) + \gamma v_{\psi'}(s') - v_{\psi'}(s)) + \pi(a|s)} \tag{2.44}$$

where u_ψ and $v_{\psi'}$ are functions approximating the *advantage* and π is the current state of the policy player. In contrast to GAIL, AIRL is considered to be an IRL method, due to the fact that it recovers a stationary reward. Specifically, Fu, Luo, et al. show that AIRL recovers the ground truth reward function up to an additive constant.

2.3.4 *Optimal transport*

The optimal transport formalism allows the definition of a distance metric between two densities defined on the appropriate measure spaces $\rho \in \Omega(\mathcal{X})$ and $\rho' \in \Omega(\mathcal{X})$.

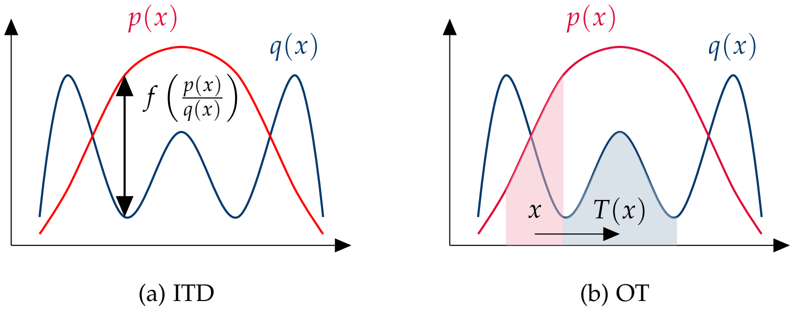

FIGURE 2.2: **Comparison of information theoretic divergences (ITD) and optimal transport (OT) metrics** in terms of how differences between distributions are computed.

The original formulation dates back to (Monge, 1811), who, tasked with finding the optimal way of transporting a pile of construction material from the extraction to the construction site, introduced the famous earth-mover problem. Figure 2.2 illustrates the difference in estimating the discrepancy between two probability measures using f-divergences and using an optimal transport map.

The *Monge formulation* of optimal transport is stated as follows:

$$W_p(\rho, \rho') = \left(\inf_{T_*(\rho)=\rho'} \int_{\mathcal{X}} c(x, T(x)) d\rho(x) \right)^{\frac{1}{p}} \tag{2.45}$$

where $c(x, y) : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is the measurable cost of moving a unit of mass and $T : \mathcal{X} \to \mathcal{Y}$ denotes the *transport map* defining the push-forward operator $T_*(\cdot)$. The cost function $c(x, y)$ plays a central role in the OT formalism as it is representative of the underlying geometry of the measurable space $\mathcal{X}$ over which the measure pairs ρ, ρ' are defined.

UNIQUENESS OF MONGE SOLUTIONS The *Brenier theorem* (Brenier, 1991) has established the uniqueness of the solution to eq. (2.45), contingent upon specific conditions related to the measurable spaces and cost functions associated with the compared measures. Its corollary states that for the cost function $c(x, y) = |x - y|$ and a convex function $\varphi : \mathcal{X} \to \mathbb{R}$, the optimal transport map is given by:

$$T^*(x) = \nabla_x \varphi(x) \tag{2.46}$$

The main issue with the Monge formulation of OT is the feasibility of the constraint $T_*(\rho) = \rho'$, requiring a particular structure of the transport map T. Moreover, the functional in eq. (2.45) is non-convex, making optimization challenging (Santambrogio, 2015).

To overcome this issue, the *Kantorovich formulation* of optimal transport, proposed by (L. V. Kantorovich, 1960), is a relaxation of the push-forward constraints by allowing for mass splitting to occur during transport. As a consequence, instead of computing a transport map T, a *transport plan* $\nu \in \Gamma(\rho, \rho')$ is computed over the set of all couplings between the densities ρ and ρ', denoted $\Gamma(\rho, \rho')$. For any $p \geq 1$, of the p-Wasserstein distance W_p is defined as follows:

$$W_p(\rho, \rho') = \left(\inf_{\nu \in \Gamma(\rho, \rho')} \int_{\mathcal{X} \times \mathcal{X}'} c(x, x') d\nu(x, x') \right)^{\frac{1}{p}} \tag{2.47}$$

When applied to empirical measures $(\hat{\rho}, \hat{\rho}')$, the Kantorovich primal OT formulation results in the following linear program:

$$\mathcal{L}_W(\hat{\rho}, \hat{\rho}') = \min_{\boldsymbol{\nu} \in \bar{\Gamma}(\hat{\rho}, \hat{\rho}')} C^T \boldsymbol{\nu} \text{ s.t. } \sum_{i=1}^{m} \nu_{ij} = \hat{\rho}_j \; \forall j \leq n \quad \sum_{j=1}^{n} \nu_{ij} = \hat{\rho}'_i \; \forall i \leq m \tag{2.48}$$

where $C \in \mathbb{R}^{m \times n}$ is the pairwise cost matrix and $\boldsymbol{\nu} \in \mathbb{R}^{m \times n}$ the transport plan.

By applying the entropic regularization on the transport plan, Cuturi, 2013 propose a computational method based on the Sinkhorn algorithm that significantly speeds up the calculation of optimal transport solutions. This method effectively balances the trade-off between computational efficiency and the accuracy of the transport plan, making it feasible for large-scale applications.

KANTOROVICH-RUBINSTEIN DUALITY The primal form of the OT problem outlined in eq. (2.47) has a variational dual formulation which avoids the search over the coupling space $\Gamma(\rho, \rho')$ between two measures, as this search process can often be restrictive from both computational and memory perspectives.

The duality (L. Kantorovich et al., 1958) states that the 1-Wasserstein distance can also be expressed as:

$$W_1(P, Q) = \sup_{f \in \mathrm{Lip}_1(\mathcal{X})} \left| \int_X f(x)\, dP(x) - \int_X f(x)\, dQ(x) \right|, \tag{2.49}$$

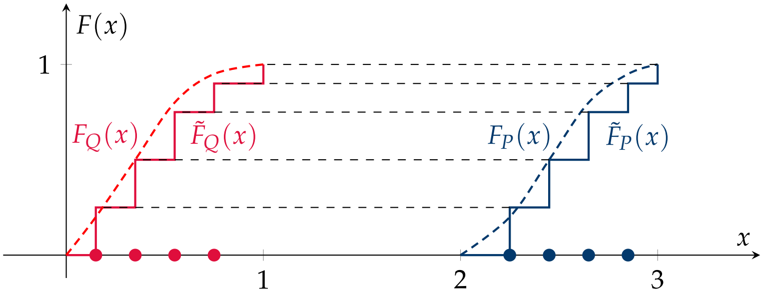

FIGURE 2.3: **Illustration of *increasing arrangements* transport map in 1D**. $F(x)$ denotes the cumulative density function (CDF) and $\tilde{F}(x)$ the sample-based approximation thereof.

where $\text{Lip}_1(\mathcal{X})$ is the space of all 1-Lipschitz functions on $\mathcal{X}$. As we have seen previously in table 2.2, the Kantorovich metric belongs to the class of IPMs. This formulation is used in generative models (Arjovsky, Chintala, et al., 2017) and imitation learning (H. Xiao et al., 2019).

SLICED OPTIMAL TRANSPORT In the one-dimensional case, the Wasserstein distance in eq. (2.47) admits a closed form corresponding to the *increasing arrangements* optimal transport map $T(x) = (F_\rho^{-1} \circ F_{\rho'})(x)$ (Santambrogio, 2015)

$$W_2(\rho, \rho') = \int_0^1 |F_\rho^{-1}(\mu) - F_{\rho'}^{-1}(\mu)|^2 d\mu \tag{2.50}$$

where $F_\rho(\cdot)$ and $F_{\rho'}^{-1}(\cdot)$ are the cumulative distribution function (CDF) and quantile (inverse CDF) function respectively. The sliced Wasserstein distance $\mathcal{SW}_2(\rho, \rho')$ is defined as the integral over 1-dimensional random slices defined via the Radon transform $\mathcal{R}\rho(t, \theta) = \int_{\mathcal{X}} \rho(x)\delta(t - \langle x, \theta \rangle)dx$ of the density $\rho(x)$:

$$\mathcal{SW}_2(\rho, \rho') = \int_{\mathcal{S}^{d-1}} W_2(\theta_\# \rho, \theta_\# \rho')d\theta \tag{2.51}$$

In practice, the increasing arrangements OT map is very efficient to compute for a pair of empirical measures $\hat{\rho}_i, \hat{\rho}_j$. It is done by sorting the samples of empirical distributions and calculating the total distance between them. Figure 2.3 illustrates the mechanism for a pair of empirical measures. We will make use of this property in chapter 5 to derive an imitation learning method based on sliced Wasserstein distances.

DYNAMICAL FORMULATION OF OPTIMAL TRANSPORT Another way to formulate the OT problem is by considering the dynamics of the measures in the space of distributions endowed with the Wasserstein metric $\mathcal{P}(\Omega, W_2)$. The objective introduced by (Benamou et al., 2000) formulates the OT problem as the minimization the kinetic energy of the mass moving along the geodesic in $\mathcal{P}(\Omega, W_2)$ over the time interval from $t = 0$ to $t = 1$.

$$\begin{aligned} W^2(\mu_0, \mu_1) = & \inf_{(\rho,v)} \int_{\mathcal{X}} \int_0^1 |v(x,t)|^2 \rho(x,t) dt dx \\ & \text{s.t. } \partial_t \rho + \operatorname{div}(\rho v) = 0 \\ & \mu_{t=0} = \mu_0 \quad \mu_{t=1} = \mu_1 \end{aligned} \tag{2.52}$$

where (ρ, v) are solutions to the continuity equation $\partial_t \rho + \operatorname{div}(\rho v) = 0$. This formulation presents a duality to the Hamilton-Jacobi equation $\partial_t \phi + |\nabla \phi|^2 = 0$ for $v = \nabla \phi$ (Gigli et al., 2020), given by the following expression

$$\int \phi(\cdot, 0) d\mu_1 - \int \phi(\cdot, 0) d\mu_0 \leq \frac{1}{2} \iint_0^1 |v(x,t)|^2 \rho(x,t) dt dx \tag{2.53}$$

where $\phi(x, t)$ denotes the time-dependent Kantorovich potential. This connection is particularly interesting in the context of reinforcement learning algorithms.

JKO SCHEME The celebrated Jordan-Kinderlehrer-Otto (JKO) (R. Jordan et al., 1998) scheme presents a variational approach to solve the OT problem. The JKO scheme demonstrates that the gradient flow of a Wasserstein functional admits a variational formulation which yields an iterative solution to a Fokker-Planck type of equations:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\mathbf{F}\rho) + \nabla \cdot (D \nabla \rho)$$

where ρ is the probability density function, $\mathbf{F}$ represents the drift vector field (often corresponding to forces acting on the particles), and D is the diffusion tensor, which describes how quickly the probability spreads out in space. The JKO scheme obtains solutions using the following iteration:

$$\rho_{t+1} = \underset{\rho \in \Omega(\mathbb{R}^d)}{\operatorname{argmin}} \, \mathcal{E}(\rho) + \frac{1}{2h} \mathcal{W}^2(\rho, \rho_t) \tag{2.54}$$

where $\mathcal{E}$ is the energy functional and h the step size. This minimization process systematically evolves the probability density ρ over iterations, leveraging the Wasserstein distance $\mathcal{W}^2(\rho, \rho_t)$ to measure the cost of moving from the current distribution ρ_t to a new distribution ρ.

The tools derived from dynamical optimal transport (OT) offer a robust and principled foundation that can be used to describe the time evolution of state occupancy measures. Specifically, policy optimization can be viewed as a variant of a Wasserstein gradient flow (R. Zhang et al., 2018).

2.4 CAUSAL INFERENCE

With the emergence of modern machine learning techniques that utilize highly parameterized models, there has been a renewed interest in the challenge of developing models, which accurately reflect the causal structure. This resurgence is driven by the growing recognition that causal inference can significantly enhance the interpretability and effectiveness of predictive models, especially in complex domains like healthcare, economics, and social sciences. Consequently, techniques such as causal discovery algorithms and structural causal models are being integrated with traditional machine learning frameworks to not only predict outcomes but also to understand and quantify the underlying causal mechanisms. This integration is promising for building robust models that can generalize better across varied environments and potentially adapt more dynamically to changes and interventions.

2.4.1 *Structural causal models*

Structural causal models are a class of directed probabilistic graph models which describe a structured generative model of the observed data. They form the basis for most graph-based causal inference approaches.

Formally, a structural causal model (SCM) (Pearl, 2009) is defined as a tuple $\mathfrak{G} = (S, P(\varepsilon))$, where $P(\varepsilon) = \prod_{i \leq K} P(\varepsilon_i)$ is a product distribution over exogenous latent variables ε_i and $S = \{f_1, ..., f_K\}$ is a set of structural mechanisms where $\text{pa}(i)$ denotes the set of parent nodes of variable $x_i := f_i(\text{pa}(x_i), \varepsilon_i)$ for $i \in |S|$. $\mathfrak{G}$ induces a directed acyclic graph (DAG) over the variables nodes x_i. The SCM entails a joint observational distribution $P_{\mathfrak{G}} = \prod_{i \leq K} p(x_i | \text{pa}(x_i))$ over variables x_i conditioned on the parents of x_i for some probability distribution $p(\cdot | \text{pa}(x_i))$ describing the mechanism f_i. Interventions on $\mathfrak{G}$ constitute modifications of one or more structural mechanisms f_i yielding interventional distributions $\tilde{P}_{\mathfrak{G}}$.

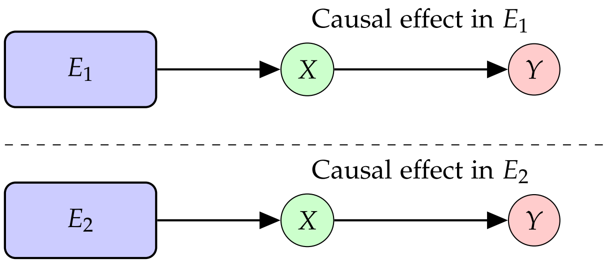

FIGURE 2.4: **Illustration of the Causal Invariance Principle.** This diagram depicts the consistent causal relationship between variables X and Y across two different environments, E_1 and E_2. Each environment influences variable X, which in turn affects Y, demonstrating the principle of causal invariance. The environments are shown as distinct but maintain the same causal structure, underscoring the robustness of the causal effect despite environmental changes.

2.4.2 *Distribution shift robustness and causal invariance*

In order to use the majority of causal inference methods such as (Pearl, 2009; Peters, Janzing, et al., 2017), one requires explicit access to the SCM describing the joint distribution over the relevant variables. This requirement rarely holds in practice due to a number of reasons including unobserved confounders, complexity of the causal graph or lack of interventional data. In general, identifying the causal graph structure from data is limited to obtaining Markov equivalence classes of solutions, which limits the applicability of these methods.

Over the recent years, the weaker concept of *causal invariance* (Peters, Bühlmann, et al., 2015; Heinze-Deml et al., 2017) has gained traction. It refers to the property that causal relations between variables remain consistent across different environments or contexts. Its main appeal is that it allows to bypass the estimation of a causal graph to identify certain causal properties of the prediction or learning task. An illustration is provided in fig. 2.4. In machine learning, causal invariance is sought after for building robust models that generalize well across different datasets and real-world scenarios. The pursuit of causally invariant features within data helps in developing predictive models that are not only accurate but also reliable under changing conditions.

We will utilize this principle in chapter 4 to infer reward functions which reduce their reliance on spurious correlations arising from diverse demonstrations.

2.5 SUMMARY

In this chapter, we have given a broad overview of the fundamentals of adaptive decision making when learning from demonstrations. We presented the distribution matching perspective as the central tool to perform model-free optimization of decision making agents, shedding light on two main mathematical tools: information theoretic divergences and integral probability metrics. We gave an overview of different variants of optimal transport methods, which are closely related to IPMs and provide a rich mathematical framework to describe state occupancy matching. In order to numerically solve problems posed by these frameworks, we presented an adversarial optimization framework which allows to instantiate imitation learning methods in practice. Finally we gave a short overview of the principles of causal inference.

3

FASTRL: SURGICAL ASSISTANCE PIPELINE

Homines, dum docent, discunt.

— Seneca

This chapter introduces FASTRL, an algorithmic pipeline for surgical assistance based on reinforcement learning agents trained to distill expert knowledge given by demonstrations. It features a benchmark set of instrument manipulation tasks derived from the Fundamentals of Arthroscopic Surgery Training (FAST) program. We have restructured this set of tasks to support exploration of reinforcement learning methodologies. This adaptation is aimed at fostering the development of human-centric RL agents tasked with assisting and evaluating human trainees in simulated surgical environments.

In particular, we have adapted several key simulation tasks from the FAST program to fit the reinforcement learning control loop that enables the training of virtual agents. These agents are designed not only to solve the tasks but also to enhance the training process of surgical trainees by providing real-time feedback and evaluation. To this end, we have developed a skill performance assessment protocol that leverages the capabilities of these virtual agents, offering a structured and quantifiable measure of trainee progress and skill acquisition.

We present several algorithmic baselines that apply both forward and inverse reinforcement learning techniques to this setting. Additionally, we explore customized heuristic reward structures tailored specifically for the benchmark tasks, which are crucial for guiding the behavior of the RL agents toward clinically relevant and surgically correct actions.

An extensive discussion on reward shaping is also included, highlighting its critical role in eliciting surgically accurate behaviors within state-of-the-art model-free reinforcement learning algorithms. Reward shaping, in this context, proves to be indispensable for aligning the learning objectives of the RL agents with the precise requirements of surgical procedures.

The proposed benchmark suite presents an API for training reinforcement learning agents in the context of arthroscopic skill training. The evaluation scheme based on both heuristic and learned reward functions robustly recovers the ground truth ranking on a diverse test set of human trajectories.

The presented benchmark enables the exploration of a novel reinforcement learning-based approach to skill performance assessment and in-procedure assistance for simulated surgical training scenarios. The evaluation protocol based on the learned reward model demonstrates potential for evaluating the performance of surgical trainees in simulation.

Overall, this chapter lays the foundation for a novel intersection of reinforcement learning and medical training, setting the stage for future innovations that could transform how surgical training is approached, optimized, and implemented.

3.1 INTRODUCTION

Training procedures of surgical residents heavily depend on accumulated operative time to document proficiency. The limited availability of in-vivo training opportunities, i.e., patient cases, can result in a suboptimal or even deficient training process of surgical novices (Chiasson et al., 2003) with far-reaching consequences. Surgical simulators provide training scenarios which allow users to alleviate the bottleneck of low patient case numbers, but often use rudimentary evaluation metrics such as total procedure time and instrument path length to assess surgical performance of practitioners. Intra-operative performance assessment (Stauder et al., 2017) plays a central role in the question of how to evaluate a sequence of decision steps taken by a surgeon. Furthermore, it also provides valuable feedback on how to improve on a given surgical task both for trainees as well as for experienced practitioners in continuing education.

In order to adapt the training process accordingly, we propose to model the behavior of surgical trainees on simulator hardware in a principled way by considering a sequential decision making agent interacting with a surgical environment. This interaction is formalised by the reinforcement learning (RL) framework. We envision a scenario where virtual agents trained using RL and inverse RL methods are utilised to provide real-time scoring and prediction of the trainees' performance as well as to generate visual cues aimed at improving the performance or procedure guidance. Inverse RL (IRL) methods are particularly appealing due to their outstanding data efficiency and robustness to distribution shift (Ng, Russell, et al., 2000) when compared to simpler imitation learning methods such as behavioral cloning (BC). Contrary to commonly studied surgical robotics settings employing RL (Tagliabue et al., 2020; Kazanzides et al., 2014), this work primarily focuses on the study of training and evaluation signals characterised by a reward function that the agent receives on interaction with the simulated environment.

The simulated environment is realised as a set of surgical tasks derived from the Fundamentals of Arthroscopic Surgery Training (FAST) suite with the purpose of training dexterous manipulation of arthroscopic instruments by a surgical trainee. The FAST suite was designed in collaboration with various orthopedic surgeon associations [1] and is implemented in Unity3D by VirtaMed AG [2]. The current work extends the existing implementation to be compatible with an RL framework which enables the interaction of a virtual agent with the simulation hardware. It has been shown (Cychosz et al., 2018), that the training of fundamental skills involving dexterous manipulation of instruments using the FAST suite improves the performance of arthroscopic trainees in subsequent procedures. Hence, the FAST framework provides a surgically relevant skill basis for the purposes of simulated training. We demonstrate the utility of our algorithmic pipeline by evaluating a diverse set of human recorded trajectories. The pipeline also suggests a selection of scoring functions for surgical performance.

In this chapter, we develop the following scientific ideas and research solutions:

- A configurable interface for the FAST surgical training simulation (FASTRL) [3] which allows easy deployment of standard RL and inverse RL algorithms. To our knowledge, this is the first benchmark aimed at improving both evaluation and assistance of surgical training in the context of RL, in contrast to robotic surgical environments.

- A new data driven approach based on expert demonstrations for evaluation of surgical skill performance using RL methods. We demonstrate the effectiveness of this evaluation on datasets recorded by experts and novice users of the simulation hardware.

---

1 ABOS, AAOS, AANA

2 virtamed.com, the environments are provided in binary format and accessible via the `FASTRL` API

3 Fundamental of Arthroscopic Surgery Training using Reinforcement Learning (FASTRL). The API and the executables for the presented environments are made public and are available under: `fastrl.ethz.ch`. The executables are distributed under the CC BY-NC-ND license. Video material and further details are available under the same URL.

3.2 RELATED WORK

RL FRAMEWORKS FOR SURGICAL ROBOTICS. Automation of surgical procedures holds promise of improving the surgical outcome, and hence, defines a long standing goal in robotics for medicine. A number of benchmarks have previously been proposed which allow to perform the training of RL agents in a surgical robotics setting. In particular, the daVinci Research Kit (dVRK) (Kazanzides et al., 2014) has been adopted as a popular platform by the surgical community. In (Tagliabue et al., 2020), the use of RL is demonstrated for manipulation of soft tissues using a robot arm. The authors of (Pore et al., 2021) demonstrate the use of an adversarial imitation method (Ho et al., 2016) in the context of surgical soft-tissue retraction using the dVRK platform (Kazanzides et al., 2014). The authors of (Xu et al., 2021) propose an open-source platform simulator to replicate a series of manipulation tasks using the dVRK telerobotic platform that is specifically geared towards robotic learning. In contrast to previous work focused on training robotic policies, our benchmark design specifically suggests a scenario of surgical teaching assistance of human trainees. Furthermore, in our approach, we explicitly propose to use the reward function recovered via IRL methods for the purposes of trainee evaluation and guidance.

SURGICAL SKILL ASSESSMENT USING ML METHODS The use of machine learning for the analysis and performance of surgical procedures has previously been explored in robotic surgery (Lam et al., 2022; Kassahun et al., 2016; Garrow et al., 2021), where both kinematic and visual data is readily available at procedure time. The demonstrated approaches typically apply various forms of supervised learning methods which exhibit a trade-off between how expressive models are and how many labeled examples they require for training. A number of methods ranging from score regression to deep neural network based classification and segmentation of surgical procedures from videos have been presented in the literature (Lavanchy et al., 2021; D. Liu et al., 2021). Typically, surgical skills are assessed post-operatively with data recorded during the procedure.

Contrary to established approaches, our system provides a dynamic real time assessment of the surgical performance in simulated tasks as well as real time feedback which can be queried by the trainee in the course of a procedure.

3.3 ALGORITHMIC PIPELINE FOR SURGICAL ASSISTANCE

In this section, we motivate the design and development of the FASTRL set of tasks by presenting (i) the algorithmic pipeline for surgical training assistance and (ii) the realization of the FAST suite as an RL benchmark.

3.3.1 *Simulated surgical training scenario*

Arthroscopic surgical skills training in a computer simulation provides a demanding and scalable setup to investigate an educational training scenario for medical interventions. The trainee is challenged with a number of basic instrument handling tasks which are performed in an augmented reality (AR) simulator. The instruments, typically consisting of an arthroscope, a palpation hook and a grasper, are inserted into the half-dome structure through portals and the output of the arthroscopic camera is simulated on a screen (fig. 3.2) depicting the surgery environment. The simulated setting has a number of benefits, including potentially unlimited interaction capacity between the trainee and the surgical environment as well as access to a variety of data modalities for analysis purposes, that are often difficult to obtain in real world settings with patients. We aim to enhance the simulated training scenario with evaluation and assistance based on the formalization of the interaction between simulator and trainee as an RL problem.

3.3.2 *RL formalism*

We model the simulation environment for surgical procedures as a *Markov decision process* (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, P_0, R)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{T} : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ is the transition function, P_0 is the initial state distribution, and $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function. A *policy* $\pi : \mathcal{S} \to \mathbb{R}_+, s \mapsto \pi(a|s)$ is a conditional probability distribution of actions $a \in \mathcal{A}$ given states $s \in \mathcal{S}$ with state features $\varphi(s)$.

In the *inverse reinforcement learning* (IRL) setting, the reward is unknown and a suitable parametric reward function r_ψ is estimated based on a dataset of *expert trajectories* $\mathcal{D}_E = \{\xi_i\}_{i \leq K}$ where $\xi_i = (s_{1:T}^{(i)}, a_{1:T}^{(i)})$ is a sequence of states and actions of expert i of length T. To achieve this goal, various distribution matching methods can be used (Ho et al., 2016; Fu, Luo, et al., 2018; Ziebart, Maas, et al., 2008). Under the assumption of fixed transition dynamics $\mathcal{T}$, the reward function defines a succinct representation of the task to be performed and it elicits desired behaviors guided by RL algorithms (Ng, Russell, et al., 2000). In the context of a simulated environment, this learning control enables reward functions to serve as a central part of evaluation schemes for human trainees.

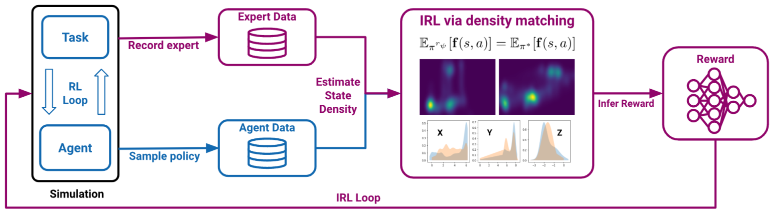

FIGURE 3.1: **Algorithmic pipeline for surgical assistance**: a handcrafted reward heuristic allows the training of the agent, forming the *RL loop*. The recorded trajectories (synthetic or human) are then used as an input to the inverse RL algorithm which enables recovery of a reward function based on state density matching between the expert and the learning agent (*IRL loop*). The resulting reward and policy are then utilised for training assistance.

We consider two learning modalities in our pipeline:

- **Forward RL**: model-free RL methods based on reward *heuristics* designed by *human experts*.
- **Inverse RL**: inverse setting where the reward function is *directly inferred* from a set of expert demonstrations.

In the first modality, we define heuristic reward functions that enable training of virtual agents to learn the completion of benchmark tasks. The benchmark tasks require a level of sophistication which depends on the design of multiple reward components in order to elicit correct behaviors. This modality enables the training of agents which learn to mimic surgical behavior via the definition of a suitable reward function and implementations of standard model-free RL methods. This modality mainly intends to explore the underlying search space of surgically correct procedures by specifying heuristic evaluation schemes. It can also enable experts to generate virtual agents by specifying a set of preference weights for the behavioral components of the heuristic reward function via the provided API.

In the inverse modality, an extrinsic reward function is not explicitly defined; instead, a reward function is inferred from the union set of the recorded expert demonstrations on the respective task. This modality directly allows users to be evaluated with respect to a *learned* reward function, as opposed to a hand-designed heuristic. The combination of these two modalities allows us to obtain a reward, value function tuple (r_ψ, V_ϕ). The policy π_{r_ψ} is then trained to optimise r_ψ derived from a set of expert demonstrations $\mathcal{D}_e$.

Both the policy and the reward function can be utilised as *feedback* mechanisms for human learners. These functions support real time contextual evaluation of the trainee's performance as well as a quantitative prediction of the procedure outcome respectively.

In addition, the trainee can query the virtual agent policy at any point in the procedure to generate a demonstration of the next steps based on the current trainee state using the simulated dynamics. This assistance modality is illustrated in fig. 3.3.

In order to explore this setting, we propose a standard RL benchmark set to train virtual agents aimed at surgical assistance of arthroscopic procedures. An example pipeline consisting of forward and inverse RL algorithms is illustrated in fig. 3.1.

3.3.3 *Benchmark environment*

We use the Arthros®FAST (Fundamentals of Arthroscopic Surgery Training) simulator implemented in Unity3D and provided by Virtamed AG to evaluate the RL pipeline. The simulation provides a number of educational navigation and manipulation tasks that have to be performed within a hollow dome structure (Sawbones®FAST workstation) in accordance with the FAST (Fundamentals of Arthroscopic Surgery Training) training problem [4]. We consider three tasks for our benchmark environment: (i) image centering and horizoning (`ImageCentering`), which trains basic endoscope manipulation and monocular depth estimation, (ii) periscoping (`Periscoping`), which trains the use of angled optics and (iii) line tracing (`TraceLines`), which trains steady instrument movement. The tasks consist of guiding the tip of the virtual arthroscope to various locations marked by an avatar that displays visual cues. The trainee or the virtual agent pursues the goal of orienting the arthroscope in such a way that it complies with the cues and

4 developed by American orthopaedic associations ABOS (American Board of Orthopaedic Surgery), AAOS (American Academy of Orthopaedic Surgeons) and AANA (Arthroscopy Association of North America)

that it centers the image of the avatar in the field of view of the endoscope camera.

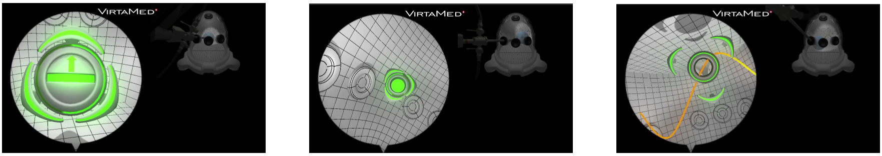

Figure 3.2: **Benchmark environment tasks**: (i) `ImageCentering` (ii) `Periscoping` (iii) `TraceLines`. The avatar highlighted in green displays visual cues to guide the training agents. The cues are encoded as part of the heuristic reward structure used for forward RL training.

MDP FORMULATION In order to solve the tasks using the RL framework, we formalise it as the following continuous control problem: a state s_t in the continuous *state space* $\mathcal{S}$ is defined as the concatenation of the three-dimensional position $\mathbf{x}_t^{\mathrm{arth}}$ and rotation quaternion Q_t^{arth} of the arthroscope as well as the respective time derivatives $\dot{\mathbf{x}}_t^{\mathrm{arth}}$ action space and $\dot{Q}_t^{\mathrm{arth}}$. Additionally, we provide the position and rotation of the target avatar $\mathbf{x}_t^{\mathrm{tgt}}, Q_t^{\mathrm{tgt}}$ as well as the cross-product between the forward vector of the arthroscope camera and the forward vector of the avatar transform $\alpha_t^{\times(\mathrm{arth,tgt})}$, which describes the alignment between the arthroscope camera and the avatar. The full state specification is given as the following tuple: $s_t = (\mathbf{x}_t^{\mathrm{arth}}, Q_t^{\mathrm{arth}}, \mathbf{x}_t^{\mathrm{arth}}, \dot{\mathbf{x}}_t^{\mathrm{arth}}, \dot{Q}_t^{\mathrm{arth}}, \mathbf{x}_t^{\mathrm{tgt}}, Q_t^{\mathrm{tgt}}, \alpha_t^{\times(\mathrm{arth,tgt})}) \in \mathcal{S}$. We propose two versions of the action space. The seven-dimensional continuous *action space* $\mathcal{A}$ consists of stacked acceleration and angular acceleration vectors as well as the light cable rotation angle ω_c of the arthroscope, which realises an independent camera angle control $a_t = (\ddot{\mathbf{x}}_t, \ddot{\phi}_t, \omega_c) \in \mathcal{A}$. The alternative five-dimensional *discrete action space* $\mathcal{A}_d$ is defined as the translation of the arthroscope along its axis x_t, changes in pivoting angle around the portal $\Delta\alpha_p, \Delta\beta_p$, rotation around the translation axis $\Delta\delta_p$ and the rotation of the camera ω_c. The *transition function* $f(s_t, a_t) \in \mathcal{T}$ is implemented by the Unity physics engine, which simulates the friction coefficients of the arthroscopic entry point. The physical interaction is deterministic for most use cases.

REWARD SHAPING Eliciting successful behavior crucially requires a correct specification of the reward. During the procedure, the simulation provides visual avatar cues to the agent which specifies a sparse reward scheme. In order to increase the sample efficiency of the training process, we additionally define a dense linear reward. The heuristic reward function $r_{\text{heur}} = \mathbf{w}^T \phi(s_i)$ is defined as a weighted sum of the reward features $\phi(s_i)$ [5]. The reward features and corresponding reward weights are configurable at runtime and are exposed to the user. In particular, this allows the experimentalist to explore the generation of behaviors based on various combinations of reward features as well as multi-objective optimization methods.

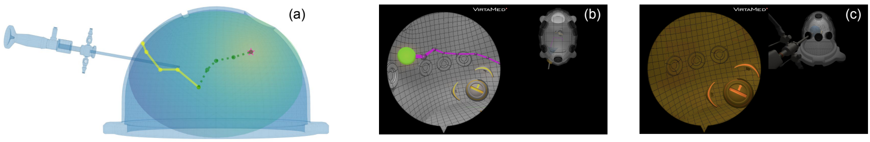

FIGURE 3.3: **Assistant illustration.** (a) path suggestion (green line) based on performed trajectory (yellow line) in the FAST dome with reward potential around target (star), (b) path suggestion from camera POV, (c) color cue for the trainee's current state.

3.3.4 *Learning algorithms*

The virtual agents are parameterised using an actor-critic architecture which consists of the policy function π_θ and either the value function $V_\phi(s)$ (on-policy) or state-action value function $Q_\phi(s, a)$ (off-policy). $V_\phi(s)$ approximates the discounted empirical returns $\hat{V}(s) = \sum_{t \leq T} \gamma^t r(s_t)$ obtained on rollouts of the policy. The agents are trained employing a variant of the *proximal policy gradient* (PPO) (Schulman, Wolski, et al., 2017) algorithm. When learning from human trajectories, we utilise an off-policy algorithm, *soft actor-critic* (SAC) (Haarnoja et al., 2018), for purposes of sample efficiency.

To solve the inverse RL problem with maximum entropy regularization as outlined in section 3.3.2, we utilise the adversarial imitation learning approach (Ho et al., 2016).

Adversarial inverse RL (AIRL) methods yield a reward function by learning to distinguish between the transitions sampled from the dataset of expert trajectories $\xi \sim \mathcal{D}_E$ and transitions continually sampled from the rollout buffer of the improving policy $\xi \sim \mathcal{D}_\pi$. The policy π_θ maximises an

5 A detailed overview of the features is provided in the appendix

expected cumulative reward objective based on the discriminator output. The discriminator is optimised using the binary cross-entropy loss. We utilise both the standard discriminator structure described in (Ho et al., 2016) as well as the shaped structure from (Fu, Luo, et al., 2018) which separates into the state dependent reward function r_ψ and the state dependent shaping term. The reward function specifically is used for downstream evaluation of novel trajectories. The *off-policy formulation* additionally necessitates two algorithmic additions. The first is spectral norm regularization (Yoshida et al., 2017), a method which enforces the Lipschitz smoothness of the discriminator and stabilises the adversarial training procedure. The second is a modified sampling procedure for discriminator training, only utilising samples from the current policy rollout as opposed to mixture policy samples from the replay buffer. We demonstrate the impact of these additions in the experimental section.

3.4 EXPERIMENTS

The algorithmic pipeline for surgical assistance (fig. 3.1) allows us to perform an evaluation of human demonstrations using heuristic and learned rewards. In this section, we demonstrate its applicability on a test set of human trajectories gathered using the simulator. Furthermore, we measure the performance of various state-of-the-art methods in forward and inverse RL to illustrate the learning complexity of the benchmark tasks.

3.4.1 *Learning from demonstrations*

In the first set of experiments, we demonstrate the ground truth task performance of the algorithms used to recover the reward and value functions on the proposed benchmark suite.

We obtain the demonstration set by sampling trajectories from the best performing policy trained using the heuristic reward formulation described in section 3.3.3. In fig. 3.4 we observe that inverse methods (GAIL, AIRL) converge to the desired solution faster for the `Periscoping` and `TraceLines` tasks than the forward RL method (PPO) but require more time to converge to the optimal solution in the `ImageCentering` case. This behavior can be attributed to the degree of alignment between heuristic reward components and state space specification, which varies across tasks, making `ImageCentering` easier to solve in the forward learning scenario (PPO). Furthermore, the IRL method (AIRL) outperforms the pure imitation learning method (GAIL) in the `Periscoping` task. The discrepancy between GAIL and AIRL is ascribed to different divergence measures realized by the algorithms. Furthermore, the smaller number of parameters of the GAIL discriminator may explain faster initial convergence. Both policies trained using the forward RL method (PPO) and the ones obtained via inverse methods (GAIL, AIRL) can be used to generate trajectory suggestions in the assistance setting. The exact hyperparameters and experimental settings can be found in appendix A.2.

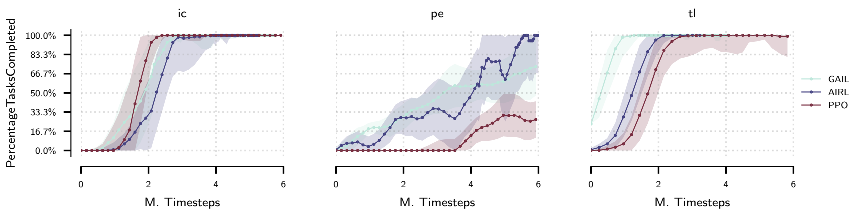

FIGURE 3.4: **Comparison of forward and inverse RL experiments**: median of percentage of tasks completed for `ImageCentering` (ic), `Periscoping` (pe) and `TraceLines` (tl) environments.

In addition to evaluating the pipeline using synthetic demonstrations sampled from a policy trained on heuristic reward, we learn a policy and reward function from a set of human trajectories recorded using the keyboard and mouse user interface in the `FASTRL` API. We perform the evaluation on the `TraceLines` task. In fig. 3.5 we can observe, that in order to attain sample efficient training and successfully complete all subtasks, the algorithmic modifications outlined in section 3.3.4 are necessary.

3.4.2 *Performance evaluation*

In this section, we evaluate two sets of human trajectories using the reward and value functions obtained by both AIRL and GAIL algorithms as scoring functions. The first set of trajectories consists of three novices (`io`, `mk` and `mv`) and two expert practitioners `fm` and `an`. These trajectories have been recorded using the physical hardware platform[6]. The second set of trajectories has been recorded using the keyboard-mouse interface and features three distinct skill levels defined based on the percentage of completed subtasks.

6 We defer the details of data gathering procedure to the appendix.

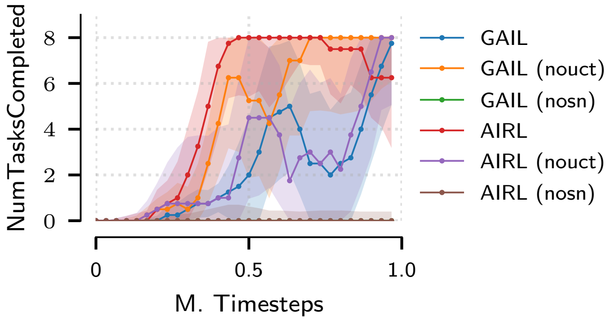

Figure 3.5: **Ablation experiments for off-policy version of the presented algorithms**: number of tasks completed for the `TraceLines` environment. The label `nouct` denotes the use of samples from entire replay buffer for specification of the density matching objective, `nosn` corresponds to the removal of spectral normalization of the discriminator.

In table 3.1, we observe that both normalised value function and reward function recover the correct ranking among the five sets of trajectories. The learned reward and value functions are consistent with both the heuristic reward baseline as well as the trajectory length metric used in the simulator as the standard evaluation option. In fig. 3.6, we plot the spatial x and y components of the states evaluated using the learned value functions. We observe a clear distinction between the novice and expert practitioners with the expert agent outperforming the human expert in terms of the score. In fig. 3.7, we show a comparison of the agents from fig. 3.6 in terms of the deviation from the expert agent reference over the course of the procedure. Again, a clear distinction between the skill levels of agents is observed. The distinction is consistent with respect to the ground truth level of expertise of the agents.

Table 3.2 summarises the results obtained on the reward learned on human trajectories. We observe a similar result in terms of the recovered ranking. The validation of learned reward functions remains an open question. Evaluation of synthetic policies typically relies on rollout performance on the ground truth reward, which is not directly applicable to the setting of human evaluation due to lack of a ground truth metric which would capture all relevant aspects of the behavior.

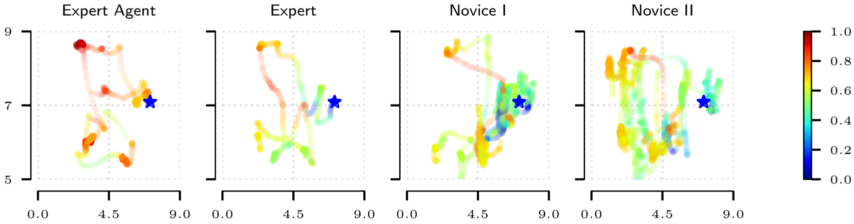

FIGURE 3.6: **Trajectory projection comparison for the `Periscoping` task** (XY coordinate plane). Colors indicate the normalised value output by the learned evaluation function on the full states of the recorded trajectories. The star symbol indicates the start and end point of the trajectory. The color transparency (alpha) value encodes the time the agents spend in the evaluated states. *Expert Agent* outperforms human subjects. *Expert* human subject has a shorter and smoother trajectory compared to the novices which is reflected in the accumulated reward. The demonstrations were all recorded under the same conditions, from the same starting point. The end point might be different as different agents have different ways of solving the task.

Agent ID	r_ψ	V_ϕ	r_{heur}	Trajectory length
fw	$0.993_{\pm 0.004}$	$0.876_{\pm 0.076}$	$0.999_{\pm 0.0002}$	$1600.6_{\pm 971.0}$
fm	$0.732_{\pm 0.019}$	$0.728_{\pm 0.016}$	$0.742_{\pm 0.016}$	$1821.5_{\pm 104.8}$
an	$0.720_{\pm 0.042}$	$0.716_{\pm 0.040}$	$0.729_{\pm 0.039}$	$1929.2_{\pm 277.4}$
mk	$0.617_{\pm 0.117}$	$0.627_{\pm 0.121}$	$0.634_{\pm 0.106}$	$2578.5_{\pm 745.8}$
mv	$0.518_{\pm 0.094}$	$0.521_{\pm 0.094}$	$0.529_{\pm 0.095}$	$3313.5_{\pm 665.5}$
io	$0.374_{\pm 0.271}$	$0.339_{\pm 0.249}$	$0.384_{\pm 0.278}$	$4290.8_{\pm 1981}$

TABLE 3.1: **Trajectory scores** (heuristic and learned rewards and values) for human (fm, an, io, mk, mv) and virtual (fw) agents on the `Periscoping` task. Here, r_ψ denotes the learned reward, V_ϕ the learned value and r_{heur} the reward heuristic.

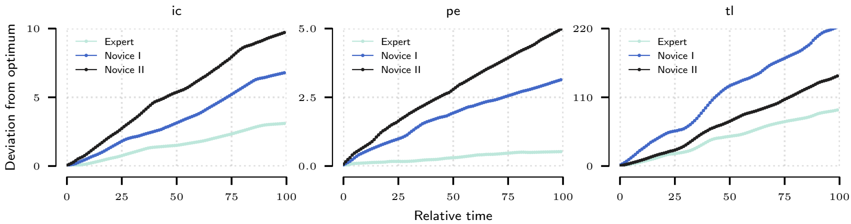

FIGURE 3.7: **Comparison of deviation from the optimum by the expert and novice agents over the course of procedure.** The error is integrated for every percent of procedure completed.

Subject	r_ψ	V_ϕ	r_{heur}	Trajectory length
Expert	$0.79_{\pm 0.20}$	$0.91_{\pm 0.05}$	$0.91_{\pm 0.035}$	$2928.2_{\pm 167.67}$
Intermediate	$0.30_{\pm 0.03}$	$0.64_{\pm 0.02}$	$0.55_{\pm 0.022}$	$2126.0_{\pm 90.48}$
Poor	$0.17_{\pm 0.035}$	$0.33_{\pm 0.035}$	$0.23_{\pm 0.027}$	$1111.2_{\pm 122.37}$

TABLE 3.2: **Normalised trajectory scores for human trajectories** (heuristic and learned rewards and values) *Expert*, *Intermediate* and *Poor* on `TraceLines` task.

As future work, validation by an independent expert using standardised evaluation protocols (Melina C. Vassiliou et al., 2005a; Goh et al., 2012) could be explored to assess the validity of the proposed evaluation metrics.

3.4.3 *Extension to laparoscopic setting*

In this section, we provide a demonstration of the proposed method on a more challenging task, the diagnostic tour of the abdomen performed by surgeons in the context of laparoscopic surgery. The task goal consists of visualising a fixed sequence of anatomical landmarks in the abdominal area. The simulated version of this task is realised using a simplified version of the VirtaMed Laparos® [7] hardware platform. The main goal of this experiment is to demonstrate the applicability of the algorithmic pipeline described in fig. 3.1 to a more sophisticated surgical task, which features a larger cohort of demonstrations. In particular, this task features both realistic textures as well as significantly more complex anatomical constraints inherent to the

7 `https://www.virtamed.com/en/products-and-solutions/simulators/laparos`

Model	Reward-Dst. (p)	Reward-Sft. (p)	Value-Dst. (p)	Value-Sft. (p)
GAIL	0.73±0.03 (0.00)	0.25±0.02 (0.01)	0.78±0.01 (0.00)	0.24±0.01 (0.02)
AIRL	-0.82±0.01 (0.00)	-0.25±0.01 (0.01)	0.80±0.00 (0.00)	0.24±0.00 (0.02)

Table 3.3: **Correlation coefficients (p-values in brackets) between laparoscopic trajectories evaluated using recovered rewards and two ground truth metrics** (*Dst*: total instrument distance and *Sft*: total safety distance).

abdominal area. Due to the more challenging nature of the full diagnostic tour, we use a subset of the tour which focuses on visualizing both liver lobes and the falciform ligament. Furthermore, we restrict the state space of the MDP $\mathcal{S} \subseteq \mathbb{R}^3$ to correspond to the position of the endoscope camera.

We evaluate the scoring functions obtained using the proposed method on a selection of human trajectories recorded as part of a surgical proficiency course [8]. We train our method using 10 trajectories, which were rated best using the internal heuristic score provided by the simulator. We use two *ground-truth* metrics: (i) *the economy score:* the total distance covered by the endoscope camera within the abdominal cavity and (ii) *the safety score:* the sum of the Euclidean distances between the endoscope camera position and the closest anatomy vertices.

We evaluate a total of 100 trajectories using the recovered reward and value functions from both GAIL and AIRL formulations and summarize the results in fig. 3.8 and table 3.3. We can observe a strong correlation in terms of the Spearman rank correlation coefficient between both the recovered reward function and the ground truth metrics as well as the recovered value function and the ground truth metrics. The total instrument path length metric is strongly correlated with both the learned reward and the learned value. The safety metric exhibits a weaker correlation which is significant according to the observed p-values.

3.5 HARDWARE PLATFORM

The hardware platform that serves as basis for the RL benchmark is the VirtaMed FAST simulator as shown in fig. 3.9. The simulator consists of a hollow dome structure with instrument entry portals for a selection of arthroscopic surgical tools (scope, hook, grasper). The output of the virtual arthroscopic camera as well as an optional third-person view of

8 Details are provided in Appendix A.2

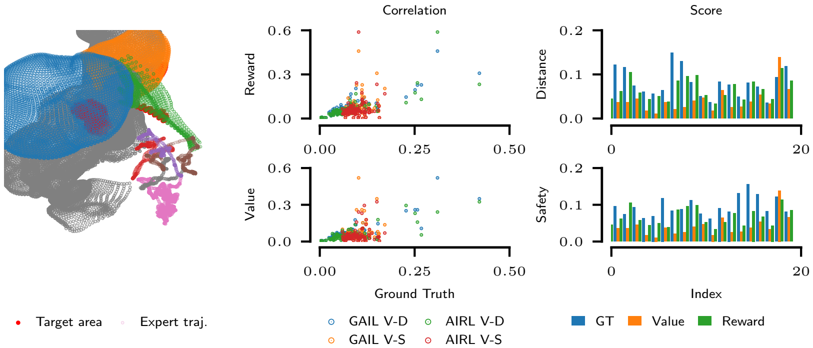

FIGURE 3.8: **Laparoscopic trajectory evaluation.** The left figure depicts the part of the anatomy to be visualized (liver lobes and falciform ligament) and corresponding trajectories. The plots on the right illustrate the trajectory scores and correlations w.r.t. ground truth metrics (V-D: Value-Distance, V-S: Value-Safety).

the dome structure are simulated on screen. The evaluation dataset for the performance ranking experiments (section 3.4.2) has been recorded using the hardware platform. Figure 3.10 depicts Laparos, the simulator platform adapted for laparoscopic surgery, which was used to record the expert and user demonstrations in section 3.4.3.

3.6 ADDITIONAL RESULTS

This section provides additional results obtained using the algorithmic pipeline in fig. 3.1 and partially described in section 3.4.1.

3.6.1 *Ablation study on reward components*

In this set of experiments, we demonstrate the dependence of the forward RL methods on the reward shaping scheme. Due to the intricate task structure which includes a combination of multiple optimization objectives, we introduce a number of additional reward components in order to improve the sample efficiency of the algorithms solving the tasks. We perform an ablation study on the reward components as shown in fig. 3.11 for the three tasks in order to demonstrate the necessity of reward shaping. The evaluation is carried out using the on-policy version of the policy algorithm (PPO).

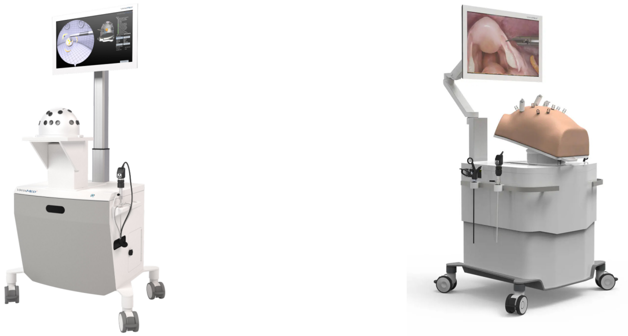

FIGURE 3.10: VirtaMed FAST (left) and Laparos (right) simulation platforms.

The five ablation settings comprise the baseline reward structure (`baseline`), the removal of distance and angle potentials respectively (`noDist`, `noAngle`, `noDistNoAngle`) and the sparse reward setting where a reward is only given if the target is visualised correctly (`onlyTaskCompleted`). The plots report the median of tasks completed over 5 randomly seeded repetition experiments per ablation setting. We can observe that potential based reward shaping on both Cartesian and angular components is crucial to obtain a policy which solves all targets successfully. We can observe a similar behavior on the `Periscoping` task. In the case of `TraceLines`, the gradation is less distinct due to the specifics of the task. The visual components of the reward are statistically more prevalent in the `TraceLines` task and allow for a more dense reward structure also in absence of the distance and angle penalties. The sparse reward signal (`onlyTaskCompleted`) fails to solve a single target.

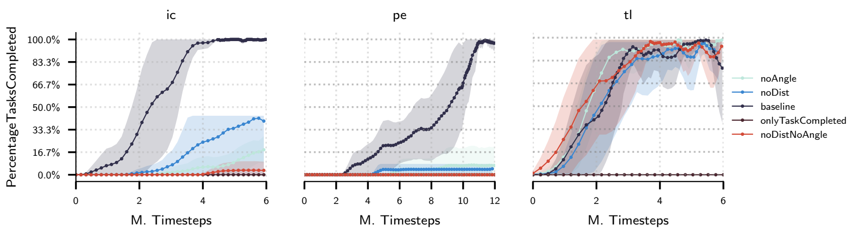

FIGURE 3.11: Ablation experiments: number of tasks completed for `ImageCentering` (ic), `Periscoping` (pe) and `TraceLines` (tl) environments.

3.7 DISCUSSION

In this chapter, we have presented FASTRL, an RL benchmark for training virtual assistant agents in the arthroscopic surgery domain. We have investigated the performance of a number of state-of-the-art forward and inverse RL approaches and illustrated a prototypical solution to a surgical evaluation and assistance framework for arthroscopic procedures using model-free RL techniques. We have performed a feasibility study using our API extension of the simulation software provided by VirtaMed AG.

LIMITATIONS Currently, the benchmark is limited to a subset of the tasks available as part of the FAST training program. More sophisticated scenarios involving additional instruments such as the palpation hook or the grasper that provide additional complexity to the trainees are conceptually equivalent to the presented tasks, but worthy of further investigation. In its current form, the pipeline has solely been evaluated on kinematic data but the possibility exists to train image-based models on the rendered arthroscope camera images.

4

LEARNING CAUSALLY INVARIANT REWARD FUNCTIONS FROM DIVERSE DEMONSTRATIONS

The first principle is not to fool yourself – and you are the easiest person to fool.

— Richard Feynman

This chapter delves into the first significant methodological enhancement to the algorithmic pipeline introduced in chapter 3. As we have discussed earlier, inverse reinforcement learning methods are designed to infer the reward function of a Markov decision process from a dataset of expert demonstrations. The inherent challenges associated with these demonstrations –namely their scarcity and the heterogeneity of their sources – often result in the learned reward function capturing spurious correlations present in the data. Such issues can significantly distort the reward function's fidelity. This adaptation frequently leads to behavioral overfitting when a policy, trained on the derived reward function, is subjected to shifts in environmental dynamics. Such overfitting manifests as policies that perform well on training distributions but poorly when transferred to new, unseen environments.

In response to these challenges, this chapter proposes a novel regularization technique for IRL that is grounded in the principle of causal invariance. This principle aims to enhance the generalization of the reward function across different contexts by mitigating the influence of spurious correlations. We explore how incorporating this causal invariance into both exact and approximate formulations of IRL can help in developing more robust reward functions.

By applying this regularization approach, we demonstrate that it is possible to achieve superior policy performance. Specifically, when policies are trained using the refined reward functions and then transferred to different settings, they exhibit enhanced adaptability and effectiveness. This chapter provides both theoretical insights and empirical evidence to support the efficacy of our proposed approach.

4.1 LEARNING REWARDS FROM DIVERSE DEMONSTRATIONS

In the domain of reinforcement learning, the formulation of a suitable reward function plays a pivotal role in shaping the behavior of decision making agents. This is commonly justified by the widely adopted belief that the reward function is a succinct representation of a task goal in a given environment specified as a Markov decision process (MDP) (Ng, Russell, et al., 2000). Eliciting the correct behavioral policies via the optimization of a reward function is of paramount importance for the deployment of RL agents to real world domains such as various robotics scenarios (Billard et al., 2008; Pomerleau, 1991) or expert behavior forecasting (Kitani et al., 2012). However, the challenge of designing such a function typically entails a cumbersome and error-prone process of handcrafting a heuristic reward signal which accounts for all the intricacies of the task at hand.

Inverse reinforcement learning (IRL) methods aim to solve the problem of inferring the reward function of an MDP based on a dataset of temporal behaviors. These trajectories are typically obtained from an agent that is assumed to demonstrate near-optimal performance in the respective MDP. There are multiple benefits to learning an explicit reward function compared to alternatives such as behavioral cloning (Pomerleau, 1991) or other imitation approaches (Ho et al., 2016; Fu, Luo, et al., 2018) including the the ability to transfer the reward function across problems and enhanced robustness to compounding errors (Swamy, Choudhury, J. A. Bagnell, et al., 2021).

The IRL problem is challenging due to a number of factors. The problem of recovering the reward from the statistics of the expert trajectories is generally ill-posed, as there typically exist many reward functions which satisfy the optimization constraints (Ng, Harada, et al., 1999). While this property is effectively tackled by regularization in the form of a maximum entropy objective (Ziebart, J. A. Bagnell, et al., 2010; Ziebart, Maas, et al., 2008), scaling IRL to large scale problems remains a challenge requiring a variational formulation dependent on non-linear function approximation methods (Fu, Luo, et al., 2018; Finn, P. Christiano, et al., 2016). Since the amount of available expert data is typically limited, this can lead to overfitting phenomena, which are particularly pronounced in high capacity models such as neural networks (Ying, 2019). This type of observational overfitting has been shown in the context of forward reinforcement learning (Song et al., 2019). An additional difficulty arises when there is a significant discrepancy between expert demonstrations originating from different experts. Adversarial imitation methods fail to recover accurate behavior from diverse experts as shown in (Li et al., 2017), since such learning results in ambiguities when inferring reward functions. We show that pooling expert demonstrations in one dataset under the same label introduces spurious correlations which are absorbed in the representation of the reward function. By optimizing the expected cumulative reward defined by such functions, the agent might fail to learn meaningful behaviors due to optimization of the spurious correlations present in the reward model.

In order to circumvent this issue, we consider causal properties of the reward learning problem. The causal invariance principle (Peters, Bühlmann, et al., 2015; Heinze-Deml et al., 2017; Arjovsky, Bottou, et al., 2019) studies the generalization problem of supervised learning models through the lens of causality. It postulates that the conditional distribution of the target variable must be asymptotically stable across samples obtained under observational and interventional settings of the data generating process. By only considering causally invariant representations of the input, this ensures avoiding the reliance on spurious correlations in the model predictions. We propose to adapt this principle for the context of reward function learning, where we aim to elicit behavioral policies without exploiting spurious reward features, i.e. features which would prevent the reward from providing a meaningful policy training signal under distribution shift. To achieve this goal, we make the assumption that variations in expert demonstrations are a product of causal interventions on the data generating process of trajectories and the optimality label conditional must be stable for experts demonstrations gathered on a specific task. By applying the causal invariance principle under this assumption, we show that we can recover reward functions which are invariant w.r.t. certain types of distribution shift across a population of experts.

CONTRIBUTIONS. Our contributions are as follows: (i) we formulate the assumption that the variations between experts that are considered to perform near-optimally on the same task can be seen as interventional settings of the underlying trajectory distribution; (ii) we propose a regularization principle for inverse reinforcement learning methods based on the principle of causal invariance. This modelling choice allows us to learn reward functions that are invariant to spurious correlations between the transitions and optimality label present in the expert data; (iii) we demonstrate the efficacy of this approach in both tractable, finite state-spaces, which we refer to as the exact setting as well as large continuous state-spaces, which we denote as the approximate setting. In the first setting, we visually analyze the recovered reward functions and verify their invariance properties w.r.t. the input data. In the second setting, we demonstrate improved ground truth performance when a policy is trained using the regularized reward in MDPs with perturbed dynamics.

4.2 METHOD

In this section, we outline the proposed method. Section 4.2.1 presents the problem setting of learning rewards in the maximum entropy IRL setting (Ziebart, Maas, et al., 2008) in both primal and dual form and reviews the connection to a class of adversarial optimization methods based on f-divergences chosen as a numerical solution strategy for the problem. In section 4.2.2, we describe how spurious correlations arise in the context of the IRL problem and review the causal invariance principle. Section 4.2.3 shows how to incorporate this principle in the context of reward learning as a regularization strategy.

4.2.1 *Problem setting*

We begin by giving an overview of the problem setting. We consider environments modelled by a *Markov decision process* $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mu_0, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{T}$ is the family of transition distributions on $\mathcal{S}$ indexed by $\mathcal{S} \times \mathcal{A}$ with $p(s'|s,a)$ describing the probability of transitioning to state s' when taking action a in state s, μ_0 is the initial state distribution, $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function and $\gamma \in (0,1)$ is the discount factor. A policy $\pi : \mathcal{S} \times \mathcal{A} \to \Omega, (s,a) \to \pi(a|s)$[1] is a map from states $s \in \mathcal{S}$ to distributions $\pi(\cdot|s)$ over actions, with $\pi(a|s)$ being the probability of taking action a in state s.

In absence of a given ground truth reward function, inverse reinforcement learning methods aim to estimate a suitable reward function based on a dataset of expert trajectories $\mathcal{D}_E = \{\xi_i\}_{i \leq K}$ where $\xi_i = (s^{(i)}_{1:T_i}, a^{(i)}_{1:T_i})$ is a sequence of states and actions of expert i of length T_i. To achieve this goal, a number of methods based on distribution matching may be used. Such methods typically minimize a divergence measure (Csiszár, 1972) between the expert trajectories and the trajectories induced by a policy optimizing the estimated reward.

1 Ω denotes the set of probability measures over the action space

We begin by presenting the maximum entropy IRL (MaxEntIRL) (Ziebart, Maas, et al., 2008) method, which serves as a foundation for most of these methods. In MaxEntIRL, the first order feature statistics of the trajectory $\varphi(\xi)$ under the empirical state occupancy measure of the expert, $\mathbb{E}_{\xi\sim\mathcal{D}_E}[\varphi(\xi)]$, are matched with the statistics of the trajectory distribution $p(\xi|\psi,\varphi)$ induced by a policy interacting with the environment: $\mathbb{E}_{\xi\sim p(\xi|\psi,\varphi)}[\varphi(\xi)]$. The policy π_{r_ψ} is trained by optimizing the expected cumulative reward using the reward function estimate r_ψ. The model describes the fact that the expert trajectories are sampled from the Gibbs distribution:

$$p(\xi|\psi,\varphi) = \frac{1}{Z_{\varphi,\psi}} \exp(r_\psi(\varphi(\xi))) \tag{4.1}$$

which corresponds to the solution of the entropy maximization problem of the trajectory distribution under feature matching and normalization constraints.

In the more general case of stochastic dynamics $p(s_{t+1}|s_t,a_t)$ and random initial state distribution μ_0, we can define the *generative model of trajectories* $p(\xi|O_{1:T})$ conditioned on the optimality variables $\{O_t\}_{t=1:T}$ as follows (Levine, 2018):

$$p(\xi|O_{1:T}) \propto p(\xi,O_{1:T}) = \mu_0(s_0)\prod_{t=1}^{T} p(O_t=1|s_t,a_t)p(s_{t+1}|s_t,a_t) \tag{4.2}$$

where the conditional distribution $p(O_t=1|s_t,a_t) \propto \exp(r_\psi(\varphi(\xi)))$ encodes the optimality of a single timestep of the trajectory, i.e. $O_t = 1$ corresponds to an expert-level transition. The optimal reward weight solution ψ^* can be obtained by maximizing the likelihood of eq. (4.2) w.r.t. the parameter ψ. In the feature matching case, the reward $r_\psi(\varphi(\xi))$ is typically assumed to be linear in the state features $r_\psi(\xi) = \psi^T\varphi(\xi)$. Here, the state features $\varphi(\xi)$ may be specified a priori or learned using a neural network model (Wulfmeier et al., 2015). We state the *primal maximum-likelihood objective* (Ziebart, Maas, et al., 2008):

$$\max_\psi \mathbb{E}_{\xi\sim\mathcal{D}_E}\left[\log\frac{1}{Z(\psi)}e^{\psi^T\varphi(\xi)}\right] = \max_\psi \mathbb{E}_{\xi\sim\mathcal{D}_E}[\psi^T\varphi(\xi) - \log Z(\psi)] \tag{4.3}$$

VARIATIONAL DUAL FORMULATION. Due to the difficulties of computing the log-partition function in high-dimensional spaces, a dual formulation of the maximum likelihood problem is derived. The Gibbs distribution over trajectories $p(\xi|\psi,\varphi)$ obtained via the maximum entropy principle belongs to the exponential family of distributions:

$$p(\xi|\psi,\varphi) = p_0(\xi)\exp(\psi^T\varphi(\xi) - A(\psi)) \tag{4.4}$$

where $A(\psi) = \log Z_\psi = \log\int_\Xi \exp(\psi^T\varphi(\xi))p_0(\xi)d\xi$ is the log-partition function defined over the space of trajectories Ξ and p_0 is the base measure. Leveraging the strict concavity of the log-partition function $A(\psi)$ in ψ (M. J. Wainwright et al., 2008), the Fenchel-Legendre dual (Rockafellar, 2015) of $A(\psi)$ is given by

$$A(\psi) = \max_{\varphi\in\Phi}\langle\psi,\varphi(\xi)\rangle - D_{KL}(q||p_0) = \max_{q\in\mathcal{P}}\langle q(\xi), g_{\psi,\phi}(\xi)\rangle - D_{KL}(q||p_0) \tag{4.5}$$

where we use the generalization of the marginal polytope Φ over first-order statistics $\varphi(\xi)$ to the space of distributions $\mathcal{P}$ (Dai et al., 2019) and $g_{\psi,\phi}(\xi) = \psi^T\phi(\xi)$. By plugging in the resulting dual of the log-partition in eq. (4.5) into the maximum likelihood objective in eq. (4.3), we obtain the dual saddle-point objective:

$$\max_{\psi,\phi}\min_{q\in\mathcal{P}}\mathbb{E}_{\xi\sim\mathcal{D}_E}[g_{\psi,\phi}(\xi)] - \mathbb{E}_{\xi\sim q}[g_{\psi,\phi}(\xi)] + D_{KL}(q||p_0) \tag{4.6}$$

In our case, the base measure p_0 corresponds to the Lebesgue or count measure according to the continuity of the state space, i.e. $p_0(\xi) = 1$. Thus, $D_{KL}(q||p_0)$ simplifies to the negative entropy of the sampling distribution $\mathcal{H}(q)$.

CONNECTION TO f-DIVERGENCES. The resulting dual problem is closely related to f-divergence (Csiszár, 1972) minimization which allows us to obtain a numerical solution strategy for eq. (4.6). It is well known that the maximum-likelihood problem (eq. (4.3)) is asymptotically equivalent to the minimization of the KL-divergence $D_{KL}(p(\xi|\theta^*)||p(\xi|\theta))$ (Andersen, 1970), where θ^* is the parameter vector maximizing the likelihood of $p(x|\theta^*)$. The variational formulation of the more general f-divergence minimization problem, of which D_{KL} is an instance, is given by:

$$D_f(p||q) = \mathbb{E}_q\left[f\left(\frac{p}{q}\right)\right] \geq \sup_{g:\mathcal{X}\to\mathbb{R}}\mathbb{E}_p[g(\xi)] - \mathbb{E}_q[f^*(g(\xi))] \tag{4.7}$$

where f is a convex function and f^* denotes its Fenchel conjugate. In particular, for $f(x) = \frac{1}{2}|x-1|$, we obtain the *total variation distance* $D_{TV}(P,Q)$:

$$D_{TV}(p||q) = \sup_{||g||_\infty\leq 1}\mathbb{E}_p[g(\xi)] - \mathbb{E}_q[g(\xi)] \tag{4.8}$$

which is equivalent to eq. (4.6) for a restricted class of functions $g \in \{g : \mathcal{X} \to \mathbb{R}, ||g||_\infty \leq 1\}$ and an entropy regularization term for the sampling distribution q. Moreover, it is related to D_{KL} via Pinsker's inequality $D_{TV}(p||q) \leq \frac{1}{2}\sqrt{D_{KL}(p||q)}$ (Gilardoni, 2010).

In order to perform the minimization of the f-divergence objective in eq. (4.7), we can leverage a correspondence between optimal risk functions and f-divergences. In particular, for a given f-divergence D_f, there exists a corresponding decreasing convex risk function $\phi(\alpha)$ of the classification margin α, such that the optimal risk $R_\phi = -D_f$ (X. Nguyen et al., 2009, Thm. 1). This correspondence allows the objective to be described as a two-player zero-sum game and is amenable to optimization using *generative-adversarial-network*-like (Goodfellow et al., 2014; Finn, P. Christiano, et al., 2016) frameworks. When used to minimize the divergence between the expert and policy occupancy measures, this problem class includes many adversarial imitation learning algorithms (Ho et al., 2016; Fu, Luo, et al., 2018; Ni et al., 2021). We will use these algorithms as solution strategies for eq. (4.6).

Now that we have both exact and approximate formulations of the IRL problem (eq. (4.3), eq. (4.6)) we shall see how spurious correlations can arise when the expectations over the datasets $\mathcal{D}_E$ in eq. (4.3) and eq. (4.6) are evaluated over samples from diverse sources.

4.2.2 *Spurious correlations and causal invariance approaches*

We shall now outline the intuition as to what we consider spurious correlations in inverse reinforcement learning. It is necessary, at this point, to introduce structural causal models, which will help define the notion of spurious correlations. A structural causal model (SCM) (Pearl, 2009) is defined as a tuple $\mathfrak{G} = (S, P(\varepsilon))$, where $P(\varepsilon) = \prod_{i \leq K} P(\varepsilon_i)$ is a product distribution over exogenous latent variables ε_i and $S = \{f_1, ..., f_K\}$ is a set of structural mechanisms where $\text{pa}(i)$ denotes the set of parent nodes of variable x_i: $x_i := f_i(\text{pa}(x_i), \varepsilon_i) \quad \text{for } i \in |S|$. $\mathfrak{G}$ induces a directed acyclic graph (DAG) over the variables nodes x_i. The SCM entails a joint observational distribution $P_{\mathfrak{G}} = \prod_{i \leq K} p(x_i|\text{pa}(x_i))$ over variables x_i conditioned on the parents of x_i for some probability distribution $p(\cdot|\text{pa}(x_i))$ describing the mechanism f_i. Interventions on $\mathfrak{G}$ constitute modifications of structural mechanisms f_i yielding interventional distributions $\tilde{P}_{\mathfrak{G}}$. In the context of IRL, we consider interventional settings of the expert trajectory distribution $p(\xi|\psi, \varphi)$ in eq. (4.2).

In supervised learning, a correlation between the input representation and the label is considered spurious if it does not generalize under distribution shift – e.g. when the trained model is evaluated on a test set of unseen examples. More specifically, the distribution shift constitutes an intervention on the causal parents of the target label, which allows the application of invariance based approaches, that we will describe below. In the case of IRL, we consider a correlation to be spurious when a reward function does not generalize to perturbations in the initial measure or dynamics of the MDP, i.e. the policy optimizing a reward that reflects this correlation will absorb this signal and fail to perform optimally under perturbed environment dynamics.

TRANSITION SCM. To illustrate scenarios in which such spurious correlations arise, we consider the structural causal model of a transition in Figure 4.1(a). At timestep t, the model is composed of the state s_t , action a_t, next state s_{t+1}, the optimality label O_t describing the optimality of the transition at timestep t, a variable E encoding the setting index $e \in I(\mathcal{E}_{tr})$ and an unobserved latent variable C, which might vary as E changes. In this model, we would like to identify non-causal information paths between the environment index E and the optimality label. The presence of such paths corresponds to spurious correlations.

In Figure 4.1b we can observe the scenario where we do not condition the representation of the optimality conditional on the action. By conditioning on the collider node s_{t+1} and not observing the action node a, a backdoor path Pearl, 2009 is formed between the setting index E and the optimality variable O_t, resulting in the violation of their conditional independence relationship. A second scenario can be observed in Figure 4.1c. This scenario requires the assumption that the orientation of the edge from node O_t to node s_{t+1} is temporally causal, meaning that the optimality of a state s_t at time t is a causal parent of the next state. In this case, observing the collider node s_{t+1} makes nodes E and O_t conditionally dependent [2]: $E \not\perp\!\!\!\perp O_t | s_{t+1}$. Beyond these scenarios, one can further assume a more general partitioning of an arbitrary transition input (s, a, s') into the causal transition feature components $\mathbf{x}^{(c)} = f(s, a, s')$ and $\mathbf{x}^{(nc)} := (\mathbf{x}^{(c)}, \mathbf{x}^{(nc)})$, illustrated in Figure 4.1d, whereby conditioning on the $\mathbf{x}^{(nc)}$ collider introduces a spurious correlation path. Among other causes, this situation can arise due to selection bias caused by a prevalence of certain transitions in the pooled dataset, i.e., the pooling of *diverse sources of demonstrations* under the binary

2 Here, $\perp\!\!\!\perp$ denotes statistical independence

optimality label introduces a spurious correlation between states visited by expert policies as a consequence of the respective preferences and the binary optimality label.

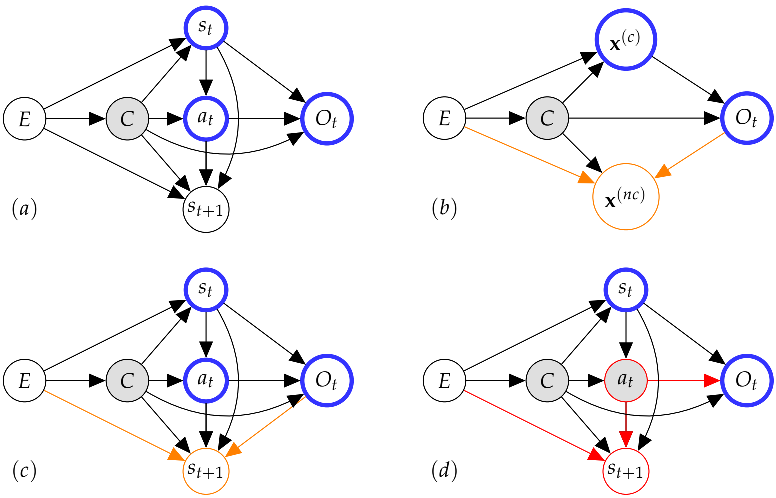

FIGURE 4.1: **Transition SCM** (a) Probabilistic graphical model of a transition under influence of the index variable E and latent variable C. The invariant conditional is highlighted in blue. (b) General setting where O_t depends on causal $\mathbf{x}^{(c)}$ and non-causal $\mathbf{x}^{(nc)}$ features of the transition. Conditioning on the collider variable (in orange) creates a spurious correlation path. (c) Collider conditioning assuming wrong edge orientation $O_t \rightarrow s_{t+1}$. This corresponds to O_t being the causal parent of s_{t+1} (d) Spurious correlations arising under assumption of state-only formulation of the reward. Since a_t is unobserved, a backdoor path (in red) is formed.

DATASET PARTITIONING AND TRAINING SETTINGS. The standard IRL procedure typically pools input demonstration data into one dataset which may lead to absorption of these correlations by the learned reward. To resolve this, we first need to make the assumption that the data is partitioned according to different sources obtained from observational and interventional settings of the trajectory distribution. This assumption is valid in three different scenarios: (i) expert demonstrations were gathered on different environment dynamics, (ii) initial states were sampled from different initial state distributions or (iii) experts have reward preferences, i.e. optimize a perturbed version of the reward associated with the true task goal. In the context of the current work, we consider the source diversity to be a consequence of the individual expert preferences expressed as reward functions parameterizing eq. (4.1). We denote the union set of samples obtained from these settings as $\mathcal{E}_{tr}$, the set of *training settings*, and assume access to multiple such settings $\mathcal{D}_e := \{(\xi_i)\}_{i=1}^{|\mathcal{E}_{tr}|}$ during training.

FROM ROBUSTNESS TO INVARIANCE. The consideration of multiple source distributions and associated target noise warrants the application of robust methods such as distributionally robust optimization (DRO) (Bashiri et al., 2021; Namkoong et al., 2016; Viano et al., 2021). Such approaches modify the loss objective by searching over the space of empirical distributions indexed by $e \in \mathcal{E}_{tr}$ under which the expected loss $\mathcal{L}_e$ is maximized. This is implemented in practice by searching over the set of training settings $\mathcal{E}_{tr}$ resulting in a min-max objective for some function class $f \in$:

$$\min_f \max_{e \in \mathcal{E}_{tr}} -\mathbb{E}_{\xi \sim \mathcal{D}_e} \mathcal{L}_e(f, \xi) \tag{4.9}$$

This effectively regularizes the model by optimizing based on its *worst-case* performance. While this modification can tackle the issue of model overfitting to scarce data, it does not address potential diversity of the data due to mode-seeking behavior of the min-max problem (Rahimian et al., 2019). This behavior describes the fact that the maximum "latches on" to the training setting with the largest likelihood loss which might constitute a spurious mode of the demonstrations with respect to the actual task goal. Arjovsky, Bottou, et al. show that the solution of robust regression is a first order stationary point of the weighted square error (in the convex case), if the variance of the loss is used as a per-setting bias (Arjovsky, Bottou, et al., 2019, Prop. 2). This limits the generalization capacity to the convex hull of the training settings $\mathcal{E}_{tr}$.

Following (Arjovsky, Bottou, et al., 2019), we consider causal properties of the robust estimation problem for the purposes of improved generalization outside of the convex hull of training settings. The *causal invariance principle* postulates that for the problem of estimating a target conditional, e.g. classification label, one should only consider variables which belong to the set of causal parents of the target. More specifically, causally invariant covariates must yield the same conditional distribution of the target in both observational and interventional settings of the SCM. Lifting the distributional restrictions of the methods described in (Peters, Bühlmann, et al., 2015), the invariant risk minimization (IRM) principle (Arjovsky, Bottou, et al., 2019) aims to identify a causally invariant data representation φ by instantiating a bi-level optimization problem for a representation function φ and a predictor function w:

$$\min_{\substack{\varphi:\mathcal{X}\to\mathcal{H} \\ w:\mathcal{H}\to\mathcal{Y}}} \sum_{e\in\mathcal{E}_{tr}} \mathcal{L}^e(w\circ\varphi) \ \text{ s.t.: } \ w\in \operatorname*{argmin}_{\bar{w}:\mathcal{H}\to\mathcal{Y}} \mathcal{L}^e(\bar{w}\circ\varphi) \quad \forall e\in\mathcal{E}_{tr} \tag{4.10}$$

This objective admits an unconstrained relaxation using the gradient norm penalty $\mathbb{D}(w = 1.0, \varphi, e) = ||\,|\nabla_w \mathcal{L}^e(w\circ\varphi)|_{w=1.0}\,||^2$ which quantifies the optimality of a fixed predictor ($w = 1.0$) at each setting e (Arjovsky, Bottou, et al., 2019). This leads to the following unconstrained formulation of the learning problem:

$$\min_{\varphi:\mathcal{X}\to\mathcal{Y}} \sum_{e\in\mathcal{E}_{tr}} \mathcal{L}^e(\varphi) + \lambda||\,|\nabla_w \mathcal{L}^e(w\circ\varphi)|_{w=1.0}\,||^2 \tag{4.11}$$

In the following section, we will show how to incorporate this principle into the IRL setting.

4.2.3 *Reward regularization using causal invariance*

Mapping the objective in eq. (4.10) to the context of reward learning, we consider data gathered by different experts to correspond to interventional settings of the trajectory distribution in eq. (4.2), where the interventions reflect the varying preferences exhibited by the policy of the respective experts. The stable conditional that we would like to identify corresponds to the conditional distribution of the optimality label $P(O_t|s_t, a_t)$. To achieve this goal, we invoke the causal invariance principle to learn reward functions that utilize features, invariant to some class of deviations exhibited by the experts. We motivate this regularization strategy by the fact that despite the discrepancies in the demonstrations, all experts are assumed to perform the task in an optimal fashion with respect to the true task goal. This assumption implies that all experts optimize the same underlying objective that we would like to recover as a reward function. In doing so, we hope to extract succinct descriptions of the underlying agreed intentions of the experts as reward functions. As a result, we expect such rewards to be more readily applicable to MDPs with distribution shift of the dynamics. As a next step, we show how to apply this principle into practice by introducing the causal invariance (CI) regularization, instantiated as the IRM penalty described in eq. (4.11).

FEATURE MATCHING REGULARIZATION. We begin by considering the maximum entropy feature matching problem. For the Gibbs distribution $p(\xi|\psi,\varphi) = \exp(\psi^T\varphi(\xi))/Z_{\varphi,\psi}$ over trajectories, we can write down the constrained optimization problem analogously to eq. (4.10):

$$\max_{\varphi,\psi} \sum_{e\in\mathcal{E}_{tr}} \sum_{\xi\in\mathcal{D}_e} \log p(\xi|\psi,\varphi) \text{ s.t. } \psi \in \underset{\bar{\psi}}{\operatorname{argmax}} \sum_{\xi\in\mathcal{D}_e} \log p(\xi|\bar{\psi},\varphi) \tag{4.12}$$

Intuitively, due to convexity of the likelihood function w.r.t. the natural parameter ψ (M. J. Wainwright et al., 2008), the constraint corresponds to penalizing the deviation ψ in $p(\xi|\psi,\phi)$ from the optimal parameter ψ^* which maximizes the likelihood $p(\xi|\psi^*,\phi)$. In an analogous fashion to the IRM approach described above (eq. (4.10)), we propose to relax the constrained optimization problem by defining a regularization term $\mathbb{D}(\psi,\varphi,e)$ which describes this deviation. Let $\mathcal{E}_{tr}$ be the set of training settings and ψ,φ be the parameters of the likelihood $p(\xi|\psi,\varphi)$. $\mathbb{D}(\psi,\varphi,e)$ is a distance function representing the violation of the constraints of eq. (4.12) in training setting $e \in \mathcal{E}_{tr}$ w.r.t. the optimal solution.

$$\mathbb{D}(\psi,\varphi;e) = |||\nabla_\psi \mathcal{L}^e(\psi,\varphi)|_{\psi=1.0}||^2 \tag{4.13}$$

In our case, the deviation is computed w.r.t. the parameters maximizing the likelihood of the Gibbs distribution due to convexity of $\mathcal{L}_{MLE}$ w.r.t. function $g_{\psi,\varphi}$. Applying the gradient of this penalty to the representation function φ effectively regularizes the representation to minimize the constraint violations. In simple tabular MDPs, where the computation of the partition function is tractable, we directly apply this regularizer to the primal maximum likelihood objective as follows:

$$\begin{aligned}\mathcal{L}_{MLE}(\psi,\phi,e) = \\ \max_{\varphi,\psi} \sum_{e\in\mathcal{E}_{tr}} \left(\mathbb{E}_{\xi\in\mathcal{D}_e}\left[\log\left(\frac{1}{Z_{\psi,\phi}}\exp(\psi^T\varphi(\xi))\right) + \lambda\mathbb{D}(\psi,\varphi,e)\right]\right)\end{aligned} \tag{4.14}$$

In the primal case, for a trajectory distribution described by an exponential family, we can derive a closed form of the gradient penalty. We summarize this result as the following proposition [3]:

Proposition 1. *Let the likelihood $p(\xi)$ belong to a natural exponential family with parameter ψ, sufficient statistics $\varphi(x)$ and the (Lebesgue) base measure p_0. Let $\mathcal{D}_E^e$ be the dataset corresponding to interventional setting e. Then, for all $e \in \mathcal{E}_{tr}$,*

3 All omitted proofs are found in appendix A.3

Algorithm 3 CI regularized Feature Matching IRL (CI-FMIRL)

Input: Expert trajectories $\mathcal{D}_E^e$ assumed to be obtained from multiple experts *by intervening on* $p(\xi|\psi, \varphi)$
Init: Initialize reward estimate r_ψ and state feature network φ_θ
for setting e in $\{1, ..., \mathcal{E}_{tr}\}$ **do**
  while $r_{\psi,\varphi}$ not converged **do**
    Compute feature matching gradient $\nabla_\psi \mathcal{L}(\psi, \varphi; e) = \mathbb{E}_{\mathcal{D}_E^e}[\varphi(\xi)] - \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi))$ and *causal invariance* penalty gradient $\nabla_\varphi \mathbb{D}(\psi, \varphi; e)$ and backpropagate the weighted sum through feature network $\varphi_\theta(s)$

    Compute policy $\pi_{r_{\psi,\varphi}}$ using value iteration on the reward estimate $r_{\psi,\varphi}$
Return: Trained reward $r_{\varphi,\psi}$

the causal invariance penalty for the maximum likelihood loss is the norm of the expectation difference of features φ*:*

$$\mathbb{D}(\psi, \varphi; e) = ||\nabla_{\psi|\psi=1.0}\mathcal{L}^e(\psi, \varphi)||^2 = ||\mathbb{E}_{\mathcal{D}_E^e}[\varphi(\xi)] - \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi))]||^2 \tag{4.15}$$

This closed form of the gradient norm penalty can be utilized in the maximum causal entropy solver (Ziebart, J. A. Bagnell, et al., 2010). We assume the state features to be the output of a neural network according to the DeepMaxEnt model (Wulfmeier et al., 2015). In order for the network to adopt invariant features, the gradient norm penalty in eq. (4.15) is used to update the feature network using backpropagation [4]. The resulting algorithm (CI-FMIRL) is presented in Algorithm 3.

PENALTY FOR DUAL FORMULATION. In large scale MDPs, where the evaluation of the log-partition function is intractable, we use the variational dual objective outlined in eq. (4.5). In order to regularize this objective, we first need to derive the causal invariance penalty for the dual formulation. By leveraging the strict concavity of *eq.* (4.6) w.r.t. q, we can straightforwardly extend the distance penalty to the dual formulation as follows:

$$\mathcal{L}_{dual}(\psi, \varphi, q, e) = \tag{4.16}$$
$$\max_{\psi,\varphi} \sum_{e \in \mathcal{E}_{tr}} \min_q \left[\mathbb{E}_{\xi\sim\mathcal{D}_E^e}[g_{\psi,\varphi}(\xi)] - \mathbb{E}_{\xi\sim q}[g_{\psi,\varphi}(\xi)] + D_{KL}(q||p_0)\right] + \lambda\mathbb{D}(\psi, \varphi, e)$$

4 We derive a closed form of the gradient estimate in appendix A.3

The numerical solution strategy for the saddle-point objective in Equation (4.17) can be realized using a two-player min-max game implemented using a GAN-like framework (Finn, Levine, et al., 2016). More specifically, an equivalence between the maximum likelihood problem for a Gibbs distribution over trajectories and the corresponding variational approximation has been shown in (Fu, Luo, et al., 2018, App. A). We state the CI gradient penalty as a result of the following proposition.

Proposition 2. *The gradient of the primal exponential family maximum-likelihood problem in Equation (4.3) w.r.t. the natural parameter ψ is equivalent to the gradient of the dual in Equation (4.5) w.r.t the parameter ψ when the density ratio $\frac{q}{p}$ is unity.*

$$||\nabla_{\psi}\mathcal{L}_{dual}(\psi,\varphi,q,e)||_2 = \\ ||\underset{q}{min}\, \mathbb{E}_{\xi\sim\mathcal{D}_E}[\varphi(\xi)] - \mathbb{E}_{\xi\sim p(\xi|\psi,\varphi)}\left[\frac{q(\xi)}{p(\xi|\psi,\varphi)}\varphi(\xi)\right]||_2 \tag{4.17}$$

Using this result, we can apply the gradient penalty to the dual. Effectively, the resulting gradient estimate requires an importance sampling estimate of the second expectation using the sampling trajectory distribution q induced by the policy. The minimum over q is attained when the importance weight is unity.

As we have seen in section 4.2.1, the dual objective is closely related to the f-divergence minimization objectives which form the basis of multiple *adversarial imitation learning* algorithms (Ho et al., 2016; Fu, Luo, et al., 2018; Ni et al., 2021). This class of algorithms leverages the correspondence between the f-divergence objectives and equivalent binary classification surrogate losses as described in (X. Nguyen et al., 2009). One such example is the optimal logistic loss of a binary classification function $g_D(x) = \frac{p(x)}{p(x)+q(x)}$ which corresponds to the Jensen-Shannon divergence (Ho et al., 2016) between distributions $p(x)$ and $q(x)$:

$$\max_{g_D} \mathcal{L}_{BCE}(g_D) = \\ \max_{g_D} \mathbb{E}_{x\sim\mathcal{D}_E}[\log g_D(x)] + \mathbb{E}_{x\sim\pi}[\log(1-g_D(x))] = D_{JS}(p||q) \tag{4.18}$$

In practice, we can make use of this equivalence in order to apply the penalty defined in section 4.2.3 solely to the discriminator function $g_D(x)$ which classifies transitions sampled from the expert datasets $\mathcal{D}_E^e$ and transitions obtained from the imitation policy.

Algorithm 4 CI regularized Adversarial IRL (CI-AIRL)

Input: Expert trajectories $\mathcal{D}_E^e$ assumed to be obtained from multiple experts *by intervening on* $p(\xi|\psi, \varphi)$
Init: Initialize actor-critic $\pi_\theta, \nu_\vartheta$ and discriminator $g_{\xi,\phi}$
for setting e in $\{1, ..., \mathcal{E}_{tr}\}$ **do**
  Collect trajectory buffer $\mathcal{D}_\pi = \{\xi_i\}_{i \leq |\mathcal{D}_\pi|}$ by executing the policy π_θ
  Update $g_{\phi,\theta}(s, a)$ via binary logistic regression by maximizing

$$\mathcal{L}(\phi, \psi; e) = \mathcal{L}_{\text{BCE}}(\xi, \varphi, \psi; e) + \lambda||\nabla_{\psi|\psi=1.0}\mathcal{L}_{\text{BCE}}(\xi, \varphi, \psi; e)||^2$$

  using dataset tuple $(\mathcal{D}_E^e, \mathcal{D}_\pi)$
  Update actor-critic $(\pi_\theta, \nu_\vartheta)$ w.r.t. the reward function of the *regularized discriminator* using the *soft-actor-critic* RL procedure
Return: Trained reward $r_{\varphi,\psi}$ and actor-critic $\pi_\theta, \nu_\vartheta$

ALGORITHM DESCRIPTION. We present the resulting adversarial algorithm (CI-AIRL) in Algorithm 4. The algorithm describes a two-player zero-sum game between an agent parameterized using a soft-actor-critic architecture and the discriminator which is used to provide a reward function by distinguishing between expert and policy samples. The adversarial training procedure generally mimics that of divergence-based methods such as (Ho et al., 2016; Fu, Luo, et al., 2018). There are three main differences compared to baseline adversarial training algorithms. The first is the fact that we use multiple experts in a distinct fashion as opposed to pooling the demonstrations into one big dataset. The second is the regularization of the discriminator objective in Equation (4.18) using the gradient norm penalty $\mathbb{D}(\xi, \phi, \psi; e) = ||\nabla_{\psi|\psi=1.0}\mathcal{L}_{\text{BCE}}(\xi, \varphi, \psi; e)||^2$ in a similar fashion to eq. (4.10), where $\psi = 1.0$ corresponds to a fixed scalar predictor. Finally, we utilize soft-actor-critic (SAC) (Haarnoja et al., 2018) as an instance of an off-policy forward RL solution method as opposed to the on-policy algorithms used in (Ho et al., 2016; Fu, Luo, et al., 2018).

4.3 EXPERIMENTS

We will now evaluate the proposed method empirically. The experiments are designed to answer the following questions: (i) What is the effect of the causal invariance penalty on the recovered reward structure? (ii) Does regularizing the reward function using causal invariance improve downstream policy performance? (iii) What is the impact of changing the loss function of the discriminator and (iv) what is the effect of increasing the perturbation magnitude? To answer these questions, we evaluate our model in two settings.

The first setting considers a tabular gridworld scenario, where the partition function is tractable. We perform reward learning experiments using variants of the maximum entropy feature expectation matching algorithm. In particular, we use the DeepMaxEnt (Wulfmeier et al., 2015) model of state features with different regularization strategies. In the second setting, we test the invariance regularization in an adversarial IRL setting on simulated robotic locomotion environments. Here, we demonstrate the generalization of the obtained reward functions by retraining policies using the recovered rewards on perturbed versions of the environment dynamics.

4.3.1 *Tractable setting: Gridworld experiments using feature matching*

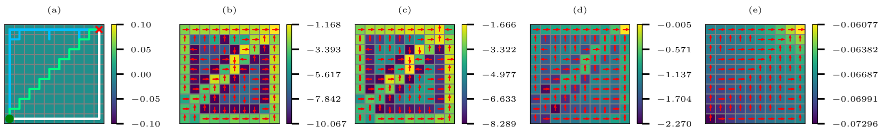

Figure 4.2: **Feature matching reward recovery on a gridworld environment.** (a) expert trajectory datasets: 1st group (blue) 40 trajectories, 2nd group (white): 10 trajectories, 3rd group (green): 1 trajectory. (b) MaxEnt IRL ERM baseline (c) MaxEnt IRL ERM baseline with L2 regularization coefficent $\lambda_{L2} = 1e - 3$ (d) MaxEnt IRL with CI penalty, $\lambda_I = 0.01$, (e) MaxEnt IRL with CI penalty, $\lambda_I = 0.05$

For the first experiment, we aim to illustrate the principle of causal invariance regularization in the tractable IRL setting using Algorithm 3. To do so, we choose a simple gridworld environment with stochastic dynamics and a sparse ground truth reward structure and corresponding trajectories illustrated in (Figure 4.2a). The goal of the agent is to navigate from the bottom left to the top right corner. Due to the tabular nature of the state space, this setting allows a direct visual comparison of the recovered reward functions for the different regularization strategies.

SETUP. In order to construct the dataset settings $\mathcal{E}_{tr}$, we generate a dataset of 3 groups of expert trajectories using a value iteration method on modified versions of the MDP. The initial and final states of the trajectories are fixed.

We introduce a selection bias into the IRL feature expectation matching problem by manipulating the expert preferences to choose different paths. This results in a trajectory dataset with different number of trajectories for each of the three paths chosen by the experts (Figure 4.2a): 40 trajectories for 1st group, 10 trajectories for 2nd group and 1 trajectory for the 3rd group.

BASELINES. Throughout this section, we compare the proposed regularization to both non-regularized and regularized versions of the DEEPMAXENT algorithm. In particular, we use an L2 penalties of the reward feature weights $\varphi(s)$ and a Lipschitz smoothness penalty (Yoshida et al., 2017) as baseline regularization strategies.

RESULTS. In Figure 4.2, we can observe that both the unregularized MaxEnt IRL algorithm (ERM) (Figure 4.2b) and L_2-regularized MaxEnt IRL algorithm (ERM-L_2) (Figure 4.2c) exhibit overfitting to the expert datasets and partially fail to recover a meaningful reward and respective policy. In contrast, the IRM-regularized version recovers a shaped reward function which takes the different optimal paths into account in a manner which demonstrates an invariance to the setting index e. In particular, increasing the regularization strength λ improves the reward significantly (Figure 4.2d - Figure 4.2e). It is easy to see that the CI-regularized reward can more straightforwardly be used in a setting where the dynamics of the MDP might be modified, e.g. when obstacles are introduced.

4.3.2 *Adversarial setting*

For the second experiment, we perform experiments in large state spaces, which require the use of Algorithm 4. Our primary goal is to investigate how well the recovered reward function allows the elicitation of a policy under change of dynamics. To do so, we first learn a reward function using the adversarial training procedure and then retrain a policy from scratch using the recovered reward as training signal.

SETUP. For our experimental setting, we choose a set of robot locomotion tasks from the MUJOCO (Todorov et al., 2012) suite. We generate the demonstration datasets by using pretrained soft-actor-critic (SAC) policies from the STABLE-BASELINES3 repository [5]. In order to diversify the demonstrations,

5 https://huggingface.co/sb3

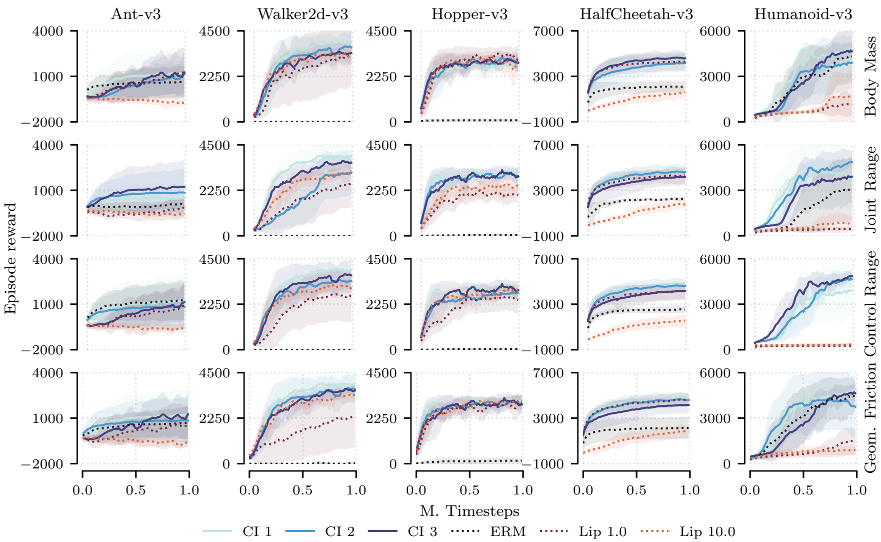

FIGURE 4.3: **Comparison of SAC policy performance w.r.t. ground truth reward when trained on inferred reward functions.** Every row depicts a different type of dynamics perturbation for the five MuJoCo tasks as described in section 4.3.2. Here, AIRL is chosen as the baseline algorithm. The variants correspond to the unregularized baseline: ERM, Lipschitz regularization: LIP and three best CI regularization parameters CI.

we perturb the policies using a structured noise approach: the optimal policy action is perturbed with Gaussian noise at every timestep with a probability $p = 0.3$ of the noise being applied. We have used 10 expert trajectories for every environment in all the experiments performed in this section. The reward function is obtained by using a number of different discriminator functions corresponding to variations of the f-divergence objective. In order to assess the quality of the recovered reward, we retrain policies on the recovered reward under distribution shift of the dynamics realized by perturbing physical parameters of the simulation. Specifically, we apply Gaussian noise to four parameters of the MuJoCo simulation: the body mass, joint range and actuator control range of the robot as well as the contact friction coefficient of the simulated surface. We evaluate the behavior of the algorithms for a variety of perturbation magnitudes.[6]

BASELINES. Throughout this section, we use three different adversarial IRL algorithms as baselines for reward learning: (i) AIRL (Fu, Luo, et al., 2018), an adversarial IRL approach which relies on a structured discriminator to recover a stationary reward function (ii) MEIRL, an adaptation of the

6 Additional experimental details are provided in section 4.5

Environment	Ant-v3	Walker2d-v3	Hopper-v3	HalfCheetah-v3	Humanoid-v3
Expert	$3168.49_{\pm 1715.68}$	$3565.33_{\pm 527.40}$	$3119.54_{\pm 524.36}$	$4340.61_{\pm 2020.14}$	$4774.17_{\pm 2063.52}$
AIRL (ERM)	$580.78_{\pm 1048.73}$	$-3.29_{\pm 0.71}$	$77.64_{\pm 88.27}$	$2046.29_{\pm 460.98}$	$4451.74_{\pm 1759.31}$
AIRL (Lip)	$1194.04_{\pm 1583.08}$	$3388.48_{\pm 1586.45}$	$3382.91_{\pm 234.02}$	$4388.94_{\pm 726.69}$	$1788.85_{\pm 1643.00}$
AIRL (CI)	$1880.42_{\pm 935.15}$	$4162.70_{\pm 517.13}$	$3334.91_{\pm 221.80}$	$4477.97_{\pm 532.72}$	$5107.54_{\pm 119.31}$
GAIL (ERM)	$-746.31_{\pm 468.03}$	$328.44_{\pm 66.02}$	$1637.50_{\pm 1419.59}$	$886.25_{\pm 404.82}$	$122.62_{\pm 71.53}$
GAIL (Lip)	$220.97_{\pm 524.83}$	$553.36_{\pm 277.24}$	$1832.39_{\pm 832.32}$	$1403.77_{\pm 1282.75}$	$77.24_{\pm 4.66}$
GAIL (CI)	$230.43_{\pm 565.68}$	$1172.57_{\pm 539.86}$	$2636.65_{\pm 1114.94}$	$2365.55_{\pm 1679.64}$	$549.63_{\pm 1692.08}$
MEIRL (ERM)	$-66.66_{\pm 112.03}$	$169.11_{\pm 344.87}$	$3.22_{\pm 0.22}$	$-177.39_{\pm 211.43}$	$55.99_{\pm 3.45}$
MEIRL (Lip)	$-365.41_{\pm 143.70}$	$917.14_{\pm 132.05}$	$1045.40_{\pm 54.76}$	$-335.10_{\pm 84.66}$	$1001.49_{\pm 1889.60}$
MEIRL (CI)	$153.43_{\pm 1134.46}$	$2520.24_{\pm 994.27}$	$2351.07_{\pm 679.37}$	$1371.59_{\pm 1469.01}$	$3099.51_{\pm 2411.21}$

TABLE 4.1: **Policy rollout results using ground truth reward for perturbed MuJoCo environments** after being trained for 1M timesteps using the rewards recovered from the different discriminators in section 4.3.2. Here, the *body mass* parameter is perturbed with a noise magnitude of $\varepsilon = 0.2$. The results are averaged over 10 rollouts and obtained by training the model using five different random seeds.

maximum entropy IRL algorithm for large state spaces *without* an importance sampling estimator (Ni et al., 2021) and (iii) GAIL (Ho et al., 2016), an imitation learning where we extract the unshaped discriminator as a reward function for policy learning. All baselines use the SAC (Haarnoja et al., 2018) algorithm as a forward RL agent for purposes of sample efficiency.

RESULTS. Figure 4.3 depicts the results of a SAC agent trained using the reward function recovered by the AIRL baseline algorithm and two regularization strategies: (i) the Lipschitz smoothness regularizer (Gulrajani et al., 2017) controlled by the penalty coefficient λ_L and the proposed regularization in section 4.2.3, controlled by the penalty coefficient λ_I. We evaluate the models on combinations of regularization coefficients drawn from the following sets: $\lambda_I \in \{0.0, 0.1, 1.0, 10.0, 100.0\}$ and $\lambda_L \in \{0.0, 1.0, 10.0\}$ and pick the three best performing λ_I coefficients for every environment. We can observe that applying the causal invariance penalty leads to superior performance compared to using non-regularized or Lipschitz regularized rewards. The choice of the λ_I hyperparameter is problem dependent – we report three best performing variants per-environment.

IMPACT OF ADVERSARIAL ALGORITHM VARIATION. We evaluate our method on a number of variants of the adversarial imitation learning algorithms based on f-divergence minimization, as outlined above. The results are presented in table 4.1. We report the ground truth reward performance attained using an SAC agent trained using the recovered rewards after 1 million timesteps. The regularization coefficients are selected on a per-environment basis. The full results tables are provided in section 4.5.4. We observe improved cumulative ground truth reward metrics for all three algorithms when compared to both the unregularized (ERM) and Lipschitz regularized (Lip) baselines. The improvement is most pronounced in the ANT, WALKER2D and HUMANOID environments, which are of higher state-space dimensionality.

PERTURBATION MAGNITUDE. In this experiment, we investigate the policy ground truth performance as a function of the perturbation magnitude applied to the physical parameters of the environment dynamics. In fig. 4.4, we observe that using the CI-penalty improves the ground truth episode reward performance of the trained policies for a large spectrum of perturbations for 3 out of 5 environments and does not suffer a performance penalty for the other two. This result demonstrates the improved generalization properties of the CI-regularized reward function. [7]

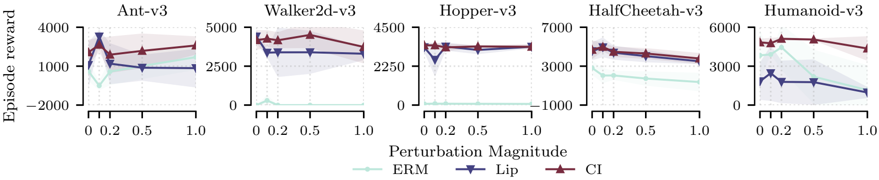

FIGURE 4.4: **Comparison of SAC policy performance w.r.t. ground truth reward when trained on recovered reward functions as a function of *perturbation magnitude*.** Here, AIRL is chosen as the baseline algorithm. ERM denotes the unregularized baseline, LIP the best Lipschitz regularization hyperparameters per environment and CI the best causal invariance regularization hyperperameter.

4.3.3 *Lunar Lander environment*

In addition to the robotic locomotion setting, we have performed a set of reward learning experiments in the context of the LUNARLANDER environment (**towers_gymnasium_2023**). Similarly to the robotic locomotion experiments described in section 4.3.2, 10 demonstrations were gathered using a pre-trained SAC baseline and a structured noise approach was used to achieve diversity of demonstrations. The reward was trained using the AIRL baseline algorithm and the same set of regularization coefficients $\lambda_I \in \{0.0, 0.1, 1.0, 10.0, 100.0\}$ and $\lambda_L \in \{0.0, 1.0, 10.0\}$ as in section 4.3.2. Subsequently, the reward was used to train an SAC policy under dynamics perturbations on three different parameters: gravity, wind power and turbulence power with noise magnitudes $\eta = 1.0$ for the gravity and wind power parameters and $\eta = 0.5$ for the turbulence power parameter.

Figure 4.5 depicts the resulting cumulative reward performance during policy training. We can observe that using the CI regularization with regularization coefficient $\lambda_I = 100.0$ allows us to train a policy which considerably improves upon both the unregularized baseline (ERM) and the Lipschitz smoothness regularized baselines (Lip) both in terms of faster convergence and asymptotic performance measured using the ground truth reward.

7 We provide a number of additional evaluations for both the tractable and approximate settings in section 4.5

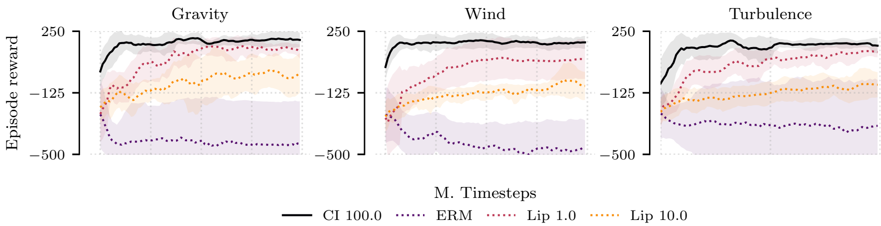

FIGURE 4.5: **Comparison of SAC policy training performance w.r.t. ground truth reward when trained using the recovered reward functions.** Here, AIRL is chosen as the baseline reward learning algorithm for the LUNARLANDER environment. Every column depicts a different type of dynamics perturbation used during training.

4.4 RELATED WORK

INVARIANCE AND CAUSALITY IN RL Following the introduction of causally invariant methods for supervised and representation learning (Peters, Bühlmann, et al., 2015; Heinze-Deml et al., 2017; Arjovsky, Bottou, et al., 2019; Ahuja, Shanmugam, et al., 2020), the concept of causal invariance has been used in a number of reinforcement learning works. Invariant causal prediction has been utilized in (A. Zhang et al., 2020) to learn model invariant state abstractions in a multiple MDP setting with a shared latent space. Invariant policy optimization (Sonar et al., 2021) uses the IRM games (Ahuja, Shanmugam, et al., 2020) formulation to learn policies invariant to certain domain variations. Haan et al. address the problem of causal confusion in imitation learning by using the causal structure of demonstrations. Causal imitation learning under temporally correlated noise has been studied in (Swamy, Choudhury, D. Bagnell, et al., 2022). In the offline RL setting without access to environment interactions, (Bica et al., 2021) propose to use the invariance principle for policy generalization. To the best of our knowledge, our algorithm is the first proposed method to use invariant causal prediction in the context of inverse reinforcement learning with the primary purpose focused on recovery of reward functions and their subsequent deployment for downstream purposes.

LEARNING FROM DIVERSE DEMONSTRATIONS Li et al. investigate the issue of imitation learning from diverse experts through the lens of identifying latent factors of variation. Zolna et al. propose a model which also addresses the issue of spurious correlations being absorbed from expert data. However, their focus is on visual features in a pure imitation learning setting in contrast to our approach, which recovers reward functions that perform favorably in a transfer setting. Diverse demonstrations have been studied in both the context of adversarial imitation learning under assumptions on latent variables in (Tangkaratt et al., 2020) as well as offline imitation learning (G.-H. Kim et al., 2021) under assumptions of explicit access to a dataset of suboptimal demonstrations. The authors of (Haug et al., 2020) propose to use suboptimal demonstrations to derive an additional supervision signal by way of matching optimality profiles for preference learning.

COMPARISON TO OTHER REGULARIZATION TECHNIQUES The divergence-based dual objective used in this work admits a number of regularization strategies, which result as a the restriction of the critic function class $\mathcal{F}$. A commonly used regularization is the Lipschitz smoothness penalty (Yoshida et al., 2017; Gulrajani et al., 2017) which restricts the class of functions $\mathcal{F}$ in eq. (4.6) to the class of Lipschitz smooth functions. Contrary to methods which penalizes the estimated gradient norm w.r.t. the input, the causal invariance penalty restricts the norm of the gradient w.r.t. to the predictor parameters of the model which vary across settings $\mathcal{E}_{tr}$. K.-E. Kim et al. restrict the function class $\mathcal{F}$ to belong to an RKHS space, resulting in an imitation learning method based on the Maximum Mean Discrepancy (MMD) metric. The authors of (Bashiri et al., 2021) present a distributionally robust imitation learning method which generalizes the maximum entropy IRL robustness properties from logistic loss to arbitrary losses. In comparison, our method remedies the issue of learning rewards as opposed to imitation policy learning and improves generalization due to the properties of the causal invariance penalty.

4.5 ADDITIONAL RESULTS

In this section, we present additional experimental evidence to support the claims made in the main text.

4.5.1 *Additional gridworld results*

Figure 4.6 depicts an additional experimental setting using algorithm 3. Here, we choose 5 sets of trajectories which solve a horizontal navigation problem illustrated in fig. 4.6a. We compare this to a set of baselines as described in section 4.3.1 with the addition of a spectral norm (Yoshida et al., 2017) regularizer which imposes the Lipschitz smoothness penalty on the reward representation. We motivate this choice by the fact that Lipschitz smoothness is a successful regularization techniques in the approximate setting. We can observe a similar pattern to the one reported in section 4.3.1, where the ERM MaxEnt baseline overfits to reward to the observed trajectories (fig. 4.6(b,c)). The Lipschitz regularization in fig. 4.6(d) provides a more succinct reward representation but fails to capture the horizontal reward gradient which is indicative of the shared intent of the expert. The CI penalty with both regularization strengths (fig. 4.6(e,f)) recovers this aspect of the ground truth reward.

4.5.2 *Adversarial training results*

For reference, in fig. 4.7, we present the results of the adversarial training procedure used to recover the reward functions for the experiments in section 4.3.2. We can observe that when used to regularize the discriminator in an adversarial setting, the CI penalty does not produce a significant regularization effect as has been established in the transfer setting.

4.5.3 *Alternative discriminator training*

Similar to the training dynamics reported in fig. 4.3 for the AIRL discriminator structure, we provide the results for the other two algorithms – GAIL in fig. 4.8 and MEIRL in fig. 4.9. We observe that for GAIL, the regularization is beneficial in all settings except for HalfCheetah where it is outperformed by the Lipschitz regularized baseline. For the MEIRL setting, we also observe a significant improvement with the exception of the Ant environment, where all algorithm fail to achieve movement using the recovered reward.

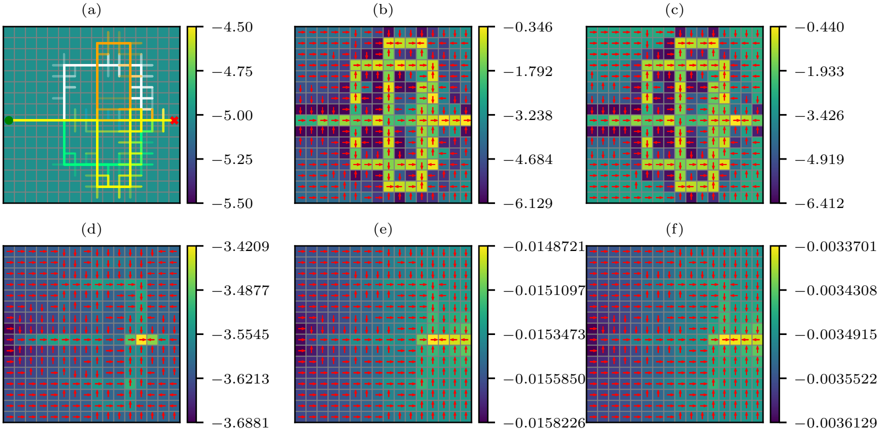

FIGURE 4.6: **Feature matching reward recovery on a gridworld environment.** (a) expert trajectory datasets: every color represents a modality containing 50 trajectories (b) MaxEnt IRL ERM baseline (c) MaxEnt IRL ERM baseline with L2 regularization coefficent $\lambda_{L2} = 1e - 3$ (d) MaxEnt IRL baseline with spectral norm (Lipschitz) regularization (e) MaxEnt IRL with CI penalty, $\lambda_I = 0.1$, (f) MaxEnt IRL with CI penalty, $\lambda_I = 0.5$

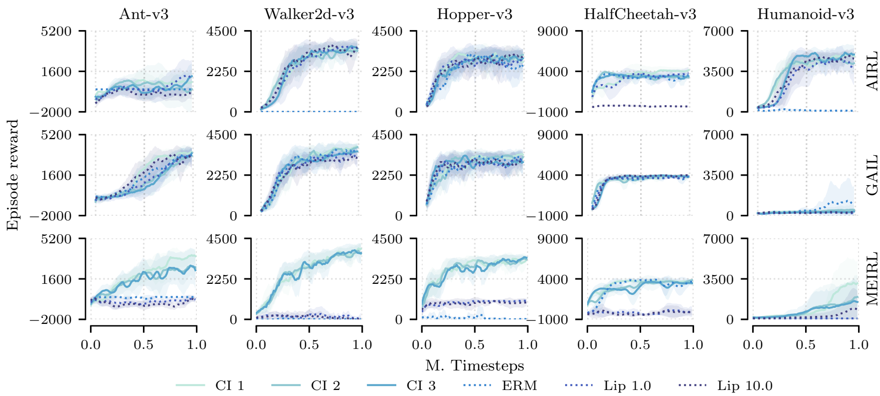

FIGURE 4.7: **Comparison of SAC policy performance w.r.t. ground truth reward** when trained using the adversarial training procedures. Every row depicts a different type of algorithm used to train the policies.

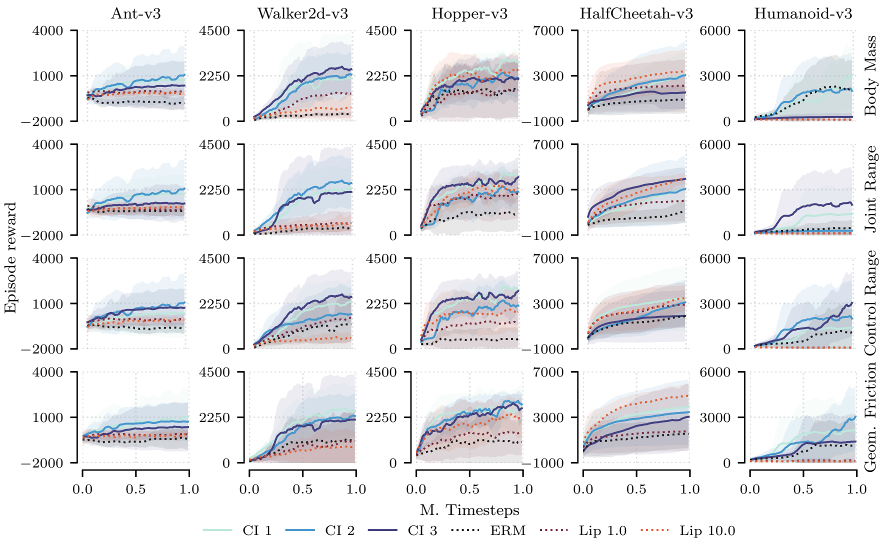

FIGURE 4.8: **Comparison of SAC policy performance w.r.t. ground truth reward** when trained on recovered reward functions. Every row depicts a different type of dynamics perturbation for the five MuJoCo tasks as described in section 4.3.2. Here, GAIL is chosen as the baseline.

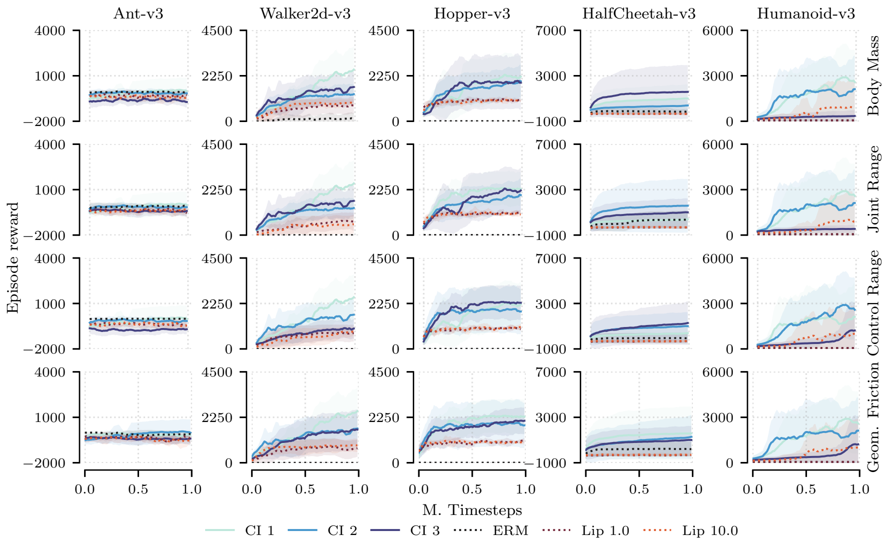

FIGURE 4.9: **Comparison of SAC policy performance w.r.t. ground truth reward** when trained on recovered reward functions. Every row depicts a different type of dynamics perturbation for the five MuJoCo tasks as described in section 4.3.2. Here, MEIRL is chosen as the baseline.

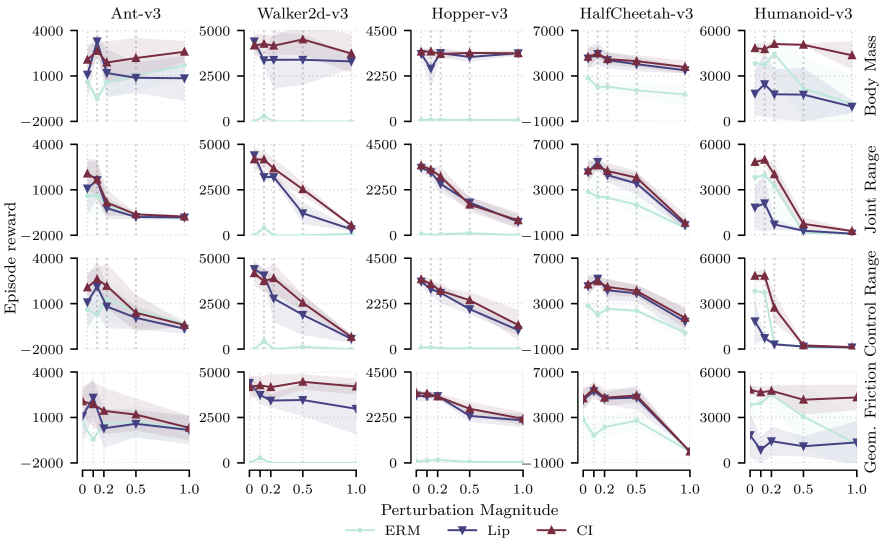

FIGURE 4.10: **Comparison of SAC policy performance w.r.t. ground truth reward** when trained on recovered reward functions as a function of the perturbation strength. Every row depicts a different type of dynamics perturbation for the five MuJoCo tasks as described in section 4.3.2. Here, AIRL is chosen as the baseline.

Environment	ANT	WALKER2D	HOPPER	HALFCHEETAH	HUMANOID
Expert	3168.49 $\pm$ 1715.68	3565.33 $\pm$ 527.40	3119.54 $\pm$ 524.36	4340.61 $\pm$ 2020.14	4774.17 $\pm$ 2063.52
AIRL (ERM)	-18.73 $\pm$ 255.23	-3.50 $\pm$ 1.66	48.82 $\pm$ 76.24	2310.93 $\pm$ 118.75	3261.17 $\pm$ 2272.68
AIRL (Lip)	-213.89 $\pm$ 738.75	3202.09 $\pm$ 185.98	2544.28 $\pm$ 445.86	4293.70 $\pm$ 666.33	710.88 $\pm$ 356.04
AIRL (CI)	155.62 $\pm$ 875.01	3670.21 $\pm$ 599.00	2906.77 $\pm$ 490.33	4653.89 $\pm$ 762.09	4022.43 $\pm$ 671.37
GAIL (ERM)	-486.05 $\pm$ 388.06	359.23 $\pm$ 254.85	1047.76 $\pm$ 871.98	1160.78 $\pm$ 1134.26	508.86 $\pm$ 601.99
GAIL (Lip)	-208.97 $\pm$ 252.99	577.40 $\pm$ 550.86	2339.87 $\pm$ 465.97	4168.13 $\pm$ 472.22	122.49 $\pm$ 82.43
GAIL (CI)	1021.26 $\pm$ 1845.56	3479.99 $\pm$ 1242.39	2976.49 $\pm$ 417.33	5581.43 $\pm$ 1442.39	2170.96 $\pm$ 2425.51
MEIRL (ERM)	-57.30 $\pm$ 93.23	-3.69 $\pm$ 0.18	3.17 $\pm$ 0.52	347.55 $\pm$ 790.54	54.70 $\pm$ 1.92
MEIRL (Lip)	-554.25 $\pm$ 82.05	622.45 $\pm$ 464.25	1136.95 $\pm$ 209.97	-297.84 $\pm$ 71.23	1014.42 $\pm$ 1915.24
MEIRL (CI)	-337.17 $\pm$ 1310.57	2292.94 $\pm$ 1521.43	2800.37 $\pm$ 666.09	1650.50 $\pm$ 1683.38	2935.90 $\pm$ 2262.13

TABLE 4.2: **Policy rollout results using ground truth reward for perturbed MuJoCo environments** after being trained for 1M timesteps using the rewards recovered from the different discriminators in section 4.3.2. Here, the *joint range* parameter is perturbed with a noise magnitude of $\varepsilon = 0.2$. The results are averaged over 10 rollouts and obtained by training the model using five different random seeds.

4.5.4 *Tables of results*

In addition to the results reported on the body mass parameter of the MuJoCo simulation, we provide the results tables for other three perturbation parameters: *joint range* in table 4.2, *actuator control range* in table 4.3 and *geometry friction* in table 4.4. We observe a similar effect in terms of the reward generalization properties as described in the main text.

Environment	Ant	Walker2d	Hopper	HalfCheetah	Humanoid
Expert	$3168.49_{\pm 1715.68}$	$3565.33_{\pm 527.40}$	$3119.54_{\pm 524.36}$	$4340.61_{\pm 2020.14}$	$4774.17_{\pm 2063.52}$
AIRL (ERM)	$1279.87_{\pm 1281.66}$	$-0.41_{\pm 3.07}$	$37.62_{\pm 62.71}$	$2536.90_{\pm 212.27}$	$357.19_{\pm 135.66}$
AIRL (Lip)	$809.60_{\pm 1425.76}$	$2779.73_{\pm 1303.91}$	$2784.28_{\pm 510.50}$	$4175.49_{\pm 918.92}$	$312.24_{\pm 90.05}$
AIRL (CI)	$2166.58_{\pm 1471.70}$	$3897.58_{\pm 831.24}$	$2884.34_{\pm 130.60}$	$4470.17_{\pm 731.81}$	$2730.60_{\pm 982.13}$
GAIL (ERM)	$-641.93_{\pm 284.65}$	$1180.57_{\pm 1413.21}$	$491.27_{\pm 565.63}$	$1862.67_{\pm 1026.74}$	$1219.94_{\pm 1784.26}$
GAIL (Lip)	$-92.74_{\pm 363.44}$	$1672.06_{\pm 1263.95}$	$2028.07_{\pm 1004.96}$	$3638.51_{\pm 1164.42}$	$93.01_{\pm 24.01}$
GAIL (CI)	$2486.00_{\pm 2078.11}$	$2660.74_{\pm 866.36}$	$2985.44_{\pm 280.70}$	$3979.90_{\pm 2494.31}$	$2986.70_{\pm 2389.78}$
MEIRL (ERM)	$-10.63_{\pm 3.35}$	$-3.83_{\pm 0.45}$	$3.17_{\pm 0.27}$	$-36.66_{\pm 489.57}$	$58.40_{\pm 0.15}$
MEIRL (Lip)	$-411.65_{\pm 244.20}$	$832.80_{\pm 311.20}$	$1073.22_{\pm 138.00}$	$-261.74_{\pm 164.78}$	$1064.41_{\pm 2011.91}$
MEIRL (CI)	$133.50_{\pm 969.39}$	$2286.80_{\pm 1040.59}$	$2551.59_{\pm 1131.76}$	$3303.82_{\pm 2332.89}$	$3058.78_{\pm 2286.41}$

Table 4.3: **Policy rollout results using ground truth reward for perturbed MuJoCo environments** after being trained for 1M timesteps using the rewards recovered from the different discriminators in section 4.3.2. Here, the *actuator control range* parameter is perturbed with a noise magnitude of $\varepsilon = 0.2$. The results are averaged over 10 rollouts and obtained by training the model using five different random seeds.

Environment	Ant	Walker2d	Hopper	HalfCheetah	Humanoid
Expert	$3168.49_{\pm 1715.68}$	$3565.33_{\pm 527.40}$	$3119.54_{\pm 524.36}$	$4340.61_{\pm 2020.14}$	$4774.17_{\pm 2063.52}$
AIRL (ERM)	$603.00_{\pm 909.86}$	$-3.87_{\pm 0.44}$	$149.17_{\pm 151.96}$	$2141.85_{\pm 942.19}$	$4507.23_{\pm 659.04}$
AIRL (Lip)	$283.73_{\pm 1294.15}$	$3429.17_{\pm 372.29}$	$3311.25_{\pm 128.82}$	$4659.44_{\pm 533.32}$	$1432.57_{\pm 952.87}$
AIRL (CI)	$1434.01_{\pm 1530.65}$	$4167.15_{\pm 721.26}$	$3288.86_{\pm 149.76}$	$4737.72_{\pm 749.52}$	$4756.93_{\pm 368.15}$
GAIL (ERM)	$-421.27_{\pm 752.40}$	$910.12_{\pm 951.80}$	$939.05_{\pm 959.38}$	$1563.17_{\pm 1245.51}$	$871.61_{\pm 1002.21}$
GAIL (Lip)	$-172.69_{\pm 196.92}$	$1065.02_{\pm 1672.12}$	$2541.08_{\pm 906.67}$	$4795.59_{\pm 1018.65}$	$89.51_{\pm 16.87}$
GAIL (CI)	$1148.32_{\pm 1938.45}$	$2395.43_{\pm 1282.70}$	$3068.00_{\pm 459.50}$	$4037.08_{\pm 983.32}$	$3385.58_{\pm 2279.28}$
MEIRL (ERM)	$-103.10_{\pm 188.90}$	$-3.43_{\pm 0.15}$	$3.36_{\pm 0.19}$	$191.23_{\pm 630.81}$	$56.41_{\pm 2.24}$
MEIRL (Lip)	$-252.29_{\pm 184.32}$	$865.84_{\pm 308.25}$	$1128.52_{\pm 125.06}$	$-284.92_{\pm 204.61}$	$991.70_{\pm 1871.72}$
MEIRL (CI)	$-191.76_{\pm 912.46}$	$2546.37_{\pm 1073.22}$	$2425.38_{\pm 1070.78}$	$1724.44_{\pm 2057.07}$	$2155.55_{\pm 2281.06}$

Table 4.4: **Policy rollout results using ground truth reward for perturbed MuJoCo environments** after being trained for 1M timesteps using the rewards recovered from the different discriminators in section 4.3.2. Here, the *geometry friction* parameter is perturbed with a noise magnitude of $\varepsilon = 0.2$. The results are averaged over 10 rollouts and obtained by training the model using five different random seeds.

4.6 MODEL ARCHITECTURE AND TRAINING DETAILS

GRIDWORLD. For the gridworld experiments, we use a DeepMaxEnt (Wulfmeier et al., 2015) formulation of the IRL problem. The state features are parametrized by a 2-layer MLP with a 1-dimensional hidden layer. We use RMSProp as an optimizer with a learning rate of 10^{-3}.

MUJOCO. For the adversarial learning experiments, we use an actor and critic network with two hidden layers of size 256 and a discriminator network with two hidden layers of size 128. We use Adam (Kingma et al., 2014) as the optimizer with an initial learning rate of $\eta = 3e - 4$. We use the gradient penalty strategy described in (Gulrajani et al., 2017) to impose the Lipschitz smoothness constraint on the discriminator networks in section 4.3.2. We perform one update of the discriminator network for every 3 updates of the actor-critic networks. We use an extension of the CLEANRL (S. Huang et al., 2022) library for all the experiments to train the policy networks both using the ground truth reward as well as during adversarial training and using the recovered reward for the transfer experiments.

4.6.1 *Regularization strength*

The causal invariance (CI) penalty term used in eq. (4.14) and eq. (4.17) bears resemblance to the Lipschitz smoothness gradient penalty (Gulrajani et al., 2017) used as the main regularized baseline througout the experiments presented in this work. The magnitude of both penalty types is controlled by the respective regularization coefficients λ_L and λ_I. The regularization strength of the CI gradient penalty controls the deviation of the learned solution from the optimum on the respective training setting in the convex case (Arjovsky, Bottou, et al., 2019). In contrast to penalizing the smoothness of the function with respect to the input, the tuning of the regularization is largely dependent on the diversity of training settings. Identifying a data-driven strategy for regularization tuning is a challenge we defer to future work.

4.7 CONCLUSION

In this chapter, we have presented a regularization objective for inverse reinforcement learning to recover reward functions which are robust to spurious correlations present in expert datasets which feature diverse demonstrations. The robustness manifests itself as improved policy performance in a transfer setting in both the maximum entropy IRL case based on feature expectation matching as well as the adversarial setting.

LIMITATIONS AND FUTURE WORK. The hyperparameter λ_I is strongly dependent on the data and environment – finding an automatic tuning procedure would overcome the main limitation in terms of the applicability of the method. Currently, the proposed method relies on a linear formulation of the causal invariance penalty. Successor methods of (Arjovsky, Bottou, et al., 2019) which introduce a nonlinear formulation (Lu et al., 2022) or include a larger class of distribution shifts (Rothenhäusler et al., 2021) could be considered in order to improve the overly conservative nature (Ahuja, Caballero, et al., 2021) of causally invariant features.

APPLICATION TO SURGICAL ASSISTANCE PIPELINE The algorithmic development presented in this chapter has a number of implications in the context of the surgical assistance pipeline described in chapter 3. Surgical demonstrations are notoriously multimodal - style preferences of individual practitioners may vary dramatically in the context of performing the same procedure. Identifying causally invariant rewards are a promising avenue that can be used for the purpose of distilling task-relevant components of the demonstrations.

5

IMITATION LEARNING USING GENERALIZED SLICED WASSERSTEIN DISTANCES

I visualize a time when we will be to robots what dogs are to humans. And I am rooting for the machines.
— Claude Shannon

In contrast to the previous chapter, which largely focused on inferring succinct reward functions for use in downstream tasks, here, we focus on directly learning a policy via imitation, thereby bypassing an explicit reward learning step. As we have outlined in chapter 2, imitation learning methods enable one to train reinforcement-learning-style policies by way of mimicking the state occupancies of a given expert agent. Most approaches are divergence-based, which can result in optimization objectives that, empirically, are brittle and difficult-to-solve. As an alternative, we explore an approach based on the sliced Wasserstein distance, with the goal of using its optimal-transport-based formulation and favorable computational properties to improve performance. To do so, we formulate a per-state reward function based on the approximate differential of the sliced Wasserstein distance, which allows one to apply standard forward reinforcement learning methods to solve the imitation learning policy optimization problem. We demonstrate that the proposed method exhibits improved performance compared to established imitation learning frameworks on a number of benchmark tasks from the MuJoCo robotic locomotion suite.

5.1 INTRODUCTION

Imitation learning methods aim to recover a policy based on a dataset of trajectories obtained from an expert assumed to perform the task at a proficient level. These methods are of paramount importance for obtaining policies in settings where the specification of a suitable reward function is difficult—this includes a wide variety of robotics scenarios (Billard et al., 2008) such as autonomous driving (Pomerleau, 1991; Bojarski et al., 2016) or dexterous manipulation (Rajeswaran et al., 2018).

Most existing methods can be cast in the framework of distribution matching—they minimize some notion of distance between the imitation learning policy's distribution of states and actions, and those of the expert. The classes of f-divergences (Ho et al., 2016; Fu, Luo, et al., 2018; Csiszár, 1972; Ni et al., 2021) and integral probability metrics (Sriperumbudur et al., 2009), such as the Wasserstein distance (H. Xiao et al., 2019; Dadashi et al., 2021; Fickinger et al., 2022), are particularly popular.

Among possible metrics, the Wasserstein distance is an appealing distribution matching objective due to its metric properties and intuitive optimal transport interpretation. Unfortunately, its computational cost scales poorly in the dimensions of the state and action spaces. This is because formulating a Wasserstein-based per-step reward function generally involves computing a full transport plan. While this can be circumvented by using greedy approximations (Dadashi et al., 2021), or by reformulating the problem in terms of the Kantorovich–Rubinstein dual (Arjovsky, Chintala, et al., 2017; H. Xiao et al., 2019), the resulting optimization problems are generally challenging to solve.

Sliced optimal transport methods (Santambrogio, 2015) are a class of techniques for defining computationally-less-expensive alternatives to the Wasserstein distance. They replace the original, high-dimensional optimal transport problem with a collection of much-simpler one-dimensional optimal transport problems defined via random projections. The obtained problems are solved via fast sorting-based algorithms. While the resulting performance can be limited by how well random projections capture differences between states (Kolouri et al., 2019), a number of extensions such as the max sliced Wasserstein distance (Deshpande et al., 2019) or distributional sliced Wasserstein distance (K. Nguyen et al., 2020) have been shown to be effective in a number of practical settings.

CONTRIBUTIONS We study imitation learning via the generalized sliced Wasserstein distances of Kolouri et al. (2019). We use the projection-based formulation of these distances to derive a per-state reward function, which we show can be computed straightforwardly via sorting and a simple array insertion and replacement scheme, and is amenable to maximization via standard reinforcement learning procedures. We evaluate the resulting algorithm on an array of benchmark tasks from the MuJoCo (Todorov et al., 2012) suite, and observe improved sample efficiency and asymptotic performance compared to a number of adversarial imitation learning baselines. In total, our algorithm provides a simpler and more computationally straightforward alternative to other transport-based imitation learning techniques, with improved reliability compared to divergence-based approaches.

5.2 IMITATION LEARNING AND OPTIMAL TRANSPORT

To study imitation learning via generalized sliced Wasserstein distances, we first define the setting. Let $(\mathcal{S}, \mathcal{A}, r, \mathcal{T}, \gamma)$ be a discounted time-homogeneous Markov decision process (MDP), where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function, $\mathcal{T}$ is the transition kernel, and $\gamma \in (0, 1)$ is the discount factor. Let $\Omega(\mathcal{X})$ denote the space of probability measures over $\mathcal{X}$. We equip our MDP with an initial state distribution $\mu_0 \in \Omega(\mathcal{S})$ and work with Markov policies $\pi(\cdot|s) \in \Omega(\mathcal{A})$, which are transition kernels that map states to distributions over actions.

5.2.1 *Imitation learning via distribution matching*

In *imitation learning*, we are given a set of N *expert trajectories* assumed generated by an expert policy π_E. We denote these by $\zeta^{(i)}_{1:N}$, where the subscript $(\cdot)_{1:N}$ denotes a sequence of length N, and each trajectory is defined as $\zeta_i = (s^{(i)}_{1:T_I}, a^{(i)}_{1:T_i})$ with T_i being the length of each trajectory. The algorithm's goal is to produce a policy π which imitates the expert policy π_E as effectively as possible, according to some suitable notion which quantifies the difference between policies, and varies by algorithm. Note that we do not have explicit access to π_E, and can only rely on a finite number of its trajectories for training. We now review typical approaches to imitation learning.

A natural way to perform imitation learning is by *distribution matching*. Here, the idea is to (i) assemble the expert's trajectories into a (discounted) *expert occupancy measure* $\hat{\rho}_E \in \Omega(\mathcal{S} \times \mathcal{A})$, and (ii) compute a policy π whose (discounted) occupancy measure ρ_π is close to $\hat{\rho}_E$. To quantify this, one introduces a *divergence* $D : \Omega(\mathcal{S} \times \mathcal{A}) \times \Omega(\mathcal{S} \times \mathcal{A}) \to \mathbb{R}$, which is non-negative and satisfies $D(\rho, \rho') = 0$ if and only if $\rho = \rho'$. To perform imitation learning, one minimizes this divergence numerically, thereby obtaining a policy π which produces similar trajectories to those of the expert.

The most common choices for D used in practice (Ho et al., 2016; Fu, Luo, et al., 2018; Ni et al., 2021) lie in the class of *f-divergences* $D_\phi(\rho, \rho') = \mathbb{E}_\rho \left[\phi\left(\frac{\rho}{\rho'}\right)\right]$, where $\frac{\rho}{\rho'}$ is the density of ρ with respect to ρ'. One generally proceeds by (i) computing rollouts using π, thereby obtaining an empirical occupancy measure $\hat{\rho}_\pi$, and (ii) comparing $\hat{\rho}_\pi$ to $\hat{\rho}_E$ using the divergence. To perform the comparison in a computationally meaningful way, one typically expresses the divergence in its dual form, which transforms divergence minimization into a min-max problem akin to the one which appears in generative adversarial networks (GANs) (Goodfellow et al., 2014). This min-max problem is approximated using neural networks, and solved numerically via gradient-based methods. The resulting procedures are generally known as *generative adversarial imitation learning* (GAIL) (Ho et al., 2016), and their computational properties mirror those of GANs. One can also work with optimal-transport-based distances (H. Xiao et al., 2019; Dadashi et al., 2021) rather than f-divergences: since these are of key interest in this work, we proceed to introduce these now.

5.2.2 *Optimal transport: Wasserstein distances and their projection-based extensions*

We now review the *optimal transport* formalism, which will be key to the imitation learning algorithms we study. Here, we are interested in defining distances between probability distributions $\rho, \rho' \in \Omega(\mathcal{X})$ over a metric space $\mathcal{X}$. To define these distances, one can imagine each probability measure as representing the distribution of some physical material across space, and consider the distance of transporting material from one distribution to another. The smallest possible total transportation distance, in an L^2-type sense, defines the Wasserstein-2 distance

$$\mathcal{W}_2(\rho, \rho') = \left(\inf_{\nu \in \Gamma(\rho,\rho')} \int_{\mathcal{X}^2} d(x, x')^2 d\nu(x, x') \right)^{\frac{1}{2}} \overset{d=1}{=} \left(\int_0^1 |F_\rho^{-1}(x) - F_{\rho'}^{-1}(x)|^2 dx \right)^{\frac{1}{2}}$$

where $\Gamma(\rho, \rho')$ is the set of all couplings between ρ and ρ', and d is the metric on X. Here, the second inequality illustrates how the one-dimensional case, namely $\mathcal{X} = \mathbb{R}$ with $d(x, x') = |x - x'|$, simplifies: it can be expressed in terms of the cumulative distribution functions F_ρ and $F_{\rho'}$ of ρ and ρ'. If ρ and ρ' are empirical measures, this expression can efficiently be computed by sorting the atoms of both measures, and summing up the sorted distances between atoms.

In higher-dimensional settings, solving the optimization problem defining the Wasserstein distance generally becomes more difficult. One can address this by applying projections, either randomly or according to an optimization problem, and considering the one-dimensional Wasserstein distances that result. This results in, respectively, the *sliced Wasserstein dis-*

tance (Santambrogio, 2015) and *max sliced Wasserstein distance* (Deshpande et al., 2019)

$$\mathcal{SW}_2(\rho,\rho') = \left(\int_{\mathcal{S}^{d-1}} \mathcal{W}_2^2(\mathcal{P}_*^{(\psi)}\rho, \mathcal{P}_*^{(\psi)}\rho')d\psi\right)^{\frac{1}{2}} \tag{5.1}$$

$$\mathcal{MSW}_2(\rho,\rho') = \sup_{\psi\in\mathcal{S}^{d-1}} \mathcal{W}_2(\mathcal{P}_*^{(\psi)}\rho, \mathcal{P}_*^{(\psi)}\rho') \tag{5.2}$$

where $\mathcal{S}^{d-1} \subseteq \mathbb{R}^d$ is the $(d-1)$-dimensional sphere embedded within $\mathbb{R}^d$, $\mathcal{P}^{(\psi)} : \mathbb{R}^d \to \mathbb{R}$ is the projection onto the line induced by $\psi \in \mathcal{S}^{d-1}$, $(\cdot)_*$ denotes the pushforward of a measure, and integration is performed with respect to the probabilistic Haar measure on the sphere. Like $\mathcal{W}_2$, both expressions can be shown to define distances on the space of probability measures, but are significantly more efficient to compute due to their ability to leverage fast sorting-based algorithms for the one-dimensional Wasserstein distance.

One can improve performance of the max sliced Wasserstein distance in an empirically-motivated manner by replacing linear projections with non-linear ones defined by a neural network. This gives the *max generalized sliced Wasserstein distance* (Kolouri et al., 2019)

$$\mathcal{MGSW}_2(\rho,\rho') = \sup_{\psi\in\Psi} W_2(g_*^{(\psi)}\rho, g_*^{(\psi)}\rho')$$

where $g^{(\psi)} : X \to \mathbb{R}$ is a neural network with weight space Ψ. While here we no longer necessarily obtain an actual metric, this expression has shown promising performance in flow-based generative modeling (Kolouri et al., 2019), and we proceed to investigate its use in imitation learning.

5.3 SLICED WASSERSTEIN IMITATION LEARNING

In imitation learning, our algorithm's goal is to learn a policy π which induces a state occupancy measure ρ_π that is as close as possible to the empirical state occupancy measure of the expert $\hat{\rho}_E$. We now study how to do so in the setting at hand.

5.3.1 *A reward-based formulation for imitation learning*

To perform imitation learning via one of the sliced Wasserstein variants, the most straightforward approach would be to directly minimize the chosen

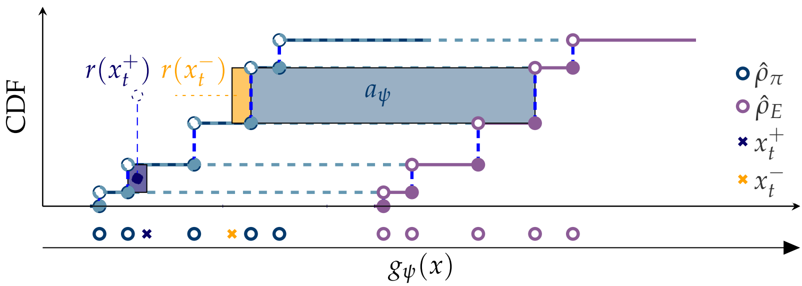

FIGURE 5.1: **Illustration of the sliced Wasserstein imitation learning proxy reward function.** First, two state transition atoms x_t^+ and x_t^- are sampled from the environment. Then, we find the closest projected atom, replace it with x_t^+ or x_t^-, respectively, and compute the difference of rectangles a_ψ. Here, x_t^+ is an example of an atom which incurs positive reward and x_t^- an example of an atom which incurs negative reward when inserted into the sorted array corresponding to the policy occupancy measure $\hat{\rho}_\pi$. We also show the sorted atoms corresponding to the expert occupancy measure $\hat{\rho}_E$.

distance via gradient descent. Unfortunately, doing so would require one to (automatically) differentiate through the transition dynamics, which we only have black-box access to. To circumvent this, we introduce a proxy reward function, which is constructed using the chosen distance and designed to bring the state occupancy measure closer to the expert's measure when optimized. We do so as follows.

Let $\mathcal{X} = \mathcal{S} \times \mathcal{A}$ be the space of state-action pairs—one can also work with state-action-state tuples, but we present the most basic variant to ease notation—and let $\hat{\rho}_\pi \in \Omega(\mathcal{X})$ be the empirical occupancy measure of the current policy, which we compute via a rollout. For a given distance d on the space of probability measures, define the *distance-based proxy reward function*

$$r_d(x_t; \hat{\rho}_E, \hat{\rho}_\pi) = d(\hat{\rho}_E, \hat{\rho}_\pi) - d(\hat{\rho}_E, \mathrm{rpl}(\hat{\rho}_\pi, x_t))$$

where rpl is defined as the *atom replacement function*, which removes the closest atom (with respect to the Euclidean distance on $\mathcal{X}$) from $\hat{\rho}_\pi$ and replaces it with x_t.

The idea behind this definition is to think of $r_d(x_t; \hat{\rho}_\pi, \hat{\rho}_E)$ as a variant of a discrete directional derivative, where the objective being differentiated is $d(\hat{\rho}_E, \cdot)$ and the direction is the Dirac measure $\delta_{x_t} \in \Omega(X)$. By performing a rollout, we obtain a trajectory ξ. We collect rewards at locations where shifting the empirical measure $\hat{\rho}_\pi$ in the direction δ_{x_t}, via atom-replacement, reduces the distance d. Maximizing these rewards over the whole trajectory will therefore tend to reduce d throughout all parts of state-action space visited by the trajectory.

To maximize r_d, or practical modifications thereof, we apply off-the-shelf reinforcement learning algorithms such as *soft actor-critic* (Haarnoja et al., 2018). Note that, unlike most rewards used in standard reinforcement learning, r_d is policy-dependent: the algorithms we consider extend to this setting without modifications, and we discuss them further in the sequel. Our formulation mirrors general use of policy-dependent or non-stationary rewards in other imitation learning settings such as those of Ho et al. (2016) and Fu, Luo, et al. (2018) which generally require a density ratio estimate between the expert and policy measures. By maximizing the respective rewards numerically, we aim to bring $\hat{\rho}_\pi$ closer to $\hat{\rho}_E$. We now examine how to select and compute the distance d in our setting.

5.3.2 *Computing sliced-Wasserstein-based rewards*

To perform imitation learning by way of maximizing the distance-based proxy reward function, we need to choose a distance: for this, we explore the use of sliced Wasserstein distances, and study the properties of the resulting algorithms. To do so, we now examine computational properties of these distances and the resulting imitation learning procedures defined by them.

The first step in calculating the proxy reward $r_{\mathcal{SW}_2}(x_i; \hat{\rho}_E, \hat{\rho}_\pi)$ is to replace the closest atom in $\hat{\rho}_\pi$ with x_i. To calculate the respective sliced Wasserstein distances, we project the atoms in each measure onto a set of one-dimensional subspaces defined either using $\psi \sim \mathcal{U}(\mathcal{S}^{d-1})$, or via the respective maximization problem. Then, we calculate the respective Wasserstein distance $\mathcal{W}_2$ by sorting, namely

$$\frac{1}{N} \sum_{n=1}^{N} |x_{i[n]} - y_{j[n]}|$$

where $i[n]$ and $j[n]$ are the sorted indices of samples of both respective empirical measures—see Santambrogio (2015) for additional details on one-dimensional Wasserstein distances and sorting-based algorithms for computing them in an automatically-differentiable (Blondel et al., 2020) manner. In practice, the overall procedure takes $\mathcal{O}(NM \log N)$ time, where

N is the number of atoms, and M is the number of projection directions. Figure 5.1 gives an illustration of this process.

Given the ability to compute the reward function, the next step is to maximize it. For this, we apply an off-the-shelf reinforcement learning algorithm. Specifically, starting from an arbitrary initial policy π_0, we perform a rollout to obtain the empirical occupancy measure $\hat{\rho}_{\pi_0}$. We then perform a reinforcement learning step using the reward $r_{\mathcal{SW}_2}(\cdot; \hat{\rho}_E, \hat{\rho}_{\pi_0})$, and obtain an improved policy π_1.

In practice, one can apply minor modifications of this procedure, if needed, to account for differences between on-policy and off-policy algorithms. For example, for off-policy algorithms such as soft actor-critic (Haarnoja et al., 2018) which use replay buffers, one can randomly sample from $\hat{\rho}_E$ instead of using the whole measure. This simplifies computation by ensuring the size of the sampled measure matches the number of elements sampled from the replay buffer at every step.

5.3.3 *Comparison of different sliced Wasserstein variants*

The key differences between $\mathcal{SW}_2$, $\mathcal{MSW}_2$, and $\mathcal{MGSW}_2$ come down to their formulations: (i) the use of a large set of random projections vs. a single, optimized projection, and (ii) the use of linear projections, vs. non-linear analogues defined by a neural network. These differences have important consequences in practice: the cost of $\mathcal{SW}_2$ is $\mathcal{O}(NM \log N)$ whereas for $\mathcal{MSW}_2$ is $\mathcal{O}(N \log N)$, and for $\mathcal{MGSW}_2$ it is $\mathcal{O}(NP \log N)$ where P is the number of neural network parameters. Compared to $\mathcal{SW}_2$, $\mathcal{MSW}_2$ is therefore generally cheaper: however, it is more difficult to capture differences between measures with only a single projection direction. This gives the main motivation behind considering $\mathcal{MGSW}_2$: its use of a neural network defines a function $g^{(\psi)} : X \to \mathbb{R}$, which one can hope will better capture differences through its non-linear nature.

To understand this distinction further, observe that the distribution matching optimization problem defined by $\mathcal{MGSW}_2$ can be written as a min-max problem, namely

$$\inf_{\pi \in \Pi} \mathcal{MGSW}_2(\hat{\rho}_E, \hat{\rho}_\pi) = \inf_{\pi \in \Pi} \sup_{\psi \in \Psi} W_2(g_*^{(\psi)} \hat{\rho}_E, g_*^{(\psi)} \hat{\rho}_\pi)$$

which illustrates what $g^{(\psi)}$ does: it acts as a critic which finds a curve that seeks to concentrate on the part of the space where the measures are the most different.

Algorithm 5 Sliced Wasserstein Imitation Learning (SWIL)

1: **require** Reinforcement learning algorithm with a policy improvement step for π_θ.
2: **input** Expert trajectories $(\xi_i)_{i=1}^N$, initial policy π_θ (and any other initial state needed by the reinforcement learning algorithm), initial $\mathcal{MGSW}_2$-critic $g^{(\psi)} : X \to \mathbb{R}_+$, policy and expert replay buffers with batch size B.

3: **while** below maximum number of iterations **do**
4: Compute a rollout $(x_R^{(t)})_{t=1}^T$ using the policy π_θ, add each element to the policy replay buffer.
5: Sample a minibatch of state-action pairs $(x_\pi^{(b)})_{b=1}^B$ and $(x_E^{(b)})_{b=1}^B$ from the policy and expert replay buffers and compute $\mathcal{G}_\pi = (g^{(\psi)}(x_\pi^{(b)}))_{b=1}^B$ and $\mathcal{G}_E = (g^{(\psi)}(x_\pi^{(b)}))_{b=1}^B$, which we implicitly interpret as empirical measures.
6: For each t, replace the closest atom in $\mathcal{G}_\pi$ with x_t, obtaining

$$\mathcal{G}_R(x_t) = \mathrm{rpl}(\mathcal{G}_\pi, x_t).$$

7: Update $g^{(\psi)}$ via gradient-based supervised learning using the cross-entropy loss

$$\mathcal{L} = \sum_{b=1}^{B} \log g^{(\psi)}(x_\pi^{(b)}) - \sum_{b=1}^{B} (1 - \log g^{(\psi)}(x_E^{(b)}))$$

with respect to the sampled minibatches.
8: Perform a reinforcement learning step using the rewards

$$r_{\mathcal{MGSW}_2} = W_2(\mathcal{G}_E, \mathcal{G}_\pi) - W_2(\mathcal{G}_E, \mathcal{G}_R(x_t))$$

for each $t = 1, .., T$.
9: **return** learned policy π_θ.

As formulated, combining equation 5.3.3 with the proxy reward function of equation 5.3.1 requires one to solve $2T$ inner maximization problems at each policy optimization step, where T is either the length of the respective trajectory, or size of the replay buffer sample. To avoid this, in practice we learn a single critic function $g^{(\psi)}$ which is used for all inner maximization problems simultaneously, thereby replacing the sequence of min-max problems with a single min-max problem as a practical approximation.

This gives Algorithm 5, our proposed practical sliced Wasserstein imitation learning algorithm. The Wasserstein distances used involve only one-dimensional spaces, and are therefore straightforward to compute using sorting-based algorithms. To ensure $g^{(\psi)}$ is non-negative, we parameterize its log using neural networks. We now compare this algorithm with its primary current alternatives.

5.3.4 *Comparison with other imitation learning algorithms*

In our framework, using generalized sliced Wasserstein distances for imitation learning leads to the minimax problem equation 5.3.3. This minimax problem bears some resemblance to the ones used by GAIL-like methods, but differs in crucial ways that we now examine.

The main correspondence between equation 5.3.3 and GAIL-like methods is that both involve a two-player zero-sum game played by an actor optimizing over policies Π, and a critic optimizing over neural networks $g^{(\psi)} : \mathcal{X} \to \mathbb{R}$. However, there are two key difference between these approaches:

1. *What the mapped state-action pairs correspond to*. In ϕ-divergence-based GAIL variants, the output of $g^{(\psi)}$ models the mapped density ratio $\phi\left(\frac{\hat{\rho}_E}{\hat{\rho}_\pi}\right)$. In contrast, SWIL's $g^{(\psi)}$ can be interpreted as a critic which tries to make different inputs in $\mathcal{X}$ produce different outputs.

2. *How mapped state-action pairs are compared*. GAIL variants do so by averaging the mapped values across the minibatch, whereas SWIL uses the one-dimensional Wasserstein distance to compare the full distributions.

Though both GAIL and SWIL result in minimax problems, one can expect the above distinctions to lead to different training dynamics. First, SWIL's critic targets an intuitively easier-to-learn objective, since density ratios can be difficult to estimate from samples, even in spaces of moderate dimension. Second, since SWIL compares the full distributions rather than computing averages, one can expect it to provide a stronger training signal. The main cost of these changes is that they yield a moderately more-complicated algorithm, as SWIL requires the use of sorting-based computations.

5.4 EXPERIMENTS

We now evaluate sliced Wasserstein imitation learning empirically by conducting experiments on a set of robot locomotion tasks from the MuJoCo (Todorov et al., 2012) suite, using expert demonstrations obtained by rolling out pretrained policies from STABLE-BASELINES3 (Raffin et al., 2021). In all cases, we apply a minimum ground-truth cumulative reward threshold to ensure valid expert behavior. The state space dimension of these tasks is $\dim(\mathcal{S}) \in \{8, 11, 17, 111, 376\}$. We evaluate two $\mathcal{MGSW}_2$-variants of SWIL: the one presented in Section 5.3, which we denote by SWIL (QL1), and a variant which averages rewards with respect to the 10 previous critic functions in a replay-buffer-inspired manner, which we denote by SWIL (QL10) and describe in full detail in Section 5.5.

We compare to a number of imitation learning baselines: (i) *Discriminator-Actor-Critic* (DAC) (Kostrikov, Agrawal, et al., 2019), an off-policy GAIL-like algorithm which uses a Lipschitz regularizer for the discriminator networks, (ii) *Primal Wasserstein Imitation Learning* (PWIL) (Dadashi et al., 2021), which uses a greedy approximation of the transport plan, (iii) on-policy GAIL (Ho et al., 2016) with spectral normalization (Yoshida et al., 2017), (iv) off-policy GAIL, with and without spectral normalization, denoted by (SN) if present, (v) *adversarial inverse reinforcement learning* (AIRL) (Fu, Luo, et al., 2018), an off-policy algorithm which uses a shaped discriminator structure and (vi) *IQ-Learn* (IQL) (Garg et al., 2021), an IL method which foregoes adversarial optimization by leveraging a reformulation of the IL objective in terms of the Q-function (Kostrikov, Nachum, et al., 2019). We omit on-policy GAIL due to poor performance. We use soft actor-critic (SAC) as the same off-policy forward RL algorithm for all baselines with the exception of on-policy GAIL which uses proximal policy optimization (PPO) (Schulman, Wolski, et al., 2017). Further information is in Section 5.5.

5.4.1 *Sample-efficiency and asymptotic performance*

For the first experiment, we aim to evaluate how effectively SWIL can recover the performance demonstrated in the expert dataset, compared to baselines. To do so, we examine how the expert dataset size, in terms of the number of trajectories, affects each algorithm's final obtained rewards and convergence rate. We work with expert dataset sizes of $|\mathcal{D}_E| \in \{1, 4, 10\}$, where all trajectories have a fixed length of 1000 timesteps.

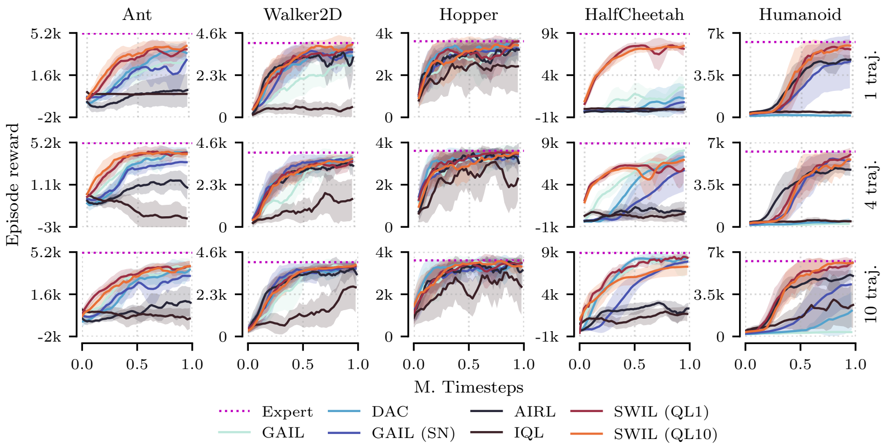

FIGURE 5.2: **Comparison of imitation learning algorithms on five benchmark tasks**, one for each column. Each row corresponds to models trained on $|\mathcal{D}_E| \in \{1, 4, 10\}$ trajectories, respectively. A description of each of the SWIL variants and baselines is given at the beginning of Section 5.4.

Figure 5.2 shows training behavior. Overall, we see that SWIL's reward curves converge to a level of performance comparable to the expert in all environments except HALFCHEETAH with ten trajectories. Table 5.1 confirms

Environment	ANT	HALFCHEETAH	HOPPER	HUMANOID	WALKER2D
Expert	5159.77	8900.85	3607.17	6249.60	4063.81
BC	978.76±10.23	221.12±48.71	504.56±154.59	347.26±32.13	299.42±15.63
DAC	3553.38±1580.64	3292.43±1365.91	3400.87±247.88	169.06±40.84	3433.62±362.73
GAIL (SAC)	4261.13±415.56	2569.26±1179.47	2866.05±760.34	147.39±59.98	2774.17±563.45
GAIL (SAC-SN)	2956.94±697.31	804.24±1441.09	3263.01±540.13	4643.93±2203.33	3296.62±417.84
AIRL (SAC)	361.58±1424.61	-9.47±382.93	3197.09±614.25	4778.57±404.82	3257.18±867.42
PWIL	1289.23±697.31	1089.76±923.19	2890.14±430.12	5252.23±156.22	2452.17±856.22
GAIL (PPO-SN)	2143.35±556.98	2863.42±1155.38	2859.98±1114.94	425.11±125.65	2648.65±1128.32
SWIL	**4338.65**±555.83	**7481.79**±769.22	**3585.28**±66.63	**5952.09**±315.92	**3936.35**±365.69

TABLE 5.1: **Cumulative rewards obtained in each environment**, using a single expert demonstration and 1m timesteps of training. Standard errors are computed over ten rollouts and five random seeds.

this: here, SWIL is shown to perform either comparably to or stronger than baselines in the one-trajectory setting. We also observe that SWIL tends to converge faster than baselines, thereby exhibiting better sample-complexity with respect to the number of reinforcement learning timesteps.

In contrast to SWIL, baselines such as DAC and GAIL work comparably-well in some environments such as ANT and WALKER2D, but fail completely on HUMANOID and the one-trajectory variant of HALFCHEETAH. On the other hand, AIRL performs best out of the baselines on HUMANOID. The only setting where SWIL does not perform at least as well as baselines is HALFCHEETAH with 10 trajectories—here, we suspect this may occur due to a more diverse set of expert demonstrations, which one could potentially handle by extending our formulation to the multi-marginal optimal transport setting (Cohen et al., 2021). We conclude that SWIL exhibits competitive imitation learning performance, particularly in settings with relatively-few expert demonstrations.

5.4.2 *Effect of trajectory sparsity*

Next, we examine how SWIL is affected by the time gap between transitions. For this purpose, we consider a single expert trajectory, and select subsets of it represented by different *sparsity factors* $\zeta \in \{1, 20, 100\}$. This procedure corresponds to taking every ζth timestep in the trajectory. We focus on the two most-difficult environments, namely ANT and HUMANOID and, as before, examine each algorithm's final obtained rewards and convergence rate. We also examine how SWIL's choice of measurable space, namely taking $\mathcal{X}$ to be $\mathcal{S}$, $\mathcal{S} \times \mathcal{A}$, or $\mathcal{S} \times \mathcal{A} \times \mathcal{S}$, affects performance.

Figure 5.3 shows results. We see SWIL performs stronger, compared to baselines, under sparsity factors $\zeta = 20$ and $\zeta = 100$, which correspond to trajectories with 50 and 10 state-action pairs. We also see that the $(\mathcal{S} \times \mathcal{A} \times \mathcal{S})$-variant of SWIL always outperforms the $\mathcal{S}$-variant except for the $\zeta = 100$ setting on HUMANOID. The $(\mathcal{S} \times \mathcal{A} \times \mathcal{S})$-variant performs the strongest under $\zeta = 1$, both variants perform comparably with $\zeta = 20$ and on ANT with $\zeta = 100$, while the $(\mathcal{S} \times \mathcal{A})$-variant performs best on HUMANOID with $\zeta = 100$, where no other method reaches comparable performance. Overall, we conclude that SWIL degrades more gracefully under trajectory sparsity compared to the baselines.

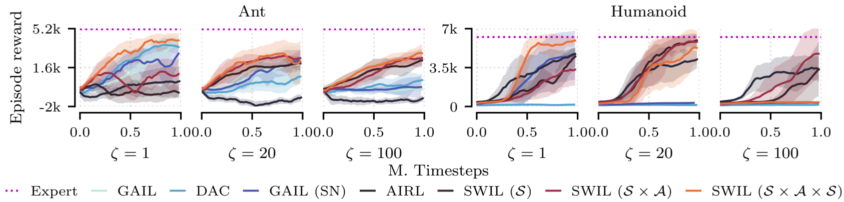

FIGURE 5.3: **Expert data sparsity ($\zeta \in \{1, 20, 100\}$) scaling behavior** of algorithms on ANT and HUMANOID. Different SWIL variants correspond to different choices of $\mathcal{X}$.

5.4.3 *Algorithmic choices, ablations, and additional results*

Since imitation learning via SWIL requires a number of algorithmic choices, we include an ablation study to quantify the individual effect of each choice. In particular, we consider non-SAC RL algorithms including proximal policy optimization (Schulman, Wolski, et al., 2017), other sliced Wasserstein variants, and vary implementation choices such as buffer lengths. We also examine several non-locomotion tasks to assess suitability in other settings. Details and results are given in Section 5.5.

5.5 IMPLEMENTATION, ABLATIONS, AND ALGORITHMIC DETAILS

We now outline a few of the design choices that need to be considered to deploy the algorithm. The precise MUJOCO environments used are ANT-V3, WALKER2D-V3, HOPPER-V3, HALFCHEETAH-V3, and HUMANOID-V3. We use expert baselines from `https://huggingface.co/sb3`. All experiments were repeated with 5 seeds to assess variability.

CHOICE OF UNDERLYING REINFORCEMENT LEARNING ALGORITHM. The algorithm presented in Algorithm 5 relies on an underlying RL algorithm. While the distribution matching problem is easiest to formulate with respect to an on-policy reinforcement learning agent, we instead adopt an off-policy actor-critic-based agent—specifically, soft actor-critic—as our main approach, due to sample efficiency.

As opposed to on-policy adversarial imitation learning methods such as GAIL, off-policy distribution matching formulations can be more challenging since the use of replay buffers means that trajectory samples come from a mixture of previous policies, instead of the latest policy. This leads to an optimization problem which matches the empirical occupancy measure of the replay buffer with the expert—rather than the original formulation of matching the current policy occupancy measure with the expert. In the learning from demonstration setting, this has been studied by Kostrikov, Nachum, et al. (2019) and Zhu et al. (2020), who derive off-policy objectives via a KL-divergence lower bound.

For sliced Wasserstein imitation learning, the standard on-policy formulation involves computing sliced Wasserstein distances using atoms sampled from the rollout of a single policy. An off-policy formulation, in contrast, involves atoms sampled from a replay buffer. Given that off-policy formulations tend to lead to better sample complexity in reinforcement learning, it is natural to also expect similar behavior in imitation learning.

To understand the differences between how on-policy and off-policy formulations affect performance, we performed a simple comparison between three algorithms—on-policy PPO (Schulman, Wolski, et al., 2017), off-policy TD3 (Fujimoto et al., 2018), and off-policy SAC—on HalfCheetah. Figure 5.4 illustrates these differences. We see that SWIL performs better when used with off-policy algorithms, especially SAC. In contrast, both GAIL baselines—with and without spectral normalization—perform worse when used with both off-policy algorithms. At the same time, even when used with on-policy PPO, SWIL either matches or exceeds the performance of both GAIL variants, depending on how many timesteps are used. In total, we conclude that the differences between on-policy and off-policy approaches manifest themselves in a similar way to the standard forward reinforcement learning problem: that is, off-policy approaches can in some cases lead to better sample-efficiency.

Sliced Wasserstein variants. To study the effect of using different variants of the sliced Wasserstein distance outlined in Section 5.2.2, we compare these methods on the one-trajectory task considered previously. Figure 5.5 shows results, from which we see that $\mathcal{MGSW}_2$ exhibits the strongest performance by a large margin on most environments. The second-most-competitive variant is $\mathcal{MSW}_2$, which, depending on the environment, either matches or significantly outperforms the remaining two variants. In turn, these variants—that is, $\mathcal{SW}_2$ and $\mathcal{GSW}_2$—perform similarly. We conclude that using a maximization-based formulation leads to improved performance in practice, with a larger degree of improvement if non-linear projections are used.

Online atom replacement. In practice, the rpl-function described in the main text is implemented as a tentative replacement operator. As the policy is rolled out, its atoms are tentatively replaced within the sample from the buffer in order to gather the transition rewards. This allows us to update the $\mathcal{GSW}_2$ estimate in an online manner. This slightly modifies the rewards obtained for subsequent policy samples, and reduces variability. To quantify how this and other modifications affect performance, we performed an ablation experiment. Figure 5.6 shows the effect of this modifications, along with others described in the sequel, illustrating that this atom-replacement strategy tends to lead to a modest performance improvement.

Varying the number of max-projections. In the training procedure outlined in algorithm 5, a single maximum projection is used to define the $\mathcal{MGSW}_2$ distance and the corresponding reward structure. In the simplest minibatch-based formulation, after the critic is updated, only the last minibatch is used to provide a reward signal—this is due to the algorithm's use of a single critic. To improve performance, one can instead use multiple critic functions, by performing the same calculation with a weighted average of previous critic functions. To implement this, the algorithm maintains a FIFO queue of adjustable length. The effect of this choice can be seen in Figure 5.6, which shows that this modification tends to modestly improve performance.

Choice of underlying measurable space. SWIL can be implemented with respect to at least four different underlying spaces $\mathcal{X}$, namely $\mathcal{S}$, $\mathcal{S} \times \mathcal{A}$, $\mathcal{S} \times \mathcal{S}$, and $\mathcal{S} \times \mathcal{A} \times \mathcal{S}$. Figure 5.6 illustrates the impact of this

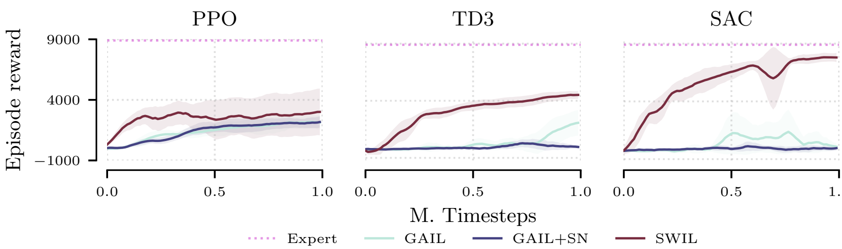

Figure 5.4: **Comparison of three model-free forward RL algorithms**: PPO (on-policy), TD3 (on-policy), and SAC (off-policy), on the HalfCheetah environment, and trained using one trajectory.

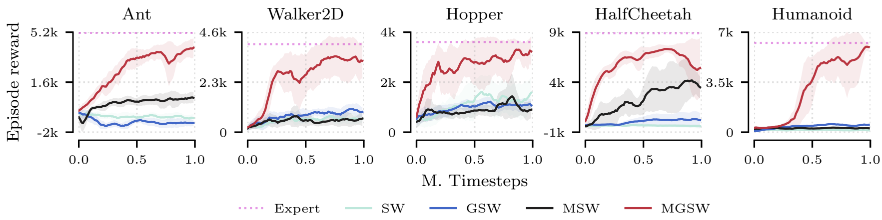

FIGURE 5.5: **Performance of different SWIL variants.** This includes the standard sliced Wasserstein distance, generalized sliced Wasserstein distance, max sliced Wasserstein distance, and max generalized sliced Wasserstein distance. The comparison has been made using a single expert trajectory.

choice. From this, we see that spaces with higher dimensionality—for instance, $\mathcal{S} \times \mathcal{A} \times \mathcal{S}$ as opposed to $\mathcal{S}$—improves the asymptotic performance significantly for ANT, but has a much more modest effect on HUMANOID, mirroring the results seen in Section 5.4.2.

ADDITIONAL ROBOTIC MANIPULATION ENVIRONMENTS. In addition to the MuJoCo locomotion suite experiments performed in the main text, we also performed a comparison of our method on two additional tasks: ADROITHANDDOOR-V1 from the Adroit Dexterous Hand manipulation (Rajeswaran et al., 2018) suite, and FRANKAKITCHEN-V1 from the set of kitchen

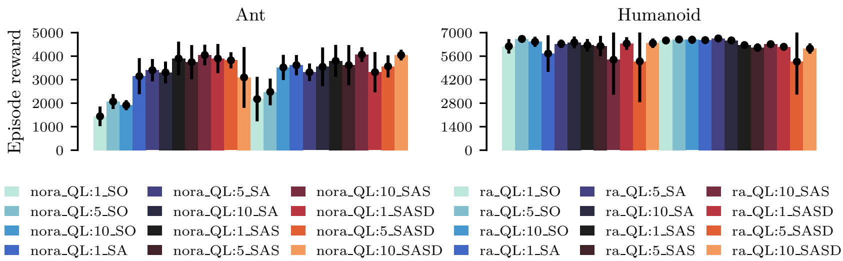

FIGURE 5.6: **Ablation of SWIL hyperparameters** for best evaluation runs on ANT-V3 and HUMANOID-V3. The legend format is as follows: *ra* denotes the atom replacement, *QL:x* denotes the number of projections used and *sa* the choice of measurable space. The models are trained using a single trajectory with subsampling factor $\varsigma = 20$.

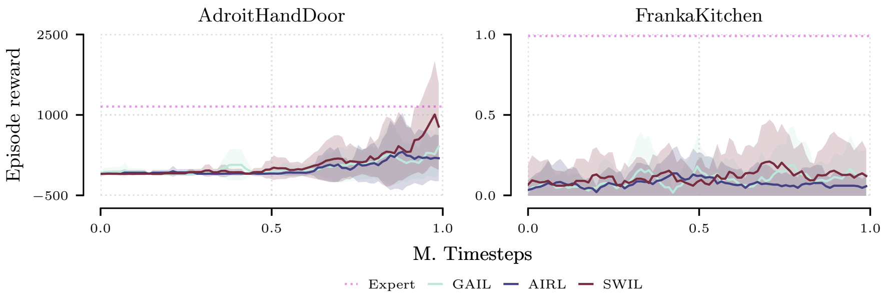

FIGURE 5.7: **Comparison of imitation learning algorithms** on ADROITHAND and FRANKAKITCHEN tasks.

object reaching tasks for the Franka robot manipulator (Gupta et al., 2020). The Adroit tasks are more difficult compared to most MUJOCO tasks, due to (a) the randomized goal state, which requires the agent to generalize over the distribution of the goal states, (b) the high action space dimensionality, and (c) the sparseness of rewards. For imitation learning, we obtained 10 expert trajectories for each of the tasks from the D4RL (Fu, Kumar, et al., 2020) offline RL dataset with a subsampling factor $\varsigma = 1$. Figure 5.7 shows performance results, where we we can observe that SWIL shows promising, near-expert performance on the ADROITHANDDOOR task, but not on the FRANKAKITCHEN task, where SWIL suffers from high variance and performs poorly in a manner that mirrors the two GAIL-based baselines.

HYPERPARAMETERS. These are as follows:

1. *Network.* We use 2-layer neural networks with a hidden layer size of 256 and Tanh activation layers for the SAC policy and Q-functions used as forward RL solvers in all the experiments. We use the same architecture with a smaller hidden size of 128 units for both the SWIL critic and ϕ-divergence based discriminators.

2. *Optimizer.* We use the Adam optimizer with a learning rate of $\eta_\pi = 3e^{-4}$ for the SAC policy updates and $\eta_Q = 1e^{-3}$ for the Q-function updates. We use $\eta_g = 3e^{-4}$ for the SWIL critic and ϕ-divergence discriminator updates.

3. *Baseline regularization parameters.* We generally use the baseline hyperparameters provided in the respective works. For DAC, we use a Lipschitz penalty with coefficient $\lambda_L = 10.0$. For spectral norm

regularization of GAIL, we use a number of power iterations equal to 1. For PWIL, we set the reward scale $\alpha = 5.0$ and reward bandwidth $\beta = 5.0$, which corresponds to the reference values provided by Dadashi et al. (2021). Similar to other imitation baselines, we do not pre-fill the replay buffer of the actor-critic algorithm for PWIL using expert demonstrations.

All algorithms considered in this work have been implemented using an extension of the CleanRL (S. Huang et al., 2022) framework. We provide our implementation at `github.com/iov9000/cleanrl`.

5.6 CONCLUSION

In this chapter, we introduced a family of distribution-matching algorithms for imitation learning using sliced Wasserstein distances, which lead to efficient computations based on sorting. We showed how to compute a proxy reward function defined using these distances, including a practical approximation suitable for minibatch-based settings. On an array of robot locomotion tasks, we found the resulting algorithm, in most cases, to both (a) either match or exceed the performance of standard imitation learning baselines, and (b) to require less interactions with the environment to achieve good performance. Overall, our results suggest that the sliced Wasserstein distances are a promising alternative to divergence-based imitation learning approaches.

APPLICATION TO DIGITAL TWIN ASSISTANCE PIPELINE When applied in the context of the surgical assistance pipeline studied in chapter 3, SWIL favorably compares to divergence-based imitation learning methods such as GAIL. The improved sample-efficiency and data-efficiency enable the training of assistive policies using a smaller number of expert trajectories and less interactions with the environment.

6 CONCLUSION

You cannot create experience. You must undergo it.
— Albert Camus

6.1 FINDINGS

In this dissertation, we have explored the development of an innovative algorithmic pipeline designed to provide surgical assistance to novice trainees through the use of reinforcement learning agents. These agents are specifically trained to encapsulate expert knowledge, which they acquire by learning to estimate both the expert's policy in a surgical environment and the underlying intentions behind the expert's actions by way of reward inference.

We have successfully demonstrated in two distinct surgical settings that the algorithms developed can effectively evaluate and assist human trainees. The reinforcement learning framework employed here offers a systematic approach to modeling trainee behavior in simulated environments. To enhance this framework further, we have introduced two novel algorithms focused on inverse reinforcement learning and imitation learning, expanding the capabilities and reach of our pipeline.

The first of these algorithms addresses the challenge of learning from diverse demonstrations. By adapting the principle of causal invariance to the inverse RL setting featuring demonstrations with intrinsic label noise, this method significantly enhances the generalization capacity of inferred reward functions across a variety of environmental conditions. In the surgical training context, this approach could translate to providing more robust training signals that are adaptable to different human anatomies, thus ensuring consistent learning outcomes despite anatomical variations among patients.

As a second methodological development, we introduced a novel imitation learning framework, named the sliced Wasserstein imitation learning algorithm. The framework introduces a novel reward structure by utilizing atom measure replacement to approximate the first-order variation of the sliced Wasserstein distance functional. By leveraging the geometric properties of the Wasserstein distance, this method achieves superior performance compared to existing divergence-based adversarial imitation learning techniques in terms of sample efficiency.

Overall, this dissertation contributes significantly to the field of computational surgery by integrating sophisticated reinforcement learning strategies into surgical training. These developments not only pave the way for more effective and adaptive training tools but also demonstrate the potential of machine learning technologies to revolutionize how surgical training is conceptualized and delivered. Through these innovations, we aim to equip medical professionals with the knowledge and skills necessary to perform surgeries with greater confidence and precision, ultimately enhancing patient outcomes.

6.2 FUTURE WORK

Looking ahead, there are several promising directions for further research based on the topics discussed in this thesis. We will outline specific avenues that could be pursued for each of the presented chapters.

6.2.1 *Surgical Assistance Pipeline*

The algorithmic pipeline for surgical assistance presents a number of opportunities for continued research and development. Here are several key areas where future studies could have significant impact:

- **Addressing Safety in IRL**: While the simulated environment is inherently safe, the policy used to generate path suggestions does not, at present, explicitly incorporate this concept. More generally, addressing safety in an IRL setting requires the definition of safe state space regions. This involves ensuring safety at training time, requiring the restriction of the policy exploration to a safe region of the state space. One such formalism is the class of Constrained Markov Decision Processes (CMDP) (Altman, 2021), where the learning problem is enhanced with a set of constraints describing violations of safe policy behaviour. In this context, it has recently been shown, that disentangling the inferred reward signal from the inferred constraint signal is a challenge (Lindner et al., 2024).

  Potential methods could include the following strategies. Restricting the base measure term in eq. (4.5) to better estimate the support of the expert would enforce a more constrained sampling distribution in the online imitation and IRL setting as presented in chapter 2. Another option would be to learn from failures – e.g. episodes which end in anatomy collision. Assuming this weak supervision signal is accessible, an empirical measure of failure states could be constructed in order to define a distance to the policy occupancy measure. By maximizing this distance, the policy is steered away from failure states. Further incorporating a constraint on the number of anatomy collisions could be tackled using Bayesian optimization methods (Shahriari et al., 2015).

- **Offline Imitation Learning**: Access to a simulated environment can often be a strong assumption, while large amounts of sequential decision data may be available from various sources. This scenario is tackled by offline RL Levine et al., 2020 methods – here, RL policies are trained on data previously gathered in a MDP setting without interacting with the environment. This RL modality is challenging due to the distribution shift issue Kumar et al., 2020 arising from lack of support estimation when compared to the online setting. Offline imitation learning methods rely on estimating (discounted) stationary distribution ratios(Kostrikov, Nachum, et al., 2019; G.-H. Kim et al., 2021) to directly infer a value function associated with expert behaviour. This approach could leverage vast amounts of historical surgical data to improve the algorithms without the need for active data generation, thus speeding up the learning process and reducing costs. Investigating the effectiveness of offline RL strategies in surgical settings could bridge the gap between simulated training and real-world applications.
- **Validation of the Pipeline Using Human Practitioners**: A critical step in the development of any medical technology is its validation with human practitioners. For the surgical assistance pipeline, it is essential to validate the system using standard clinical performance tests such as GOALS (Global Operative Assessment of Laparoscopic Skills) (Melina C Vassiliou et al., 2005b) or GEARS (Global Evaluative Assessment of Robotic Skills) (Goh et al., 2012). These tests would assess not just the alignment of the system's outputs with expert evaluations but also the interpretability and explainability of the recovered rewards. This validation would help ensure that the system is not only effective but also understandable and trustable by its human users.

6.2.2 *Learning Causally Invariant Rewards*

The causal invariance regularization proposed in chapter 4 is anchored in the invariant risk minimization (IRM) principle, as detailed by (Arjovsky, Bottou, et al., 2019). While the IRM principle aims to enhance generalization by focusing on features that are invariant across different environments, studies such as (Rosenfeld et al., 2020) and (Ahuja, J. Wang, et al., 2020) have demonstrated that it may lead to overly conservative feature representations. Additionally, IRM does not always effectively address certain types of spurious correlations, as indicated in (Ahuja, Caballero, et al., 2021), which limits its applicability in more complex causal settings.

We envision future work to include alternative methods of enforcing causal invariance in the context of IRL. Among others, anchor regression (Rothenhäusler et al., 2021) could be a viable candidate due to its distributional robustness properties. Furthermore, the regularization hyperparameter is strongly data and environment dependent. Finding an automatic tuning procedure would significantly improve the applicability of the method.

6.2.3 *Sliced Wasserstein Imitation Learning*

In chapter 5, we introduced a novel imitation learning method that leverages the sliced Wasserstein formulation of optimal transport. Specifically, we proposed an innovative approach to compute the reward as the approximate first-order variation of the sliced Wasserstein functional with respect to the occupancy measure. This method provides a more nuanced way of capturing the differences between expert and agent behaviors, enabling a deeper and more precise alignment of the agent's actions with those of the expert.

One promising direction is to extend the proposed method to a more general optimal transport framework, which estimates a transport plan. Alternatively, one could leverage the dynamical OT formulation described in chapter 2 to align the local policy updates with the Wasserstein gradient flow defined between the policy and the expert occupancy measures.

6.3 CONCLUDING REMARKS

In conclusion, this thesis has successfully demonstrated the feasibility and effectiveness of using reinforcement learning based approaches in the context of surgical training. We have proposed two novel methodological approaches which extend beyond the applied setting and may be of use in the greater scope of teaching artificial agents e.g. in robotics. This thesis further enables a number of exploration avenues, some of which we've outlined in this concluding section.

A

APPENDIX

A.1 BACKGROUND SECTION PROOFS

We restate the proofs of theorem 1 by (Schulman, Levine, et al., 2015) and theorem 2 by (Syed et al., 2008) for completeness.

Theorem 1 (Schulman, Levine, et al., 2015)**.** *Let $G(\pi)$ be the discounted empirical return of policy π.*

$$G(\pi') \geq L_\pi(\pi') - \frac{4\epsilon\gamma}{(1-\gamma)^2}\alpha^2$$

where

$$\epsilon = \max_{s,a} |A_\pi(s,a)| \quad \alpha = \max_s D_{TV}(\pi_{old}(\cdot|s)||\pi_{new}(\cdot|s))$$

$$\text{and } L_\pi(\pi') = G(\pi_{old}) + \sum_s \rho_{\pi_{old}}(s) \sum_a \pi'(a|s) A_\pi(s,a)$$

Proof. Define the returns for any stationary policy π and $\tilde{\pi}$:

$$G = (1 + \gamma P_\pi + (\gamma P_\pi)^2 + \dots) = (1 - \gamma P_\pi)^{-1},$$
$$\tilde{G} = (1 + \gamma P_{\tilde{\pi}} + (\gamma P_{\tilde{\pi}})^2 + \dots) = (1 - \gamma P_{\tilde{\pi}})^{-1}.$$

We will use the convention where ρ (a density on state space) is a vector and r (a reward function on state space) is a dual vector, thus $r\rho$ is a scalar representing the expected reward under density ρ. Note that $\eta(\pi) = rG\rho_0$, and $\eta(\tilde{\pi}) = r\tilde{G}\rho_0$. Let $\Delta = P_{\tilde{\pi}} - P_\pi$. We aim to bound $\eta(\tilde{\pi}) - \eta(\pi) = r(\tilde{G} - G)\rho_0$. Starting with some standard perturbation theory manipulations:

$$G^{-1} - \tilde{G}^{-1} = (1 - \gamma P_\pi) - (1 - \gamma P_{\tilde{\pi}}) = \gamma\Delta. \tag{A.1}$$

Left multiply by G and right multiply by $\tilde{G}$:

$$\tilde{G} - G = \gamma G \Delta \tilde{G}. \tag{A.2}$$

Substituting the right-hand side into $\tilde{G}$ gives:

$$\tilde{G} = G + \gamma G \Delta G + \gamma^2 G \Delta G \Delta \tilde{G}. \tag{A.3}$$

So we have:

$$\eta(\tilde{\pi}) - \eta(\pi) = r(\tilde{G} - G)\rho_0 = \gamma r G \Delta G \rho_0 + \gamma^2 r G \Delta G \Delta \tilde{G} \rho_0. \tag{A.4}$$

First, consider the leading term $\gamma r G \Delta G \rho_0$. Note that $rG = v$, the infinite-horizon state-value function, and $G\rho_0 = \rho^\pi$. Thus:

$$\gamma r G \Delta G \rho_0 = \gamma v \Delta \rho^\pi. \tag{A.5}$$

This expression equals the expected Trust Region Policy Optimization advantage $L^\pi(\tilde{\pi}) - L^\pi(\pi)$:

$$\begin{aligned}
L^\pi(\tilde{\pi}) - L^\pi(\pi) &= \sum_s \rho^\pi(s) \sum_a (\tilde{\pi}(a|s) - \pi(a|s)) A^\pi(s,a) \\
&= \sum_s \rho^\pi(s) \sum_a \left(\pi_\theta(a|s) - \tilde{\pi}_\theta(a|s)\right) \left[r(s) + \sum_{s'} p(s'|s,a) \gamma v(s') - v(s) \right] \\
&= \sum_s \rho^\pi(s) \sum_{s'} \sum_a \left(\pi(a|s) - \tilde{\pi}(a|s)\right) p(s'|s,a) \gamma v(s') \\
&= \sum_s \rho^\pi(s) \sum_{s'} \left(p^\pi(s'|s) - p^{\tilde{\pi}}(s'|s) \right) \gamma v(s') = \gamma v \Delta \rho^\pi.
\end{aligned}$$

Next, let us bound the $O(\Delta^2)$ term $\gamma^2 r G \Delta G \Delta \tilde{G} \rho$. First, we consider the product $\gamma r G \Delta = \gamma v \Delta$. Consider the component s of this dual vector. We have:

$$\begin{aligned}
|(\gamma v \Delta)_s| &= \left| \sum_a (\tilde{\pi}(s,a) - \pi(s,a)) Q^\pi(s,a) \right| \\
&= \left| \sum_a (\tilde{\pi}(s,a) - \pi(s,a)) A^\pi(s,a) \right| \\
&\leq \sum_a |\tilde{\pi}(s,a) - \pi(s,a)| \cdot \max_a |A^\pi(s,a)| \\
&\leq 2\alpha\epsilon
\end{aligned} \tag{A.6}$$

where the last line uses the definition of the total-variation divergence, and $\epsilon = \max_{s,a} |A^\pi(s,a)|$.

We bound the other portion $G \Delta \tilde{G} \rho$ using the ℓ_1 operator norm:

$$\|A\|_1 = \sup_\rho \left\{ \frac{\|A\rho\|_1}{\|\rho\|_1} \right\} \tag{A.7}$$

$$\text{where } \|G\|_1 = \|\tilde{G}\|_1 = \frac{1}{1-\gamma}, \quad \text{and} \quad \|\Delta\|_1 = 2\alpha. \tag{A.8}$$

$$\text{That gives } \|G \Delta \tilde{G} \rho\|_1 \leq \frac{1}{1-\gamma} \cdot 2\alpha \cdot \frac{1}{1-\gamma} \cdot 1 \tag{A.9}$$

Thus, we have that:

$$\begin{aligned}
\gamma^2|rG\Delta G\Delta\tilde{G}\rho| &\leq \gamma\|\gamma rG\Delta\|_\infty\|G\Delta\tilde{G}\rho\|_1 \\
&\leq \gamma\|\gamma v\Delta\|_\infty\|G\Delta\tilde{G}\rho\|_1 \\
&\leq \gamma\cdot 2\alpha\epsilon\cdot\frac{2\alpha}{(1-\gamma)^2} \\
&= \frac{4\gamma\epsilon\alpha^2}{(1-\gamma)^2}. \qquad \text{(A.10)}
\end{aligned}$$

□

Theorem 2 (Syed et al., 2008). *Suppose ρ is the occupancy measure of $\pi(a|s) = \frac{\rho(s,a)}{\sum_{s',a'}\rho(s',a')}$. Then, π is the only policy whose occupancy measure is ρ.*

The π-specific Bellman flow constraints are represented by the following linear system where the variables ρ_{sa} are unknown:

$$\rho_{sa} = \pi_{sa}\alpha_s + \pi_{sa}\gamma\sum_{s',a'}\rho_{s'a'}\theta_{s'a's} \qquad \text{(A.11)}$$

$$\rho_{sa} \geq 0 \qquad \text{(A.12)}$$

Lemma 1. *For any stationary policy π, the occupancy measure ρ^π of π satisfies the π-specific Bellman flow constraints eq. (A.11) - eq. (A.12).*

Proof. ρ^π_{sa} is non-negative for all s, a by definition, thereby satisfying eq. (A.12). For eq. (A.12), plugging in the definition of $\rho^\pi_{sa} = \mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^t\mathbf{1}(s_t = s\wedge a_t = a)|\mu_0,\pi,\theta\right]$ gives:

$$\begin{aligned}
\rho^\pi_{sa} &= E\left[\sum_{t=0}^{\infty}\gamma^t\mathbf{1}(s_t = s\wedge a_t = a)\right] && \text{(A.13)}\\
&= \sum_{t=0}^{\infty}\gamma^t\Pr(s_t = s, a_t = a) && \text{(A.14)}\\
&= \pi_{sa}\mu_0 + \sum_{t=0}^{\infty}\gamma^{t+1}\sum_{s',a'}\Pr(s_t = s', a_t = a')\theta_{s'a's}\pi_{sa} && \text{(A.15)}\\
&= \pi_{sa}\mu_0 + \pi_{sa}\gamma\sum_{s',a'}\mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^t\mathbf{1}(s_t = s'\wedge a_t = a')\right]\theta_{s'a's} && \text{(A.16)}\\
&= \pi_{sa}\mu_0 + \pi_{sa}\gamma\sum_{s',a'}x^\pi_{s'a'}\theta_{s'a's}. && \text{(A.17)}
\end{aligned}$$

□

Lemma 2. *The π-specific Bellman flow constraints eq. (A.11) - eq. (A.12) have at most one solution.*

Proof. Define the matrix $A(sa, s'a')$ as:

$$A(sa, s'a') = \begin{cases} 1 - \gamma\theta_{s'a's}\pi_{sa} & \text{if } (s,a) = (s',a'), \\ -\gamma\theta_{s'a's}\pi_{sa} & \text{otherwise.} \end{cases}$$

and the vector $b_{sa} = \pi_{sa}\alpha_s$. (Note that A and b are indexed by state-action pairs.) We can rewrite eq. (A.11) - eq. (A.12) equivalently as:

$$Ax = b, \tag{A.18}$$

$$x \geq 0. \tag{A.19}$$

The matrix A is column-wise strictly diagonally dominant because $\sum_{s',a'} \theta_{sas'} = 1$, $\sum_a \pi_{sa} = 1$, and $\gamma < 1$. Hence, for all s', a':

$$\sum_{s,a} \gamma\theta_{s'a's}\pi_{sa} = \gamma < 1 \Rightarrow \tag{A.20}$$

$$1 - \gamma\theta_{s'a's'}\pi_{s'a'} > \sum_{(s,a)\neq(s',a')} \gamma\theta_{s'a's}\pi_{sa} \Rightarrow \tag{A.21}$$

$$|A(s'a', s'a')| > \sum_{(s,a)\neq(s',a')} |A(sa, s'a')|. \tag{A.22}$$

This implies that A is non-singular (Horn et al., 2012), so (A.8) - (A.9) has at most one solution. □

We will now state the proof of Theorem 2.

Proof. For the first direction of the theorem, assume that x satisfies the Bellman flow constraints (5) - (6), and that $\pi_{sa} = \frac{\rho_{sa}}{\sum_a \rho_{sa}}$. Therefore:

$$\pi_{sa} = \frac{\rho_{sa}}{\sum_a \rho_{sa}} = \rho_{sa}\alpha_s + \gamma \sum_{s',a'} x_{s'a'}\theta_{s'a's}. \tag{A.23}$$

Clearly, x is a solution to the π-specific Bellman flow constraints (12) - (13), and Lemmas 2 and 3 imply that x is the occupancy measure of π. For the other direction of the theorem, assume that x is the occupancy measure of π. Lemmas 2 and 3 imply that x is the unique solution to the π-specific Bellman flow constraints eq. (A.11) - eq. (A.12). Therefore, π is given by (16). And since $\sum_a \pi_{sa} = 1$, we have:

$$\sum_a \rho_{sa}\alpha_s + \gamma \sum_{s',a'} x_{s'a'}\theta_{s'a's} = 1 \tag{A.24}$$

which can be rearranged to show that x satisfies the Bellman flow constraints, and also combined with (16) to show that $\pi_{sa} = \frac{\rho_{sa}}{\sum_a \rho_{sa}}$. $\square$

A.2 FASTRL: APPENDIX

DETAILS OF EXPERIMENTAL SETTINGS In this section we present a number of additional details pertaining the experiments performed in chapter 3.

ADDITIONAL RESULTS This section provides a number of additional evaluations pertaining to the FASTRL pipeline described in chapter 3.

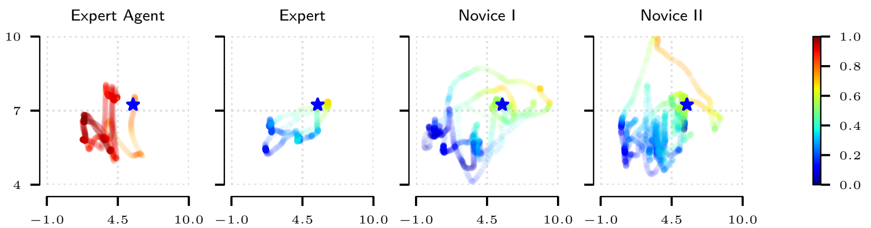

Figure A.1: **Trajectory projection comparison for `ImageCentering`.**

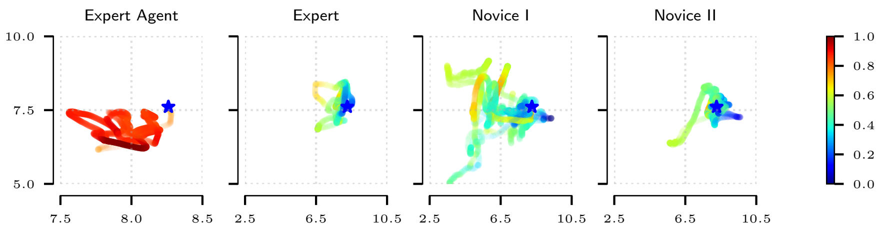

Figure A.2: **Trajectory projection comparison for `TraceLines`.**

COMPUTATIONAL RESOURCES The experiments were executed on a GPU server equipped with Intel Xeon Gold 6150 and AMD EPYC 7742 CPUs and NVIDIA GTX1080Ti GPUs. The average running time per experiment are shown in table A.6. For the experiments in section 3.4.1, we use an extension of the Unity `ml-agents` framework as algorithmic implementation.

API DESCRIPTION We provide an interface to algorithm implementations of two popular frameworks: an extension of the `ml-agents` (Juliani et al.,

Hyperparameter	`ImageCentering`	`Periscoping`	`TraceLines`
Batch size	256	1024	1024
Rollout buffer size	20480	20480	20480
Learning rate	5e-4	5e-4	5e-4
Entropy bonus β	0.01	0.01	0.01
Clipping threshold ϵ	0.2	0.3	0.3
GAE λ	0.95	0.95	0.95
Number of epochs	3	3	3
LR schedule	const	const	const
Network settings			
Normalise input	true	true	true
Number of hidden units	128	128	128
Number of layers	2	2	2
Reward signals			
Extrinsic γ	0.99	0.99	0.99
Extrinsic weight	1.0	1.0	1.0
Curiosity γ	0.99	0.99	0.99
Curiosity weight	0.15	0.2	0.1
Curiosity learning rate	3e-4	3e-4	3e-4
Curiosity hidden sise	128	128	256
IRL settings			
γ	0.99	0.99	0.99
Strength	1.0	1.0	1.0
Hidden size γ	512	512	512
Learning rate	3e-4	3e-4	3e-4
Use actions	false	false	false

TABLE A.1: PPO hyperparameters used for section 3.4.1 experiments

Hyperparameter	`ImageCentering`	`Periscoping`	`TraceLines`
Batch size	256	1024	1024
Replay buffer size	100000	100000	100000
Learning rate	3e-4	3e-4	3e-4
Initial temperature	0.2	1.0	1.0
τ	0.005	0.005	0.005
Steps per update ϵ	20	20	20
Steps per update ϵ	20	20	20
LR schedule	const	const	const
Network settings			
Normalise input	true	true	true
Number of hidden units	128	128	128
Number of layers	2	2	2
IRL settings			
γ	0.99	0.99	0.99
Strength	1.0	1.0	1.0
Hidden size γ	256	256	256
Number of layers	2	2	2
Learning rate	3e-4	3e-4	3e-4
Use actions	false	false	false

TABLE A.2: SAC hyperparameters used for section 3.4.1 experiments

Agent ID	r_ψ	V_ϕ	r_{heur}	Trajectory length
fw	$0.998_{\pm 0.002}$	$0.680_{\pm 0.104}$	$0.999_{\pm 0.0002}$	$966_{\pm 782.0}$
fm	$0.610_{\pm 0.034}$	$0.773_{\pm 0.180}$	$0.742_{\pm 0.016}$	$1180.5_{\pm 104.9}$
an	$0.654_{\pm 0.046}$	$0.408_{\pm 0.213}$	$0.729_{\pm 0.039}$	$1053.5_{\pm 130.3}$
io	$0.462_{\pm 0.302}$	$0.499_{\pm 0.254}$	$0.384_{\pm 0.278}$	$1646.5_{\pm 925.3}$
mk	$0.611_{\pm 0.079}$	$0.543_{\pm 0.235}$	$0.634_{\pm 0.106}$	$1140_{\pm 197.9}$
mv	$0.684_{\pm 0.008}$	$0.402_{\pm 0.083}$	$0.529_{\pm 0.095}$	$956_{\pm 13.45}$

TABLE A.3: **Trajectory scores for human and virtual agents** (rewards and values normalised) on the `ImageCentering` task

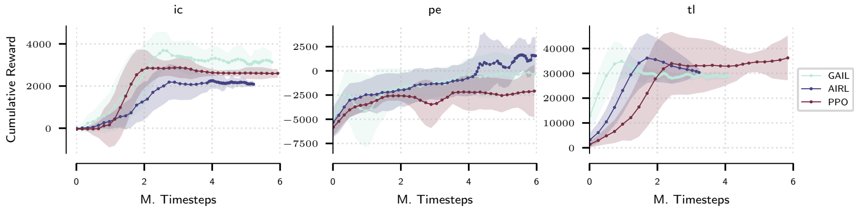

FIGURE A.3: **Comparison of forward and inverse RL experiments**: Cumulative reward for `ImageCentering` (ic), `Periscoping` (pe) and `TraceLines` (tl) environments.

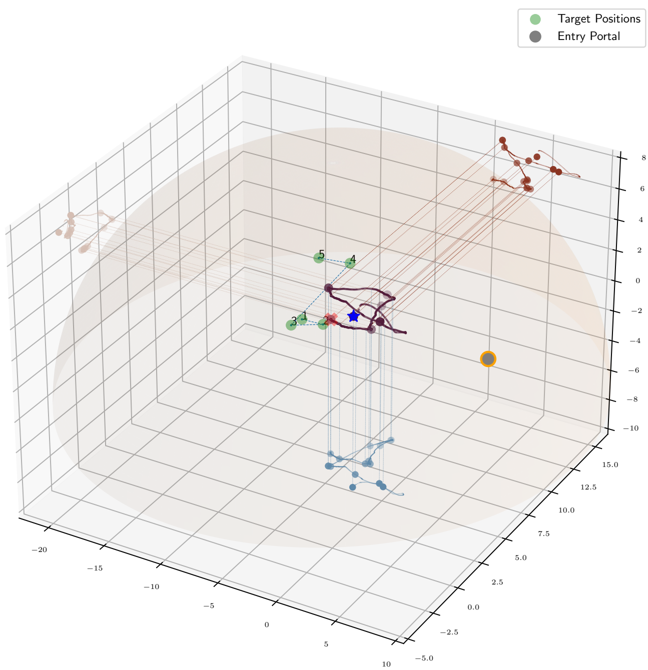

FIGURE A.4: **Example expert trajectory for `Periscoping` task**. The shaded semi-sphere indicates the dome. The green dots indicate the five target positions.

Agent ID	r_ψ	V_ϕ	r_{heur}	Trajectory length
fw	$0.985_{\pm 0.007}$	$0.975_{\pm 0.015}$	$0.999_{\pm 0.0002}$	$4548.8_{\pm 881.5}$
fm	$0.668_{\pm 0.006}$	$0.630_{\pm 0.045}$	$0.677_{\pm 0.006}$	$3705.6_{\pm 45.5}$
an	$0.495_{\pm 0.072}$	$0.281_{\pm 0.039}$	$0.476_{\pm 0.065}$	$2295.3_{\pm 450.2}$
io	$0.399_{\pm 0.269}$	$0.293_{\pm 0.232}$	$0.389_{\pm 0.266}$	$4310.8_{\pm 1854.6}$
mk	$0.518_{\pm 0.159}$	$0.544_{\pm 0.051}$	$0.542_{\pm 0.071}$	$3237.6_{\pm 506.27}$
mv	$0.363_{\pm 0.067}$	$0.439_{\pm 0.119}$	$0.407_{\pm 0.037}$	$4224.0_{\pm 255.2}$

TABLE A.4: **Trajectory scores for human and virtual agents** (rewards and values normalised) on the `TraceLines` task

Subject	r_ψ	V_ϕ	r_{heur}	Traj. length
Expert	$0.98_{\pm 0.06}$	$0.93_{\pm 0.04}$	$0.95_{\pm 0.04}$	$1475.33_{\pm 69.16}$
Intermediate	$0.81_{\pm 0.06}$	$0.76_{\pm 0.06}$	$0.34_{\pm 0.026}$	$1279.0_{\pm 128.88}$
Poor	$0.40_{\pm 0.13}$	$0.41_{\pm 0.11}$	$0.13_{\pm 0.037}$	$720.4_{\pm 192.86}$

TABLE A.5: **Normalised scores (heuristic and learned rewards and values) for human trajectories**: *Expert*, *Intermediate* and *Poor* on `ImageCentering` task

2018) framework for the PPO with curiosity exploration algorithm and the inverse RL algorithms as well as `stable-baselines3` (Raffin et al., 2021) for a more broad selection of baseline forward RL algorithms. Furthermore, the simulation parameters such as heuristic reward weights, target positions and curricula are exposed to the user as well. For the full documentation, we refer the reader the project website: `fastrl.ethz.ch`.

This section provides an overview of the data used in the context of the surgical assistance pipeline.

FAST DATASET. For the purpose of skill performance evaluation, a dataset was obtained by subjects performing the three benchmark tasks using the simulator platform. Five subjects with different levels of expertise were invited to perform each of the three benchmark tasks for a total number of 5 repetitions. The trajectories were recorded according to the state space specification described in section 3.3.3. Two subjects were already very familiar with all tasks and are considered as *Experts*, while others with little or no experience with the simulation are considered as *Novices* as shown in table A.7. Additionally, the participants are graded based on the length of their performed trajectories. This metric coincides with the default

Algorithm	`ImageCentering`	`Periscoping`	`TraceLines`
PPO	143	151	149
SAC	186	197	194
GAIL (PPO)	202	229	226
AIRL (PPO)	180	192	182
GAIL (SAC)	285	321	304
AIRL (SAC)	242	239	243

Table A.6: **Overview of the average algorithm runtimes** (in minutes per 1M environment interactions) for the different benchmark environments.

Subject ID	Expertise	Trajectory length
fm	*Expert*	$1600.6_{\pm 971.0}$
an	*Expert*	$1929.2_{\pm 277.4}$
mk	*Novice*	$2578.5_{\pm 745.8}$
mv	*Novice*	$3313.5_{\pm 665.5}$
io	*Novice*	$4290.8_{\pm 1981}$

Table A.7: **Participant performance ranking**

heuristic used by the VirtaMed simulator platform. For the evaluation performed in section 3.4.2, the subjects used the physical hardware described in section 3.5, recorded using a magnetic sensor, which tracks instrument movement. In the second evaluation, the keyboard-and-mouse interface was used to record the trajectories.

LAPAROSCOPIC DIAGNOSTIC TOUR DATASET. The diagnostic tour dataset used in section 3.4.3 was gathered using the laparoscopy simulator, where users were asked to perform a guided diagnostic tour of the abdomen with a fixed sequence of anatomical landmarks highlighted on screen. A total of 100 trajectories from a diverse set of experienced practitioners were obtained. The average reported proficiency based on the simulator evaluation (max. attainable score 150) was 128.24 ± 18.32 points.

A.3 CI-IRL: APPENDIX

A.3.1 *Proposition proofs*

Proposition 1. *Let the likelihood $p(\xi)$ belong to a natural exponential family with parameter ψ, sufficient statistics $\varphi(x)$ and the (Lebesgue) base measure p_0. Let $\mathcal{D}_E^e$ be the dataset corresponding to interventional setting e. Then, for all $e \in \mathcal{E}_{tr}$, the causal invariance penalty for the maximum likelihood loss is the norm of the expectation difference of features φ:*

$$\mathbb{D}(\psi, \varphi; e) = ||\nabla_{\psi|\psi=1.0}\mathcal{L}^e(\psi, \varphi)||^2 = ||\mathbb{E}_{\mathcal{D}_E^e}[\varphi(\xi)] - \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi))]||^2 \tag{4.15}$$

Proof. The result directly follows from the definition of the primal problem.

$$\begin{aligned}
\nabla_{\psi|\psi=1.0}\mathcal{L}^e(\psi, \varphi) &= \nabla_\psi \left(\mathbb{E}_{\xi \in \mathcal{D}_e} \left[\log \left(\frac{1}{Z_{\psi,\phi}} \exp(\psi^T \varphi(\xi)) \right) \right] \right) \\
&= \mathbb{E}_{\xi \in \mathcal{D}_e} \left[\nabla_\psi (\psi^T \varphi(\xi) - \log Z_{\psi,\phi}) \right] \\
&= \mathbb{E}_{\mathcal{D}_E^e}[\varphi(\xi)] - \nabla_\psi \log Z_{\psi,\phi} \\
&= \mathbb{E}_{\mathcal{D}_E^e}[\varphi(\xi)] - \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi))]
\end{aligned}$$

where we use the moment generating property of the log partition function $\nabla_\psi \log Z_{\psi,\varphi} = \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi)]$. □

Proposition 2. *The gradient of the primal exponential family maximum-likelihood problem in Equation (4.3) w.r.t. the natural parameter ψ is equivalent to the gradient of the dual in Equation (4.5) w.r.t the parameter ψ when the density ratio $\frac{q}{p}$ is unity.*

$$\begin{aligned}
&||\nabla_\psi \mathcal{L}_{dual}(\psi, \varphi, q, e)||_2 = \\
&\qquad ||\min_q \mathbb{E}_{\xi \sim \mathcal{D}_E}[\varphi(\xi)] - \mathbb{E}_{\xi \sim p(\xi|\psi,\varphi)} \left[\frac{q(\xi)}{p(\xi|\psi, \varphi)} \varphi(\xi) \right] ||_2
\end{aligned} \tag{4.17}$$

Proof. We begin by computing the gradient of

$$\min_q \, [\mathbb{E}_{\xi \sim \mathcal{D}_E^e}[g_{\psi,\varphi}(\xi)] - \mathbb{E}_{\xi \sim q}[g_{\psi,\varphi}(\xi)] + D_{KL}(q||p_0)]$$

w.r.t. to the parameters ψ for setting $e \in \mathcal{E}_{tr}$:

$$\begin{aligned}
&\nabla_\psi \mathcal{L}_{dual}(\psi, \varphi, q, e) \\
&\quad = \nabla_\psi (\min_q \, [\mathbb{E}_{\xi \sim \mathcal{D}_E^e}[g_{\psi,\varphi}(\xi)] - \mathbb{E}_{\xi \sim q}[g_{\psi,\varphi}(\xi)] + D_{KL}(q||p_0)]) \\
&\quad \overset{(1)}{=} \nabla_\psi (\min_q \, [\mathbb{E}_{\xi \sim \mathcal{D}_E}[g_{\psi,\varphi}(\xi)] - \mathbb{E}_{\xi \sim q}[g_{\psi,\varphi}(\xi)] + \mathcal{H}(q)]) \\
&\quad \overset{(2)}{=} \min_q \, \left[\nabla_\psi \left(\mathbb{E}_{\xi \sim \mathcal{D}_E}[g_{\psi,\varphi}(\xi)] - \mathbb{E}_{\xi \sim q}[g_{\psi,\varphi}(\xi)] + \mathcal{H}(q)\right)\right] \\
&\quad \overset{(3)}{=} \min_q \, [\mathbb{E}_{\xi \sim \mathcal{D}_E^e}[\nabla_\psi g_{\psi,\varphi}(\xi)] - \mathbb{E}_{\xi \sim q}[\nabla_\psi g_{\psi,\varphi}(\xi)]]
\end{aligned}$$

where we use: (1) the fact that we assume the base measure p_0 to be the Lebesgue measure (or count measure in the discrete case), i.e. $p_0(\xi) = 1$, (2) the envelope theorem and (3) the fact that q does not directly depend on ψ and thus, $\nabla_\psi \mathcal{H}(q) = 0$. We can rewrite the second expectation using the importance sampling trick:

$$\begin{aligned}
&\min_q \mathbb{E}_{x \sim \mathcal{D}_E}[\varphi(\xi)] - \mathbb{E}_{x \sim q}[\varphi(\xi)] \\
&= \min_q \mathbb{E}_{x \sim \mathcal{D}_E}[\varphi(\xi)] - \mathbb{E}_{x \sim p(\xi|\psi,\varphi)} \left[\frac{q(\xi)}{p(\xi|\psi,\varphi)} \varphi(\xi)\right]
\end{aligned}$$

By definition of the distribution matching problem and strict concavity w.r.t. q, the optimum is attained when $q(\xi) = p(\xi|\psi, \varphi)$, i.e. when the importance sampling ratio is 1.

□

A.3.2 *Derivation of the penalty term for feature matching IRL*

To apply the proposed regularization to the feature matching problem described in algorithm 3, we need to derive an explicit gradient term. We will do so here. The gradient of the per-environment log-likelihood loss $\mathcal{L}_{MLE}(\psi, \varphi, e) = \sum_{\xi \in \mathcal{D}_e} \log p(\xi|\psi, \varphi)$ w.r.t. to ψ is computed as follows:

$$\begin{aligned}
\mathcal{L}_{MLE} = \sum_{\xi \in \mathcal{D}_e} \log p(\xi|\psi, \varphi) &= \sum_{\xi \in \mathcal{D}_e} \log(\exp(\psi^T \varphi(\xi))) - \log Z_{\psi,\varphi} \\
&= \sum_{\xi \in \mathcal{D}_e} \psi^T \varphi(\xi) - \log Z_{\psi,\varphi}
\end{aligned}$$

Differentiating w.r.t. ψ yields:

$$
\begin{aligned}
\frac{\partial L_{MLE}}{\partial \psi} &= \mathbb{E}_{\mathcal{D}_e}[\varphi(\xi)] - \frac{1}{Z_{\psi,\varphi}} \int \exp(\psi^T \varphi(\xi))\varphi(\xi) d\xi \\
&= \mathbb{E}_{\mathcal{D}_E}[\varphi(\xi)] - \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi)]
\end{aligned}
$$

The gradient penalty term from Equation (4.15) with respect to the features φ is derived as follows:

$$
\nabla_\varphi \left\| \nabla_{\psi|\psi=1.0} L^e\left(r\left(\psi,\varphi\right)\right) \right\|^2 = \frac{\partial || \frac{\partial L^e(r(\psi,\varphi)\cdot)}{\partial \psi} |_{\psi=1.0} ||^2}{\partial \varphi}
$$

We employ the chain rule:

$$
\frac{\partial L^e\left(r\left(\psi,\varphi\right)\right)}{\partial \psi} = \frac{\partial L^e\left(r\left(\psi,\varphi\right)\right)}{\partial r} \cdot \frac{\partial\left(r\left(\psi,\varphi\right)\right)}{\partial \psi} = \frac{\partial L^e\left(r\left(\psi,\varphi\right)\right)}{\partial r} \cdot \varphi
$$

where the last equality holds because we assume a linear reward with respect to the features φ: $r\left(\psi,\varphi\right) = \psi^T\varphi$. In section 4.2.1 we showed that :

$$
\frac{\partial L^e\left(r\left(\psi,\varphi\right)\right)}{\partial r} = \mathbb{E}_{\mathcal{D}_E}[\varphi(\xi)] - \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi)]
$$

where $\mathbb{E}_{\mathcal{D}_E}[\varphi(\xi)]$ are the trajectory feature statistics of the expert and $\mathbb{E}_\pi\left[\varphi(\xi)\right]$ are the trajectory feature statistics of the imitation policy. For the sake of simplicity we define: $C := \mathbb{E}_{\mathcal{D}_E}[\varphi(\xi)] - \mathbb{E}_{p(\xi|\psi)}[\varphi(\xi))$ which is independent of φ: Then:

$$
\frac{\partial L^e\left(r\left(\psi,\varphi\right)\right)}{\partial \psi} = C\varphi
$$

We obtain the CI penalty for the feature matching maximum entropy IRL case as the following term:

$$
\begin{aligned}
\nabla_\varphi \left\| \nabla_{\psi|\psi=1.0} L^e\left(r\left(\psi,\varphi\right)\right) \right\|^2 &= \frac{\partial || \frac{\partial L^e(r(\psi,\varphi)\cdot)}{\partial \psi} |_{\psi|\psi=1.0} ||^2}{\partial \varphi} = \frac{\partial \left\| C\varphi \right\|^2}{\partial \varphi} \\
&= \frac{\partial \left[C\varphi\right]^T \left[C\varphi\right]}{\partial \varphi} \\
&= C^T C \frac{\partial \varphi^T \varphi}{\partial \varphi} = 2\left\|C\right\|^2 \varphi = 2||\rho_E - \mathbb{E}\left[\rho_\psi\right]||^2 \varphi
\end{aligned}
$$

A.4 RL ENVIRONMENT DETAILS

This section provides an overview of the environment specifications used in training the imitation and IRL agents in chapter 4 and chapter 5.

MUJOCO LOCOMOTION SUITE The MuJoCo (Todorov et al., 2012) locomotion benchmark is a performance assessment tool used in robotics and AI research, focusing on the simulation of complex movements and behaviors in robotic agents. It utilizes the MuJoCo physics engine, renowned for its accuracy and efficiency in simulating the dynamics of biomechanically-inspired robotic systems. The benchmark tasks typically involve controlling virtual robots to achieve various locomotive goals, such as walking, running, or jumping across different terrains or obstacles. These tasks are designed to test the agility, stability, and control strategies of the robots, providing valuable insights into their potential real-world applications and performance. Specifically, we use five tasks from this suite, listed in increasing order of state-space dimensionality: HOPPER, HALFCHEETAH, WALKER2D, ANT, and HUMANOID.

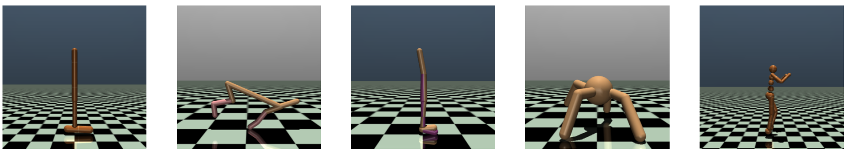

FIGURE A.5: **MuJoCo Locomotion Suite environments.** From left to right: HOPPER, HALFCHEETAH, WALKER2D, ANT, HUMANOID

ADROIT HAND MANIPULATION The Adroit dexterous hand manipulation suite of tasks (Rajeswaran et al., 2018), part of the D4RL dataset (Fu, Kumar, et al., 2020), is a comprehensive set of benchmarks designed to evaluate robotic hands' ability to manipulate and interact with various objects. The tasks encompass a range of activities from basic object handling, like picking up and moving items, to more complex actions such as tool use and object reorientation. Each task is aimed at testing the robotic hand's control algorithms, finger coordination, grip strength, and adaptability to different shapes and textures, making it a critical tool for advancing robotic dexterity in real-world applications. The tasks feature a high-dimensional action space and randomized goal positions, requiring a high generalization capacity of policies performing the task.

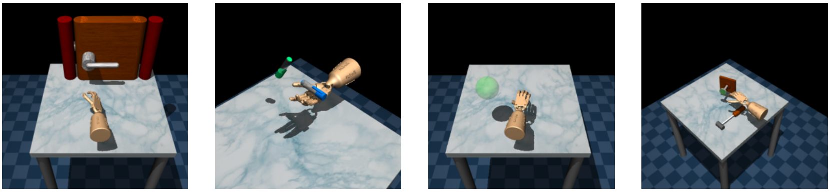

FIGURE A.6: **Adroit Hand Manipulation Suite environments.** From left to right: ADROITHANDDOOR, ADROITHANDPEN, ADROITHANDRELOCATE, ADROITHANDHAMMER

FRANKA KITCHEN MANIPULATION TASKS The Franka Kitchen benchmark suite of tasks (Gupta et al., 2020) is a specialized set of evaluations designed for testing robotic manipulation in a kitchen environment. This suite, which uses the Franka Emika Panda 7-axis robot arm, aims to simulate common manipulation tasks that require a high degree of precision and adaptability. The tasks include activities such as opening and closing drawers and cabinets, turning on appliances, and manipulating kitchen utensils and cookware. The goal of these tasks is to assess and improve the robot's ability to perform complex, multi-step actions in a dynamic, semi-structured environment, mirroring the challenges faced in real-world domestic settings. The task is challenging in the imitation learning setting due to a relatively small subdomain of success states.

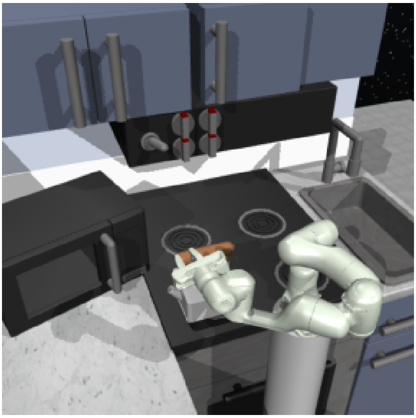

FIGURE A.7: **Franka Kitchen Robotic Manipulation environment.** The task entails performing a sequence of manipulation tasks in the kitchen environment: open the microwave door, move the kettle and turn off the switch.

BIBLIOGRAPHY

Sutton, Richard S and Andrew G Barto (2018). *Reinforcement learning: An introduction*. MIT press (cit. on pp. 1, 8, 11).

Kingma, Diederik P and Jimmy Ba (2014). "Adam: A method for stochastic optimization". In: *arXiv preprint arXiv:1412.6980* (cit. on pp. 1, 12, 81).

Mnih, Volodymyr, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. (2015). "Human-level control through deep reinforcement learning". In: *nature* 518.7540, 529 (cit. on pp. 1, 12, 14).

Silver, David, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas Baker, Matthew Lai, Adrian Bolton, et al. (2017). "Mastering the game of go without human knowledge". In: *nature* 550.7676, 354 (cit. on p. 1).

Vinyals, Oriol, Timo Ewalds, Sergey Bartunov, Petko Georgiev, Alexander Sasha Vezhnevets, Michelle Yeo, Alireza Makhzani, Heinrich Küttler, John Agapiou, Julian Schrittwieser, et al. (2017). "Starcraft ii: A new challenge for reinforcement learning". In: *arXiv preprint arXiv:1708.04782* (cit. on p. 1).

Billard, Aude, Sylvain Calinon, Ruediger Dillmann, and Stefan Schaal (2008). *Survey: Robot programming by demonstration*. Tech. rep. Springrer (cit. on pp. 1, 52, 83).

Ouyang, Long, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. (2022). "Training language models to follow instructions with human feedback, 2022". In: *URL https://arxiv. org/abs/2203.02155* 13, 1 (cit. on p. 1).

Degrave, Jonas, Federico Felici, Jonas Buchli, Michael Neunert, Brendan Tracey, Francesco Carpanese, Timo Ewalds, Roland Hafner, Abbas Abdolmaleki, Diego de Las Casas, et al. (2022). "Magnetic control of tokamak plasmas through deep reinforcement learning". In: *Nature* 602.7897, 414 (cit. on p. 1).

Kalidindi, Surya R, Michael Buzzy, Brad L Boyce, and Remi Dingreville (2022). "Digital twins for materials". In: *Frontiers in Materials* 9, 818535 (cit. on p. 1).

Jiang, Yuchen, Shen Yin, Kuan Li, Hao Luo, and Okyay Kaynak (2021). "Industrial applications of digital twins". In: *Philosophical Transactions of the Royal Society A* 379.2207, 20200360 (cit. on p. 1).

Van der Valk, Hendrik, Hendrik Haße, Frederik Möller, Michael Arbter, Jan-Luca Henning, and Boris Otto (2020). "A Taxonomy of Digital Twins." In: *AMCIS* (cit. on p. 1).

Croatti, Angelo, Matteo Gabellini, Sara Montagna, and Alessandro Ricci (2020). "On the integration of agents and digital twins in healthcare". In: *Journal of Medical Systems* 44.9, 161 (cit. on p. 2).

Laumer, Fabian, Lena Rubi, Michael Matter, Stefano Buoso, Gabriel Fringeli, Francois Mach, Frank Ruschitzka, Joachim Buhmann, and Christian Matter (2023). "Fully automated 4D cardiac modelling from 2D echocardiography for clinical applications". In: (cit. on p. 2).

Katsoulakis, Evangelia, Qi Wang, Huanmei Wu, Leili Shahriyari, Richard Fletcher, Jinwei Liu, Luke Achenie, Hongfang Liu, Pamela Jackson, Ying Xiao, et al. (2024). "Digital twins for health: a scoping review". In: *npj Digital Medicine* 7.1, 77 (cit. on p. 2).

Schaal, Stefan (1996). "Learning from demonstration". In: *Advances in neural information processing systems* 9 (cit. on p. 2).

Pomerleau, Dean A (1991). "Efficient training of artificial neural networks for autonomous navigation". In: *Neural Computation* (cit. on pp. 2, 52, 83).

Christiano, Paul F, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei (2017). "Deep reinforcement learning from human preferences". In: *Advances in neural information processing systems* 30 (cit. on p. 2).

Brown, Daniel, Wonjoon Goo, Prabhat Nagarajan, and Scott Niekum (2019). "Extrapolating beyond suboptimal demonstrations via inverse reinforcement learning from observations". In: *International conference on machine learning*. PMLR, 783 (cit. on p. 2).

Ho, Jonathan and Stefano Ermon (2016). "Generative adversarial imitation learning". In: *Advances in Neural Information Processing Systems* (cit. on pp. 2, 24, 36, 37, 41, 42, 52, 57, 64, 65, 70, 84–86, 89, 93).

Fu, Justin, Katie Luo, and Sergey Levine (2018). "Learning robust rewards with adversarial inverse reinforcement learning". In: *International Conference on Learning Representations* (cit. on pp. 2, 24, 37, 42, 52, 57, 64, 65, 69, 84, 85, 89, 93).

Ratliff, Nathan D, J Andrew Bagnell, and Martin A Zinkevich (2006). "Maximum margin planning". In: *Proceedings of the 23rd international conference on Machine learning*, 729 (cit. on pp. 2, 16).

Ziebart, Brian D, J Andrew Bagnell, and Anind K Dey (2010). "Modeling interaction via the principle of maximum causal entropy". In: *ICML* (cit. on pp. 2, 52, 63).

Kostrikov, Ilya, Ofir Nachum, and Jonathan Tompson (2019). "Imitation learning via off-policy distribution matching". In: *International Conference on Learning Representations* (cit. on pp. 2, 93, 97, 105).

Nguyen, XuanLong, Martin J Wainwright, and Michael I Jordan (2009). "On surrogate loss functions and f-divergences". In: (cit. on pp. 3, 21, 23, 57, 64).

Sriperumbudur, Bharath K, Kenji Fukumizu, Arthur Gretton, Bernhard Schölkopf, and Gert RG Lanckriet (2009). "On integral probability metrics, ϕ-divergences and binary classification". In: *arXiv:0901.2698* (cit. on pp. 3, 19, 21–23, 84).

Arora, Saurabh and Prashant Doshi (2021). "A survey of inverse reinforcement learning: Challenges, methods and progress". In: *Artificial Intelligence* 297, 103500 (cit. on p. 3).

Bellman, Richard (1957). "A Markovian decision process". In: *Journal of mathematics and mechanics*, 679 (cit. on p. 9).

Ozdaglar, Asuman E, Sarath Pattathil, Jiawei Zhang, and Kaiqing Zhang (2023). "Revisiting the linear-programming framework for offline rl with general function approximation". In: *International Conference on Machine Learning*. PMLR, 26769 (cit. on p. 10).

Williams, Ronald J (1992). "Simple statistical gradient-following algorithms for connectionist reinforcement learning". In: *Machine learning* 8, 229 (cit. on p. 12).

Kakade, Sham and John Langford (2002). "Approximately optimal approximate reinforcement learning". In: *Proceedings of the Nineteenth International Conference on Machine Learning*, 267 (cit. on p. 12).

Schulman, John, Sergey Levine, Pieter Abbeel, Michael Jordan, and Philipp Moritz (2015). "Trust region policy optimization". In: *International conference on machine learning*. PMLR, 1889 (cit. on pp. 12, 13, 109).

Sutton, Richard S, David McAllester, Satinder Singh, and Yishay Mansour (1999). "Policy gradient methods for reinforcement learning with function approximation". In: *Advances in neural information processing systems* 12 (cit. on p. 12).

Schulman, John, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel (2015). "High-dimensional continuous control using generalized advantage estimation". In: *arXiv preprint arXiv:1506.02438* (cit. on p. 12).

Schulman, John, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov (2017). *Proximal policy optimization algorithms*. Tech. rep. OpenAI (cit. on pp. 12, 13, 41, 93, 96, 97).

Tesauro, Gerald et al. (1995). "Temporal difference learning and TD-Gammon". In: *Communications of the ACM* 38.3, 58 (cit. on p. 14).

Haarnoja, Tuomas, Aurick Zhou, Pieter Abbeel, and Sergey Levine (2018). "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor". In: *International Conference on Machine Learning* (cit. on pp. 14, 41, 65, 70, 89, 90).

Fujimoto, Scott, Herke Hoof, and David Meger (2018). "Addressing function approximation error in actor-critic methods". In: *International Conference on Machine Learning* (cit. on pp. 14, 97).

Ng, Andrew Y, Stuart Russell, et al. (2000). "Algorithms for inverse reinforcement learning." In: *Icml*. Vol. 1, 2 (cit. on pp. 16, 34, 38, 52).

Ng, Andrew Y, Daishi Harada, and Stuart Russell (1999). "Policy invariance under reward transformations: Theory and application to reward shaping". In: *Icml*. Vol. 99, 278 (cit. on pp. 16, 52).

Abbeel, Pieter and Andrew Y Ng (2004). "Apprenticeship learning via inverse reinforcement learning". In: *Proceedings of the twenty-first international conference on Machine learning*, 1 (cit. on p. 16).

Ziebart, Brian D, Andrew L Maas, J Andrew Bagnell, and Anind K Dey (2008). "Maximum entropy inverse reinforcement learning." In: *Aaai*. Vol. 8. Chicago, IL, USA, 1433 (cit. on pp. 16, 37, 52, 54, 55).

Bashiri, Mohammad Ali, Brian Ziebart, and Xinhua Zhang (2021). "Distributionally Robust Imitation Learning". In: *Advances in neural information processing systems* 34, 24404 (cit. on pp. 16, 60, 73).

Jaynes, Edwin T (1957). "Information theory and statistical mechanics". In: *Physical review* 106.4, 620 (cit. on p. 16).

Rockafellar, Ralph Tyrell (2015). "Convex Analysis:(PMS-28)". In: (cit. on pp. 17, 56).

Csiszár, Imre (1972). "A class of measures of informativity of observation channels". In: *Periodica Mathematica Hungarica* (cit. on pp. 18–20, 54, 56, 84).

Donsker, MD and SR Srinivasa Varadhan (1976). "On the principal eigenvalue of second-order elliptic differential operators". In: *Communications on Pure and Applied Mathematics* 29.6, 595 (cit. on pp. 18, 20, 22).

Goodfellow, Ian J, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio (2014). "Generative Adversarial Networks". In: *Advances in Neural Information Processing Systems* (cit. on pp. 18, 23, 57, 86).

Syed, Umar, Michael Bowling, and Robert E Schapire (2008). "Apprenticeship learning using linear programming". In: *Proceedings of the 25th international conference on Machine learning*, 1032 (cit. on pp. 19, 109, 111).

Andersen, Erling Bernhard (1970). "Asymptotic properties of conditional maximum-likelihood estimators". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 32.2, 283 (cit. on pp. 22, 56).

Finn, Chelsea, Sergey Levine, and Pieter Abbeel (2016). "Guided cost learning: Deep inverse optimal control via policy optimization". In: *International conference on machine learning*. PMLR, 49 (cit. on pp. 23, 64).

Devroye, Luc (1985). "Nonparametric density estimation". In: *The L_1 View* (cit. on p. 23).

Monge, Gaspard (1811). *Géométrie descriptive*. J. Klostermann fils (cit. on p. 25).

Brenier, Yann (1991). "Polar factorization and monotone rearrangement of vector-valued functions". In: *Communications on pure and applied mathematics* 44.4, 375 (cit. on p. 25).

Santambrogio, Filippo (2015). "Optimal transport for applied mathematicians". In: *Birkäuser* (cit. on pp. 26, 27, 84, 87, 89).

Kantorovich, Leonid V (1960). "Mathematical methods of organizing and planning production". In: *Management science* 6.4, 366 (cit. on p. 26).

Cuturi, Marco (2013). "Sinkhorn distances: Lightspeed computation of optimal transport". In: *Advances in neural information processing systems* 26 (cit. on p. 26).

Kantorovich, L.V. and S.G. Rubinstein (1958). "On a space of totally additive functions". In: *Vestnik of the St. Petersburg University: Mathematics* 13.7, 52 (cit. on p. 26).

Arjovsky, Martin, Soumith Chintala, and Léon Bottou (2017). "Wasserstein generative adversarial networks". In: *International Conference on Machine Learning* (cit. on pp. 27, 84).

Xiao, Huang, Michael Herman, Joerg Wagner, Sebastian Ziesche, Jalal Etesami, and Thai Hong Linh (2019). "Wasserstein adversarial imitation learning". In: *arXiv:1906.08113* (cit. on pp. 27, 84, 86).

Benamou, Jean-David and Yann Brenier (2000). "A computational fluid mechanics solution to the Monge-Kantorovich mass transfer problem". In: *Numerische Mathematik* 84.3, 375 (cit. on p. 28).

Gigli, Nicola and Luca Tamanini (2020). "Benamou–Brenier and duality formulas for the entropic cost on RCD(K, N) spaces". In: *Probability Theory and Related Fields* 176.1, 1 (cit. on p. 28).

Jordan, Richard, David Kinderlehrer, and Felix Otto (1998). "The variational formulation of the Fokker–Planck equation". In: *SIAM journal on mathematical analysis* 29.1, 1 (cit. on p. 28).

Zhang, Ruiyi, Changyou Chen, Chunyuan Li, and Lawrence Carin (2018). "Policy optimization as wasserstein gradient flows". In: *International Conference on machine learning*. PMLR, 5737 (cit. on p. 29).

Pearl, Judea (2009). *Causality*. Cambridge university press (cit. on pp. 29, 30, 57, 58).

Peters, Jonas, Dominik Janzing, and Bernhard Schölkopf (2017). *Elements of causal inference: foundations and learning algorithms*. The MIT Press (cit. on p. 30).

Peters, Jonas, Peter Bühlmann, and Nicolai Meinshausen (2015). "Causal inference using invariant prediction: identification and confidence intervals". In: *arXiv preprint arXiv:1501.01332* (cit. on pp. 30, 53, 60, 72).

Heinze-Deml, Christina, Jonas Peters, and Nicolai Meinshausen (2017). *Invariant Causal Prediction for Nonlinear Models* (cit. on pp. 30, 53, 72).

Chiasson, Patrick M., David Pace, Christopher Schlachta, Joseph Mamazza, and Éric C. Poulin (2003). "Minimally invasive surgery training in Canada: a survey of general surgery." In: *Surgical endoscopy* 17 3, 371 (cit. on p. 34).

Stauder, Ralf, Daniel Ostler, Thomas Vogel, Dirk Wilhelm, Sebastian Koller, Michael Kranzfelder, and Nassir Navab (2017). "Surgical data processing for smart intraoperative assistance systems". In: *Innovative Surgical Sciences* 2.3, 145 (cit. on p. 34).

Tagliabue, Eleonora, Ameya Pore, Diego Dall'Alba, Enrico Magnabosco, Marco Piccinelli, and Paolo Fiorini (2020). "Soft tissue simulation environment to learn manipulation tasks in autonomous robotic surgery". In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 3261 (cit. on pp. 34, 36).

Kazanzides, Peter, Zihan Chen, Anton Deguet, Gregory S Fischer, Russell H Taylor, and Simon P DiMaio (2014). "An open-source research kit for the da Vinci® Surgical System". In: *2014 IEEE international conference on robotics and automation (ICRA)*. IEEE, 6434 (cit. on pp. 34, 36).

Cychosz, Chris C, Josef N Tofte, Alyssa Johnson, Yubo Gao, and Phinit Phisitkul (2018). "Fundamentals of arthroscopic surgery training program improves knee arthroscopy simulator performance in arthroscopic trainees". In: *Arthroscopy: The Journal of Arthroscopic & Related Surgery* 34.5, 1543 (cit. on p. 35).

Pore, Ameya, Eleonora Tagliabue, Marco Piccinelli, Diego Dall'Alba, Alicia Casals, and Paolo Fiorini (2021). "Learning from demonstrations for autonomous soft-tissue retraction". In: *2021 International Symposium on Medical Robotics (ISMR)*. IEEE, 1 (cit. on p. 36).

Xu, Jiaqi, Bin Li, Bo Lu, Yun-Hui Liu, Qi Dou, and Pheng-Ann Heng (2021). "SurRoL: An Open-source Reinforcement Learning Centered and dVRK Compatible Platform for Surgical Robot Learning". In: *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 1821 (cit. on p. 36).

Lam, Kyle, Junhong Chen, Zeyu Wang, Fahad M Iqbal, Ara Darzi, Benny Lo, Sanjay Purkayastha, and James M Kinross (2022). "Machine learning for technical skill assessment in surgery: a systematic review". In: *NPJ digital medicine* 5.1, 1 (cit. on p. 36).

Kassahun, Yohannes, Bingbin Yu, Abraham Temesgen Tibebu, Danail Stoyanov, Stamatia Giannarou, Jan Hendrik Metzen, and Emmanuel Vander Poorten (2016). "Surgical robotics beyond enhanced dexterity instrumentation: a survey of machine learning techniques and their role in intelligent and autonomous surgical actions". In: *International journal of computer assisted radiology and surgery* 11.4, 553 (cit. on p. 36).

Garrow, Carly R, Karl-Friedrich Kowalewski, Linhong Li, Martin Wagner, Mona W Schmidt, Sandy Engelhardt, Daniel A Hashimoto, Hannes G Kenngott, Sebastian Bodenstedt, Stefanie Speidel, Beat P. Müller-Stich, and Felix Nickel (2021). "Machine learning for surgical phase recognition: a systematic review". In: *Annals of surgery* 273.4, 684 (cit. on p. 36).

Lavanchy, Joël L, Joel Zindel, Kadir Kirtac, Isabell Twick, Enes Hosgor, Daniel Candinas, and Guido Beldi (2021). "Automation of surgical skill assessment using a three-stage machine learning algorithm". In: *Scientific reports* 11.1, 1 (cit. on p. 36).

Liu, Daochang, Qiyue Li, Tingting Jiang, Yizhou Wang, Rulin Miao, Fei Shan, and Ziyu Li (2021). "Towards unified surgical skill assessment". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 9522 (cit. on p. 36).

Yoshida, Yuichi and Takeru Miyato (2017). "Spectral norm regularization for improving the generalizability of deep learning". In: *arXiv:1705.10941* (cit. on pp. 42, 67, 73, 74, 93).

Vassiliou, Melina C., Liane S. Feldman, Christopher G. Andrew, Simon Bergman, Karen Leffondré, Donna D. Stanbridge, and Gerald M. Fried (2005a). "A global assessment tool for evaluation of intraoperative laparoscopic skills." In: *American journal of surgery* 190 1, 107 (cit. on p. 46).

Goh, Alvin C., David W. Goldfarb, James C. Sander, Brian J. Miles, and Brian Dunkin (2012). "Global evaluative assessment of robotic skills: validation of a clinical assessment tool to measure robotic surgical skills." In: *The Journal of urology* 187 1, 247 (cit. on pp. 46, 105).

Kitani, Kris M, Brian D Ziebart, James Andrew Bagnell, and Martial Hebert (2012). "Activity forecasting". In: *Computer Vision–ECCV 2012: 12th European Conference on Computer Vision, Florence, Italy, October 7-13, 2012, Proceedings, Part IV 12*. Springer, 201 (cit. on p. 52).

Swamy, Gokul, Sanjiban Choudhury, J Andrew Bagnell, and Steven Wu (2021). "Of moments and matching: A game-theoretic framework for closing the imitation gap". In: *International Conference on Machine Learning*. PMLR, 10022 (cit. on p. 52).

Finn, Chelsea, Paul Christiano, Pieter Abbeel, and Sergey Levine (2016). "A connection between generative adversarial networks, inverse reinforcement learning, and energy-based models". In: *arXiv preprint arXiv:1611.03852* (cit. on pp. 52, 57).

Ying, Xue (2019). "An overview of overfitting and its solutions". In: *Journal of physics: Conference series*. Vol. 1168. IOP Publishing, 022022 (cit. on p. 52).

Song, Xingyou, Yiding Jiang, Stephen Tu, Yilun Du, and Behnam Neyshabur (2019). "Observational overfitting in reinforcement learning". In: *arXiv preprint arXiv:1912.02975* (cit. on p. 52).

Li, Yunzhu, Jiaming Song, and Stefano Ermon (2017). "Infogail: Interpretable imitation learning from visual demonstrations". In: *arXiv preprint arXiv:1703.08840* (cit. on pp. 53, 72).

Arjovsky, Martin, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz (2019). *Invariant Risk Minimization* (cit. on pp. 53, 60, 61, 72, 81, 82, 106).

Levine, Sergey (2018). "Reinforcement learning and control as probabilistic inference: Tutorial and review". In: *arXiv preprint arXiv:1805.00909* (cit. on p. 55).

Wulfmeier, Markus, Peter Ondruska, and Ingmar Posner (2015). "Maximum entropy deep inverse reinforcement learning". In: *arXiv preprint arXiv:1507.04888* (cit. on pp. 55, 63, 66, 81).

Wainwright, Martin J, Michael I Jordan, et al. (2008). "Graphical models, exponential families, and variational inference". In: *Foundations and Trends® in Machine Learning* 1.1–2, 1 (cit. on pp. 56, 62).

Dai, Bo, Hanjun Dai, Arthur Gretton, Le Song, Dale Schuurmans, and Niao He (2019). "Kernel exponential family estimation via doubly dual embedding". In: *The 22nd International Conference on Artificial Intelligence and Statistics*. PMLR, 2321 (cit. on p. 56).

Gilardoni, Gustavo L (2010). "On Pinsker's and Vajda's type inequalities for Csiszár's f-divergences". In: *IEEE Transactions on Information Theory* 56.11, 5377 (cit. on p. 57).

Ni, Tianwei, Harshit Sikchi, Yufei Wang, Tejus Gupta, Lisa Lee, and Ben Eysenbach (2021). "f-IRL: Inverse reinforcement learning via state marginal matching". In: *Conference on Robot Learning* (cit. on pp. 57, 64, 70, 84, 85).

Namkoong, Hongseok and John C Duchi (2016). "Stochastic gradient methods for distributionally robust optimization with f-divergences". In: *Advances in neural information processing systems* 29 (cit. on p. 60).

Viano, Luca, Yu-Ting Huang, Parameswaran Kamalaruban, Adrian Weller, and Volkan Cevher (2021). "Robust inverse reinforcement learning under transition dynamics mismatch". In: *Advances in Neural Information Processing Systems* 34, 25917 (cit. on p. 60).

Rahimian, Hamed and Sanjay Mehrotra (2019). "Distributionally robust optimization: A review". In: *arXiv preprint arXiv:1908.05659* (cit. on p. 60).

Todorov, Emanuel, Tom Erez, and Yuval Tassa (2012). "MuJoCo: A physics engine for model-based control". In: *Intelligent Robots and Systems* (cit. on pp. 67, 84, 93, 123).

Gulrajani, Ishaan, Faruk Ahmed, Martin Arjovsky, Vincent Dumoulin, and Aaron C Courville (2017). "Improved training of wasserstein gans". In: *Advances in neural information processing systems* 30 (cit. on pp. 70, 73, 81).

Ahuja, Kartik, Karthikeyan Shanmugam, Kush Varshney, and Amit Dhurandhar (2020). "Invariant risk minimization games". In: *arXiv preprint arXiv:2002.04692* (cit. on p. 72).

Zhang, Amy, Clare Lyle, Shagun Sodhani, Angelos Filos, Marta Kwiatkowska, Joelle Pineau, Yarin Gal, and Doina Precup (2020). "Invariant causal prediction for block mdps". In: *International Conference on Machine Learning*. PMLR, 11214 (cit. on p. 72).

Sonar, Anoopkumar, Vincent Pacelli, and Anirudha Majumdar (2021). "Invariant policy optimization: Towards stronger generalization in reinforcement learning". In: *Learning for Dynamics and Control*. PMLR, 21 (cit. on p. 72).

Haan, Pim de, Dinesh Jayaraman, and Sergey Levine (2019). "Causal confusion in imitation learning". In: *arXiv preprint arXiv:1905.11979* (cit. on p. 72).

Swamy, Gokul, Sanjiban Choudhury, Drew Bagnell, and Steven Wu (2022). "Causal imitation learning under temporally correlated noise". In: *International Conference on Machine Learning*. PMLR, 20877 (cit. on p. 72).

Bica, Ioana, Daniel Jarrett, and Mihaela van der Schaar (2021). "Invariant causal imitation learning for generalizable policies". In: *Advances in Neural Information Processing Systems* 34, 3952 (cit. on p. 72).

Zolna, Konrad, Scott Reed, Alexander Novikov, Sergio Gomez Colmenarej, David Budden, Serkan Cabi, Misha Denil, Nando de Freitas, and Ziyu Wang (2019). "Task-relevant adversarial imitation learning". In: *arXiv preprint arXiv:1910.01077* (cit. on p. 72).

Tangkaratt, Voot, Bo Han, Mohammad Emtiyaz Khan, and Masashi Sugiyama (2020). "Variational imitation learning with diverse-quality demonstrations". In: *Proceedings of the 37th International Conference on Machine Learning*, 9407 (cit. on p. 73).

Kim, Geon-Hyeong, Seokin Seo, Jongmin Lee, Wonseok Jeon, HyeongJoo Hwang, Hongseok Yang, and Kee-Eung Kim (2021). "Demodice: Offline imitation learning with supplementary imperfect demonstrations". In: *International Conference on Learning Representations* (cit. on pp. 73, 105).

Haug, Luis, Ivan Ovinnikov, and Eugene Bykovets (2020). *Inverse Reinforcement Learning via Matching of Optimality Profiles* (cit. on p. 73).

Kim, Kee-Eung and Hyun Soo Park (2018). "Imitation learning via kernel mean embedding". In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 32. 1 (cit. on p. 73).

Huang, Shengyi, Rousslan Fernand Julien Dossa, Chang Ye, Jeff Braga, Dipam Chakraborty, Kinal Mehta, and João G.M. Araújo (2022). "CleanRL:

High-quality Single-file Implementations of Deep Reinforcement Learning Algorithms". In: *Journal of Machine Learning Research* (cit. on pp. 81, 101).

Lu, Chaochao, Yuhuai Wu, Jośe Miguel Hernández-Lobato, and Bernhard Schölkopf (2022). *Nonlinear Invariant Risk Minimization: A Causal Approach* (cit. on p. 82).

Rothenhäusler, Dominik, Nicolai Meinshausen, Peter Bühlmann, and Jonas Peters (2021). "Anchor regression: Heterogeneous data meet causality". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 83.2, 215 (cit. on pp. 82, 106).

Ahuja, Kartik, Ethan Caballero, Dinghuai Zhang, Jean-Christophe Gagnon-Audet, Yoshua Bengio, Ioannis Mitliagkas, and Irina Rish (2021). "Invariance principle meets information bottleneck for out-of-distribution generalization". In: *Advances in Neural Information Processing Systems* 34, 3438 (cit. on pp. 82, 106).

Bojarski, Mariusz, Davide Del Testa, Daniel Dworakowski, Bernhard Firner, Beat Flepp, Prasoon Goyal, Lawrence D Jackel, Mathew Monfort, Urs Muller, Jiakai Zhang, et al. (2016). *End to end learning for self-driving cars*. Tech. rep. Nvidia (cit. on p. 83).

Rajeswaran, Aravind, Vikash Kumar, Abhishek Gupta, Giulia Vezzani, John Schulman, Emanuel Todorov, and Sergey Levine (2018). "Learning complex dexterous manipulation with deep reinforcement learning and demonstrations". In: *Robotics: Science and Systems* (cit. on pp. 83, 99, 123).

Dadashi, Robert, Léonard Hussenot, Matthieu Geist, and Olivier Pietquin (2021). "Primal Wasserstein imitation learning". In: *International Conference on Learning Representations* (cit. on pp. 84, 86, 93, 101).

Fickinger, Arnaud, Samuel Cohen, Stuart Russell, and Brandon Amos (2022). "Cross-domain imitation learning via optimal transport". In: *International Conference on Learning Representations* (cit. on p. 84).

Kolouri, Soheil, Kimia Nadjahi, Umut Simsekli, Roland Badeau, and Gustavo Rohde (2019). "Generalized sliced Wasserstein distances". In: *Advances in Neural Information Processing Systems* (cit. on pp. 84, 87).

Deshpande, Ishan, Yuan-Ting Hu, Ruoyu Sun, Ayis Pyrros, Nasir Siddiqui, Sanmi Koyejo, Zhizhen Zhao, David Forsyth, and Alexander G Schwing

(2019). "Max-sliced Wasserstein distance and its use for GANs". In: *Computer Vision and Pattern Recognition* (cit. on pp. 84, 87).

Nguyen, Khai, Nhat Ho, Tung Pham, and Hung Bui (2020). *Distributional Sliced-Wasserstein and Applications to Generative Modeling* (cit. on p. 84).

Blondel, Mathieu, Olivier Teboul, Quentin Berthet, and Josip Djolonga (2020). "Fast differentiable sorting and ranking". In: *International Conference on Machine Learning* (cit. on p. 89).

Raffin, Antonin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann (2021). "Stable-Baselines3: Reliable Reinforcement Learning Implementations". In: *Journal of Machine Learning Research* (cit. on pp. 93, 118).

Kostrikov, Ilya, Kumar Krishna Agrawal, Debidatta Dwibedi, Sergey Levine, and Jonathan Tompson (2019). "Discriminator-actor-critic: Addressing sample inefficiency and reward bias in adversarial imitation learning". In: *International Conference on Learning Representations* (cit. on p. 93).

Garg, Divyansh, Shuvam Chakraborty, Chris Cundy, Jiaming Song, and Stefano Ermon (2021). "Iq-learn: Inverse soft-q learning for imitation". In: *Advances in Neural Information Processing Systems* 34, 4028 (cit. on p. 93).

Cohen, Samuel, Alexander Terenin, Yannik Pitcan, Brandon Amos, Marc Peter Deisenroth, and K. S. Sesh Kumar (2021). "Sliced multi-marginal optimal transport". In: *NeurIPS Workshop on Optimal Transport and Machine Learning* (cit. on p. 95).

Zhu, Zhuangdi, Kaixiang Lin, Bo Dai, and Jiayu Zhou (2020). "Off-policy imitation learning from observations". In: *Advances in Neural Information Processing Systems* (cit. on p. 97).

Gupta, Abhishek, Vikash Kumar, Corey Lynch, Sergey Levine, and Karol Hausman (2020). "Relay policy learning: Solving long-horizon tasks via imitation and reinforcement learning". In: *Conference on Robot Learning* (cit. on pp. 100, 124).

Fu, Justin, Aviral Kumar, Ofir Nachum, George Tucker, and Sergey Levine (2020). "D4RL: Datasets for Deep Data-driven Reinforcement Learning". In: *arXiv:2004.07219* (cit. on pp. 100, 123).

Altman, Eitan (2021). *Constrained Markov decision processes*. Routledge (cit. on p. 104).

Lindner, David, Xin Chen, Sebastian Tschiatschek, Katja Hofmann, and Andreas Krause (2024). "Learning safety constraints from demonstrations with unknown rewards". In: *International Conference on Artificial Intelligence and Statistics*. PMLR, 2386 (cit. on p. 104).

Shahriari, Bobak, Kevin Swersky, Ziyu Wang, Ryan P Adams, and Nando De Freitas (2015). "Taking the human out of the loop: A review of Bayesian optimization". In: *Proceedings of the IEEE* 104.1, 148 (cit. on p. 105).

Levine, Sergey, Aviral Kumar, George Tucker, and Justin Fu (2020). "Offline reinforcement learning: Tutorial, review, and perspectives on open problems". In: *arXiv preprint arXiv:2005.01643* (cit. on p. 105).

Kumar, Aviral, Aurick Zhou, George Tucker, and Sergey Levine (2020). "Conservative q-learning for offline reinforcement learning". In: *Advances in Neural Information Processing Systems* 33, 1179 (cit. on p. 105).

Vassiliou, Melina C, Liane S Feldman, Christopher G Andrew, Simon Bergman, Karen Leffondré, Donna Stanbridge, and Gerald M Fried (2005b). "A global assessment tool for evaluation of intraoperative laparoscopic skills". In: *The American journal of surgery* 190.1, 107 (cit. on p. 105).

Rosenfeld, Elan, Pradeep Ravikumar, and Andrej Risteski (2020). "The risks of invariant risk minimization". In: *arXiv preprint arXiv:2010.05761* (cit. on p. 106).

Ahuja, Kartik, Jun Wang, Amit Dhurandhar, Karthikeyan Shanmugam, and Kush R Varshney (2020). "Empirical or invariant risk minimization? a sample complexity perspective". In: *arXiv preprint arXiv:2010.16412* (cit. on p. 106).

Horn, Roger A and Charles R Johnson (2012). *Matrix analysis*. Cambridge university press (cit. on p. 112).

Juliani, Arthur, Vincent-Pierre Berges, Ervin Teng, Andrew Cohen, Jonathan Harper, Chris Elion, Chris Goy, Yuan Gao, Hunter Henry, Marwan Mattar, and Danny Lange (2018). *Unity: A General Platform for Intelligent Agents* (cit. on p. 114).